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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

An optimization problem is one in which you want to find,

moaaz

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Answer (Please select your correct option)

- Not a solution
- An algorithm
- Good solution
- The best solution

correct

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose that a graph $G = (V,E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?

page 116

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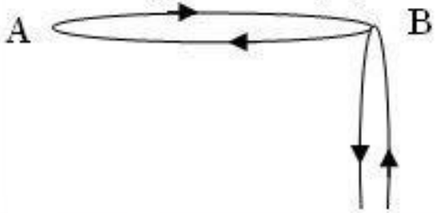
Answer (Please select your correct option)

- $O(|V|^2)$
- $O(|V| \, |E|)$
- $O(|V|^2|E|)$
- $O(|V| + |E|)$

correct

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The total degrees of the graph is



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Answer (Please select your correct option)

- 3
- 5
- 6
- 9

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Answer (Please select your correct option)

- 3
- 5
- 6
- 9

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Answer (Please select your correct option)

- 3
- 5
- 6
- 9

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Non-optimal or greedy algorithm for money change takes_____

Answer (Please select your correct option)

VuAnswers.com

- O(k)

correct

- $O(2^k)$
- O(N)
- O(kN)

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman algorithm finds a (n) _____ solution.

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Answer (Please select your correct option)

- Optimal

correct

- Non-optimal
- Exponential
- Polynomial

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Which formula is used for calculating worst case running time?

VuAnswers.com

Answer (Please select your correct option)

- $Tworst(n) = \max_{|I|=n} T(I)$ correct
- $Tworst(n) = \max_{|I|=1} T(I)$
- $Tworst(n) = \max_{|I|=1} T(n)$
- $Tworst(n) = \max_{|I|=n} T(n)$

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

34

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Answer (Please select your correct option)

- divide-and-conquer

  correct
- decrease and conquer
- greedy nature
- 2-dimension Maxima

Made by: Waqar Siddhu

Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

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Answer (Please select your correct option)

- n items

correct

- phases
- pointers
- constant

Made by: Waqar Siddhu

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

The number of nodes in a complete binary tree of height h is

VuAnswers.com

Answer (Please select your correct option)

- 2^(h+1) – 1 **correct**
- 2 * (h+1) – 1
- 2 * (h+1)
- ((h+1) ^ 2) – 1

Made by: Waqar Siddhu

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

If there are $\Theta(n^2)$ entries in edit distance matrix then the total running time is

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Answer (Please select your correct option)

- $\Theta(1)$
- $\Theta(n^2)$ correct
- $\Theta(n)$
- $\Theta(n \log n)$

Made by: Waqar Siddhu

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

When a recursive algorithm revisits the same problem over and over again, we say that the optimization problem has _____ sub-problems.

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Answer (Please select your correct option)

- Overlapping **correct**
- Over costing
- Optimized
- None of these

Made by: Waqar Siddhu

A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p . r) total entries in C and each takes _____ to compute.

VuAnswers.com

Answer (Please select your correct option)

- (q) **correct**
- (1)
- (n^2)
- (n^3)

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Matrix – Chain – Order is _____ than the exponential time method of enumerating all possible parenthesizations and checking each one.

VuAnswers.com

Answer (Please select your correct option)

- Much more efficient
- Only fractional efficient
- Worst
- Too slow

Made by: Waqar Siddhu

The recurrence relation of Tower of Hanoi is given below

$$T(n)=\begin{cases}1 & \text{if } n=1\\ 2T(n-1)+1 & \text{if } n>1\end{cases}$$

In order to move a tower of 4 rings from one peg to another, how many ring moves are required?

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Answer (Please select your correct option)

- 15
- 7
- 12
- 32

correct

Made by: Waqar Siddhu

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

Maximum number of edges in a Directed Graph may be

VuAnswers.com

Answer (Please select your correct option)

- V
- 2V
- Approximatly $|V^2|$

correct

- v/2

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An adjacency matrix for a graph

VuAnswers.com

Answer (Please select your correct option)

- always square in shape
- It is not necessary for it to be square in shape

correct

- Is square in shape for directed graphs but not for undirected graphs
- Is always diagonal matrix

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

Using ASCII standard the string "ab-#$c" will be encoded _____ bytes

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Answer (Please select your correct option)

- 16
- 7
- 6
- This string cannot be stored using ASCII standard

correct

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Using Huffman encoding technique the string "a@$a" will be encoded with _____bits

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Answer (Please select your correct option)

- 5
- 6
- 8
- Huffman encoding fail at this string

correct

Made by: Waqar Siddhu

Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

In fractional knapsack we sort the

VuAnswers.com

Answer (Please select your correct option)

- Value per unit weight in decreasing order — correct
- Weight per unit value in decreasing order
- Value per unit weight in increasing order
- Weight per unit value in increasing order

Made by: Waqar Siddhu

The greedy part of the Huffman encoding algorithm is to first find two nodes with _____ frequency.

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Answer (Please select your correct option)

- Larger
- Smallest

correct

- Balance
- Character

Made by: Waqar Siddhu

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

The codeword assigned to characters by the Huffman algorithm have the property

VuAnswers.com

Answer (Please select your correct option)

- that no codeword is the prefix of any other

correct

- that no codeword is the postfix of any other
- that no codeword is the infix of any other
- that no codeword is neither prefix nor postfix of any other

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

In undirected graphs there

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Answer (Please select your correct option)

VuAnswers.com

- are no Cross edges but have forward and back edges
- are only forward edges
- is convention of only back edges

correct

- is convention of forward edges

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In time stamp DFS for the edge (u,v) if f(u)>f(v) then

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Answer (Please select your correct option)

- the edge is cross
- the edge is back
- the edge is forward
- the edge is tree or cross or forward

correct

Made by: Waqar Siddhu

Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Kruskal's algorithm

VuAnswers.com

Answer (Please select your correct option)

- Choose the best non-cycle edge **correct**
- Choose the best tree edge
- Choose the vertex that gives the lightest weight
- Follow the dynamic programming rules for choosing edges

Made by: Waqar Siddhu

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

In Prim's algorithm we use

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Answer (Please select your correct option)

- Queue data structure
- Priority Queue data structure

correct

- Stack data structure
- Both stack and Queue data structures

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Adding any edge to a free tree

142

Answer (Please select your correct option)

VuAnswers.com

- keeps it the free tree and increases the size of the tree
- creates a unique cycle

correct

- it is not allowed to add the edge in free tree
- creates multiple cycles

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Floyd-Warshall algorithm is

161

VuAnswers.com

Answer (Please select your correct option)

- based on greedy approach and allow negative edges
- based on divide and conquer approach and allow negative edges
- based on dynamic programming approach and allow negative cycles
- based on dynamic programming approach and allow negative edges

correct

Made by: Waqar Siddhu

Dijkstra's algorithm is used for

154

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Answer (Please select your correct option)

- calculating multiple source shortest path problems
- calculating Minimum spanning tree
- shortest and Minimum Spanning tree both can be calculated by it
- single source shortest path problems

correct

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Bellman Ford algorithm applies relaxation to every

160

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Answer (Please select your correct option)

- edge of the graph and repeats exactly v-1 times

  correct

- vertex of the graph and repeats exactly E-1 times
- edge of the graph and repeats exactly E-1 times
- edge but use the back edges for the completion

Made by: Waqar Siddhu

Complexity wise the comparison based merge and quick sort algorithms fall in

VuAnswers.com

Answer (Please select your correct option)

- Deterministic Polynomial class
- Non-Deterministic Polynomial class
- Quick sort in P class and Merge sort in NP class
- Quick sort in NP class and Merge sort in P class

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Running time of Floyd-Warshall algorithm is

161

Answer (Please select your correct option)

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- $\Theta(n^2)$
- $\Theta(n^4)$
- $\Theta(n^3)$ correct
- $\Theta(3^n)$

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

If we can solve a single NP problem in P time

173

Answer (Please select your correct option)

VuAnswers.com

- All NP-problems can be solved

correct

- All P problems can be solved
- We cannot predict about the solution of other NP problems
- This can be never possible to solve the NP problem in P time

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

If a problem is NP-complete

VuAnswers.com

Answer (Please select your correct option)

- it must be in P
- it must also be in NP

correct

- there is no relation between NP and NP-complete
- it can be solved in P time

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

3-color problem is known as _____

173

VuAnswers.com

Answer (Please select your correct option)

- P
- NPC

correct

- Co-NP
- P and NP

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Clique cover problem arises in applications of _____.

176

VuAnswers.com

Answer (Please select your correct option)

- Map drawing
- Clustering

correct

- Architectural design
- Clique

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

In the 3-coloring problem, for two vertices to be in the same group, they must be not _____to each other.

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VuAnswers.com

Answer (Please select your correct option)

- Apart from
- Far from
- Near to
- Adjacent to

correct

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

What is the worst-case time for merge sort to sort an array of n elements?

VuAnswers.com

Answer (Please select your correct option)

- O(n log n) correct
- $O(n^2)$
- O(log n)
- O(n)

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Search techniques of various algorithms look at____

97

VuAnswers.com

Answer (Please select your correct option)

- Many possible solutions

correct

- Maximum 2 possible solutions
- Minimum 2 possible solutions
- Sorting solutions

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman encoding algorithm is a _____

99

Answer (Please select your correct option)

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- Dynamic and greedy algorithm
- Divide and conquer and greedy algorithm
- Geedy algorithm.

correct

- Dynamic programming algorithm

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Breadth first search is shortest path algorithm that works

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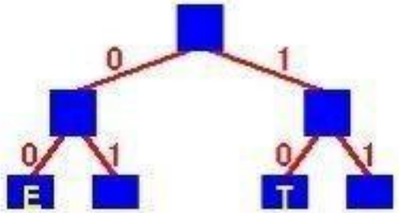
Answer (Please select your correct option)

- on un-weighted graphs — correct
- on weighted graphs
- on both weighted and un-weighted graphs
- BFS cannot be used for shortest path problems

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Huffman Tree



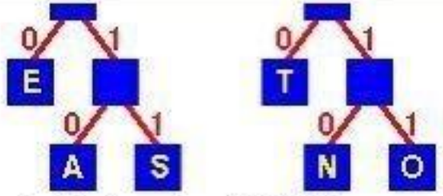
VuAnswers.com

Answer (Please select your correct option)

- 10 00 010 correct
- 011 00 010
- 10 00 110
- 11 10 110

Made by: Waqar Siddhu

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)



The binary code for the string "**TEA**" is

Answer (Please select your correct option)

VuAnswers.com

- 10 00 010 **correct**
- 011 00 010
- 10 00 110
- 11 10 110

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Total running time of BFS is

VuAnswers.com

Answer (Please select your correct option)

- O(V + E) **correct**
- O(V - E)
- O(VE)
- None of these

Made by: Waqar Siddhu

Total running time of BFS is

VuAnswers.com

Answer (Please select your correct option)

- O(V + E) **correct**
- O(V - E)
- O(VE)
- None of these

Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Using ASCII standard the string "abacdaacac" will be encoded with _____ bits.

VuAnswers.com

Answer (Please select your correct option)

- 80 **correct**
- 160
- 320
- 100

Made by: Waqar Siddhu

Consider the string "abacdaacac" if the string is coded with ASCII codes using Huffman encoding scheme, the message length would be

VuAnswers.com

Answer (Please select your correct option)

- 8 bits
- 80 bits

correct

- Less than 50 bits
- More than 50 bits

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What is the asymptotic growth of $\frac{4n^3+15n^2+11n}{6}$?

Answer (Please select your correct option)

VuAnswers.com

- $\Theta\left(\frac{4n^3+15n^2+11n}{6}\right)$
- $\Theta\left(4n^3+15n^2+11n\right)$
- $\Theta\left(15n^2\right)$ **correct**
- $\Theta\left(n^3\right)$

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The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

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Answer (Please select your correct option)

- divide-and-conquer

correct

- decrease and conquer
- greedy nature
- 2-dimension Maxima

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

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Answer (Please select your correct option)

- n items **correct**
- phases
- pointers
- constant

Made by: Waqar Siddhu

A *heap* is a left-complete binary tree that conforms to the _____

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Answer (Please select your correct option)

- (log n) order
- increasing order only
- decreasing order only
- heap order

correct

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

What is common between Bubble sort, Insertion sort, Selection sort, Quick sort, and Heap sort?

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Answer (Please select your correct option)

- All are in-place algorithms **correct**
- All are stable algorithms
- None of these
- All are unstable algorithms

Made by: Waqar Siddhu

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In in-place sorting algorithm is one that uses no _____ arrays for storage.

VuAnswers.com

Answer (Please select your correct option)

- two dimensional
- three dimensional
- n dimensional
- additional

correct

Made by: Waqar Siddhu

The main shortcoming of counting sort is that it is useful for

VuAnswers.com

Answer (Please select your correct option)

- Small Integers **correct**
- Small characters
- Floats
- None of these

Made by: Waqar Siddhu

Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

The original recursive algorithm takes $\Theta(\Phi^n)$ time, where

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Answer (Please select your correct option)

- $\Phi = 1.618$ **correct**
- $\Phi = 3.142$
- $\Phi = 1.816$
- $\Phi = 1.168$

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Maximum number of edges in a Directed Graph may be

Answer (Please select your correct option)

VuAnswers.com

- V
- 2V
- Approximatly $|V^2|$

correct

- v/2

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The Huffman algorithm finds

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Answer (Please select your correct option)

- sometime optimal some time non optimal solution
- space wise optimal and time wise non optimal solution
- a non-optimal solution
- an optimal solution

correct

Made by: Waqar Siddhu

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman codes provide a method of encoding data which

VuAnswers.com

Answer (Please select your correct option)

- is efficient and use a variable length codes
- is efficient and use fixed length codes i.e. ASCII

correct

- is efficient and both ways of variable and fixed length codes can be used
- is efficient time wise but not space wise

Made by: Waqar Siddhu

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Using ASCII standard the string "abacdaacac" will be encoded with _____ bytes.

VuAnswers.com

Answer (Please select your correct option)

- 10
- 16
- 32
- 8

correct

Made by: Waqar Siddhu

Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

In fractional knapsack we sort the

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Answer (Please select your correct option)

- Value per unit weight in decreasing order
- Weight per unit value in decreasing order

correct

- Value per unit weight in increasing order
- Weight per unit value in increasing order

Made by: Waqar Siddhu

The greedy part of the Huffman encoding algorithm is to first find two nodes with _____ frequency.

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Answer (Please select your correct option)

- Larger
- Smallest

correct

100

- Balance
- Character

Made by: Waqar Siddhu

In directed graphs the cardinality of edges |E| =

no idea

VuAnswers.com

Answer (Please select your correct option)

- Sum of out-degrees of all the vertices
- Sum of in-degrees of all the vertices
- First both are true
- There is no relation between degree of vertices and no of edges

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The codeword assigned to characters by the Huffman algorithm have the property

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Answer (Please select your correct option)

- that no codeword is the prefix of any other

  correct

- that no codeword is the postfix of any other
- that no codeword is the infix of any other
- that no codeword is neither prefix nor postfix of any other

Made by: Waqar Siddhu

In undirected graphs there

For undirected graphs, there is no distinction between forward and back edges. By convention they are all called back edges. Furthermore, there are no cross edges (can you see why not?)

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Answer (Please select your correct option)

- are no Cross edges but have forward and back edges
- are only forward edges
- is convention of only back edges

correct

- is convention of forward edges

Made by: Waqar Siddhu

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

In time stamp traversal we can calculate

Answer (Please select your correct option)

VuAnswers.com

- whether the graph has Cycles

correct

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- total number of cycles on the bases of forward edges
- total number of cycles on the bases if back edges
- total no of paths of certain length

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

In time stamp DFS for the edge (u,v) if f(u)>f(v) then

If this edge is a tree, forward or cross edge, then f[u] > f[v].

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VuAnswers.com

Answer (Please select your correct option)

- the edge is cross
- the edge is back
- the edge is forward
- the edge is tree or cross or forward

correct

Made by: Waqar Siddhu

Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

Precedence constraint graph is

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VuAnswers.com

Answer (Please select your correct option)

- non acyclic directed graph
- acyclic undirected graph
- non acyclic undirected graph
- acyclic directed graph

correct

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

In Prim's algorithm, the additional information maintained by the algorithm is

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Answer (Please select your correct option)

- the length of the shortest path from vertex v to the vertex u — correct — not sure
- the length of the shortest edge from vertex *v* to points already in the tree
- the dynamic programming rules
- the information about all adjacent vertices

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

In strongly connected components the component digraph is

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Answer (Please select your correct option)

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- necessarily cyclic

correct

- necessarily acyclic
- not necessary it can be both cyclic and acyclic
- cyclic with some other constraints

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Floyd-Warshall algorithm is

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Answer (Please select your correct option)

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- based on greedy approach and allow negative edges
- based on divide and conquer approach and allow negative edges
- based on dynamic programming approach and allow negative cycles
- based on dynamic programming approach and allow negative edges

correct

Made by: Waqar Siddhu

Dijkstra's algorithm is used for

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Answer (Please select your correct option)

- calculating multiple source shortest path problems
- calculating Minimum spanning tree
- shortest and Minimum Spanning tree both can be calculated by it
- single source shortest path problems

correct

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Kruskal's Algorithm has time complexity

149 page

VuAnswers.com

Answer (Please select your correct option)

- overall Ø (V log E)
- overall Ø (E log V) for sparse graph Ø (V log E)
- overall Ø (E log E) and for sparse graph Ø (E log V)

correct

- overall Ø (EV) and for sparse graph Ø (V^2)

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Bellman Ford algorithm applies relaxation to every

159 page

Bellman-Ford applies relaxation to every edge of the graph and repeats this V − 1 times.

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Answer (Please select your correct option)

- edge of the graph and repeats exactly v-1 times — correct
- vertex of the graph and repeats exactly E-1 times
- edge of the graph and repeats exactly E-1 times
- edge but use the back edges for the completion

Made by: Waqar Siddhu

In NP-problems "NP" represents

The term "NP" does not mean "not polynomial". Originally, the term meant " non-deterministic polynomial"

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Answer (Please select your correct option)

- Non-deterministic Polynomials

  correct

- Null-polynomials
- Negative Polynomials
- Non-polynomials

Made by: Waqar Siddhu

The recurrence represented by $T(n) = \sum_{i=0}^{n} 2 + \sum_{i=0}^{n} i / 2$ has time complexity belongs to

no idea

VuAnswers.com

Answer (Please select your correct option)

- P-Class
- NP-Class
- Co-NP Class
- Unpredictable class

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The function having complexity $O(n^n)$ belongs to

VuAnswers.com

Answer (Please select your correct option)

- NP-Class — correct — not sure
- Co-Prime Class
- P-Class
- Both P and NP Classes

Made by: Waqar Siddhu

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

3-color problem is known as _____

137

VuAnswers.com

Answer (Please select your correct option)

- P
- NPC
- Co-NP
- P and NP

correct

Made by: Waqar Siddhu

Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Generalize Coloring problem arises in various partitioning problems where there is a constraint

173

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Answer (Please select your correct option)

- that two objects can not be assigned to the same set of partitions and is belong to NP class

  correct
- that two objects can not be assigned to the same set of partitions and is belong to P class
- of that we can organize the different partitions in P time and NP space
- of colors does not effect the classifications

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

In the 3-coloring problem, for two vertices to be in the same group, they must be not _____to each other.

176 page

VuAnswers.com

Answer (Please select your correct option)

- Apart from
- Far from
- Near to
- Adjacent to

correct

Made by: Waqar Siddhu

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Sieve Technique can be applied to solve _____

35 page

VuAnswers.com

Answer (Please select your correct option)

- Selection problems — correct
- Arguement problems
- Dynamic problems
- Greedy problems

Made by: Waqar Siddhu

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

If an algorithm has a complexity of $5n + \log_2(\log_2 n) + 10$ for some model of computation (some set of assumptions) and some complexity measures (such as number of comparison operations) we could say that it has complexity

no idea

VuAnswers.com

Answer (Please select your correct option)

- $O(\log_2 n)$
- $O(n)$
- $O(3 + 1 + 3)$
- $O(\log_2(\log_2 n))$

Made by: Waqar Siddhu

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Search techniques of various algorithms look at____

97 page

VuAnswers.com

Answer (Please select your correct option)

- Many possible solutions

correct

- Maximum 2 possible solutions
- Minimum 2 possible solutions
- Sorting solutions

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Usually which type of algorithm is harder to prove the correctness?

Answer (Please select your correct option)

VuAnswers.com

- Dynamic programming
- Brute Force

correct

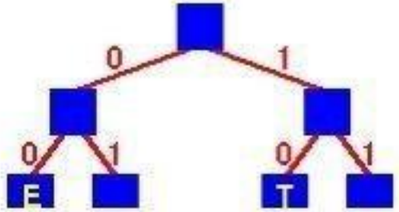
solve by comen fact.....not in the book

- Greedy
- Divide and conquer

Made by: Waqar Siddhu

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Huffman Tree



VuAnswers.com

Answer (Please select your correct option)

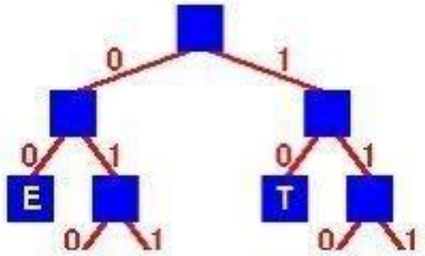
- 10 00 010

correct

- 011 00 010
- 10 00 110
- 11 10 110

Made by: Waqar Siddhu

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

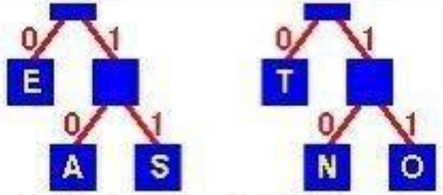


VuAnswers.com

Answer (Please select your correct option)

- 10 00 010 — correct
- 011 00 010
- 10 00 110
- 11 10 110

Made by: Waqar Siddhu



The binary code for the string **"TEA"** is

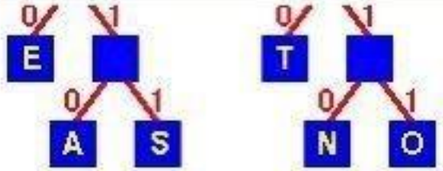
VuAnswers.com

Answer (Please select your correct option)

- 10 00 010 — correct
- 011 00 010
- 10 00 110
- 11 10 110

Made by: Waqar Siddhu

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)



The binary code for the string "**TEA**" is

Answer (Please select your correct option)

VuAnswers.com

- 10 00 010 — correct
- 011 00 010
- 10 00 110
- 11 10 110

Made by: Waqar Siddhu

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

VuAnswers.com

Answer (Please select your correct option)

- *Lists* require less space than *matrices* but take longer to find the weight of an edge (v1,v2) — correct
- *Lists* require less space than *matrices* and they are faster to find the weight of an edge (v1,v2)
- *Lists* require more space than *matrices* and they take longer to find the weight of an edge (v1,v2)
- *Lists* require more space than *matrices* but are faster to find the weight of an edge (v1,v2)

Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

Answer (Please select your correct option)

VuAnswers.com

- v edges.
- v – e + 5 edges
- v + e edges.
- None of these

correct

Made by: Waqar Siddhu

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman algorithm finds a (n) _____ solution.

Answer (Please select your correct option)

VuAnswers.com

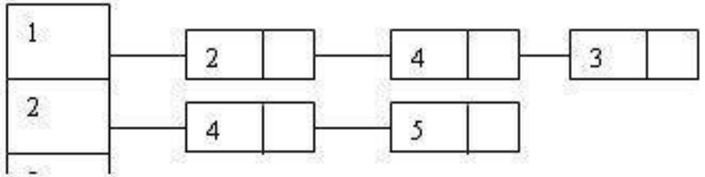
- Optimal

  correct

- Non-optimal
- Exponential
- Polynomial

Made by: Waqar Siddhu

Consider the following adjacency list:



VuAnswers.com

Answer (Please select your correct option)



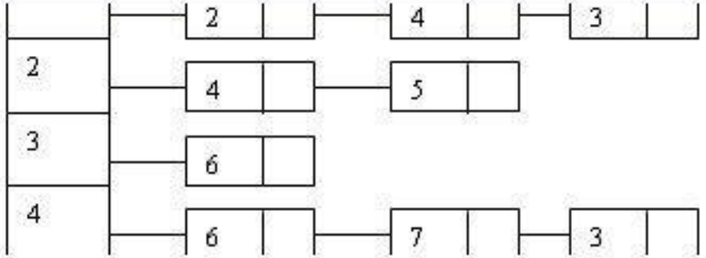






correct

Made by: Waqar Siddhu



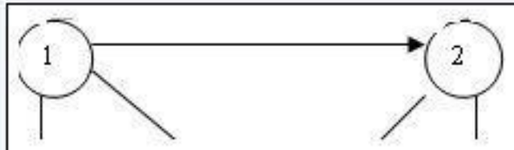
Answer (Please select your correct option)

VuAnswers.com

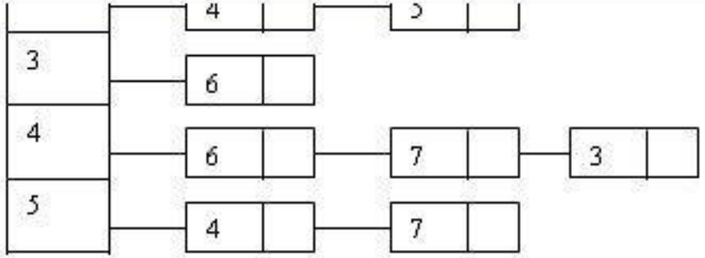








Made by: Waqar Siddhu



Answer (Please select your correct option)

VuAnswers.com

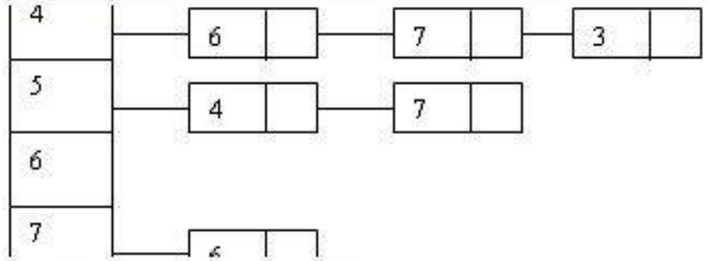








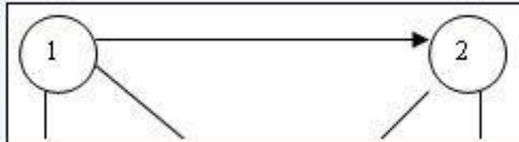
Made by: Waqar Siddhu



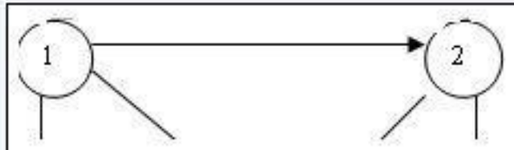
Answer (Please select your correct option)

VuAnswers.com

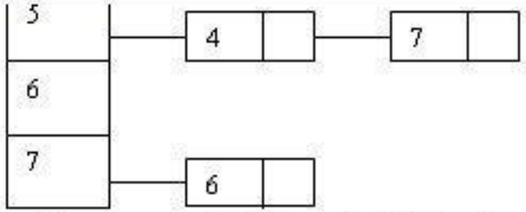








Made by: Waqar Siddhu



Which of the following graph(s) describe(s) the above adjacency list?

Answer (Please select your correct option)

VuAnswers.com









Made by: Waqar Siddhu

Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is a graphical representation of an algorithm

Answer (Please select your correct option)

VuAnswers.com

- Σ notation
- Θ notation
- Flowchart
- Asymptotic notation

correct

Made by: Waqar Siddhu

Identify the maximal points in given set, according to 2-D maxima (the points that are NOT dominated by other points).
$\{(2,5),(4,4),(4,11),(5,1),(7,7),(7,13),(9,10),(11,5),(12,12),(13,3),(14,10),(15,7)\}$

not sure

VuAnswers.com

Answer (Please select your correct option)

- $\{(7,13),(12,12),(14,10),(15,7)\}$ correct
- $\{(7,7),(7,13),(9,10),(11,5),(14,10)\}$
- $\{(2,5),(4,4),(4,11),(5,1),(14,10)\}$
- $\{(4,4),(4,11),(7,13),(9,10)(14,10)\}$

Made by: Waqar Siddhu

What will be result of the following recurrence relation?

$$T(n) = \begin{cases} 4 & \text{if } n = 1 \\ T(n/5) + 3n^2 & \text{if } n \text{ is divisible by } 5 \end{cases}$$

Then T(5) is

VuAnswers.com

Answer (Please select your correct option)

- 25
- 75
- 79
- 70

Made by: Waqar Siddhu

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

When we call heapify then at each level the comparison performed takes time

Answer (Please select your correct option)

VuAnswers.com

- It will take Θ (1)

correct

- Time will vary according to the nature of input data
- It can not be predicted
- It will take Θ (log n)

Made by: Waqar Siddhu

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Who invented Quick sort procedure?

VuAnswers.com

Answer (Please select your correct option)

- Hoare

correct

- Sedgewick
- Mellroy
- Coreman

Made by: Waqar Siddhu

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

The main shortcoming of counting sort is that it is useful for

The main shortcoming of counting sort is that it is useful for small integers. 71 page

VuAnswers.com

Answer (Please select your correct option)

- Small Integers — **correct**
- Small characters
- Floats
- None of these

Made by: Waqar Siddhu

Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

A product of matrices is _____ if it is either single matrix or the product of two matrix products, surrounded by parentheses.

not sure

VuAnswers.com

Answer (Please select your correct option)

- Fully parenthesized — correct
- Partially parenthesized
- Not parenthesized
- None of the options

Made by: Waqar Siddhu

Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Maximum number of edges in a Directed Graph may be

Answer (Please select your correct option)

VuAnswers.com

- V
- 2V
- Approximatly $|V^2|$

correct

- v/2

Made by: Waqar Siddhu

Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

If we encode and compress text using ASCII standard each character is represented by

Answer (Please select your correct option)

VuAnswers.com

- Fixed length codeword of 4 bits
- Variable length codeword up to 4 bits
- Variable length codeword up to 8 bits
- Fixed length codeword of 8 bits.

correct

Made by: Waqar Siddhu

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman Coding uses

101

VuAnswers.com

Answer (Please select your correct option)

- Prefix property that code words are not matched at ends
- No prefix property it has its own method to store the codes
- Prefix property that no code word is prefix of any other code

correct

- Prefix property that no code words at same level of tree are prefix at other levels

Made by: Waqar Siddhu

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

In directed graphs the cardinality of edges |E| =

VuAnswers.com

Answer (Please select your correct option)

- Sum of out-degrees of all the vertices
- Sum of in-degrees of all the vertices
- First both are true

correct

- There is no relation between degree of vertices and no of edges

Made by: Waqar Siddhu

A Hamiltonian cycle is a cycle

VuAnswers.com

Answer (Please select your correct option)

- that visits every vertex in the graph exactly once — correct
- that visits both vertex and edge exactly once
- that visits all vertices without any constraint
- that visits every edge in the graph exactly once

Made by: Waqar Siddhu

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

In generic graph traversal algorithm we

Answer (Please select your correct option)

VuAnswers.com

- put vertices in the bag data structure
- put edges in the bag data structure

correct

- put edges in stack data structure
- put vertices in the stack data structure

Made by: Waqar Siddhu

Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

In time stamp traversal we can calculate

VuAnswers.com

Answer (Please select your correct option)

- whether the graph has Cycles **correct**
- total number of cycles on the bases of forward edges
- total number of cycles on the bases if back edges
- total no of paths of certain length

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Precedence constraint graph is

VuAnswers.com

Answer (Please select your correct option)

- non acyclic directed graph
- acyclic undirected graph
- non acyclic undirected graph
- acyclic directed graph

correct

Made by: Waqar Siddhu

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

In Prim's algorithm, the additional information maintained by the algorithm is

not sure

Answer (Please select your correct option)

VuAnswers.com

- the length of the shortest path from vertex v to the vertex u

  correct

- the length of the shortest edge from vertex *v* to points already in the tree
- the dynamic programming rules
- the information about all adjacent vertices

Made by: Waqar Siddhu

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Bellman Ford algorithm is for the

Answer (Please select your correct option)

VuAnswers.com

- single source shortest path finding problem and does allow negative cycles
- single source shortest path finding problem and does allow negative edges and negative cycles
- multiple-source shortest path finding problem and does allow negative edges

correct

- single source shortest path finding problem and does allow negative edges

Made by: Waqar Siddhu

Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is not true about Dijkstra's algorithm?

VuAnswers.com

Answer (Please select your correct option)

- The length of the shortest path to the start vertex is always zero
- It can find the shortest paths to all other vertices in the same worst case time that it needs to find the shortest path to a single vertex

correct

- It will work on any weighted graph with positive weights
- The running time of Bellman - Ford Algorithm is greater than Dijkstra's algorithm

Made by: Waqar Siddhu

Kruskal's Algorithm is used for

VuAnswers.com

Answer (Please select your correct option)

- calculating shortest path problem
- calculating Minimum spanning tree

correct

- shortest and Minimum Spanning tree both can be calculated by it
- single source shortest path problems

Made by: Waqar Siddhu

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Dijkstra's algorithm is used for

VuAnswers.com

Answer (Please select your correct option)

- calculating multiple source shortest path problems
- calculating Minimum spanning tree
- shortest and Minimum Spanning tree both can be calculated by it
- single source shortest path problems

correct

Made by: Waqar Siddhu

Kruskal's Algorithm has time complexity

VuAnswers.com

Answer (Please select your correct option)

- overall Ø (E log E) and for sparse graph Ø (E log V) **correct**
- overall Ø (EV) and for sparse graph Ø (V^2)
- overall Ø (V log E)
- overall Ø (E log V) for sparse graph Ø (V log E)

Made by: Waqar Siddhu

Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Bellman Ford algorithm applies relaxation to every

VuAnswers.com

Answer (Please select your correct option)

- edge of the graph and repeats exactly E-1 times
- edge but use the back edges for the completion
- edge of the graph and repeats exactly v-1 times

correct

- vertex of the graph and repeats exactly E-1 times

Made by: Waqar Siddhu

Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Complexity wise the comparison based merge and quick sort algorithms fall in

VuAnswers.com

Answer (Please select your correct option)

- Deterministic Polynomial class
- Non-Deterministic Polynomial class
- Quick sort in P class and Merge sort in NP class
- Quick sort in NP class and Merge sort in P class

Made by: Waqar Siddhu

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

In NP-problems "NP" represents

VuAnswers.com

Answer (Please select your correct option)

- Non-deterministic Polynomials

correct

- Null-polynomials
- Negative Polynomials
- Non-polynomials

Made by: Waqar Siddhu

Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Floyd-Warshall algorithm dates back to the early _____.

VuAnswers.com

Answer (Please select your correct option)

- 70's
- 90's
- 60's

correct

- 50's

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Space used by Floyd-Warshall algorithm is

the running time is □(n3). The space used by the algorithm is □(n2).

Answer (Please select your correct option)

VuAnswers.com

- $\Theta(n^4)$
- $\Theta(n^3)$
- $\Theta(n^2)$ **correct**
- $\Theta(2^n)$

Made by: Waqar Siddhu

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

In the clique cover problem, for two vertices to be in the same group, they must be _____each other.

176

VuAnswers.com

Answer (Please select your correct option)

- Apart from
- Far from
- Near to
- Adjacent to

correct

Made by: Waqar Siddhu

Polynomial time certificates

nai pta

VuAnswers.com

Answer (Please select your correct option)

- indicate there are polynomial solutions for NP –class problems
- are the tools to solve the problems in P class in P time
- use in reductions to verify for the NP-problems classes
- use in Polynomial classes to interchange the problems

Made by: Waqar Siddhu

What is the solution to the recurrence T(n) = T(n/2)+n?

VuAnswers.com

Answer (Please select your correct option)

- $O(\log n)$
- $O(n)$ correct
- $O(n \log n)$
- $O(n^2)$

Made by: Waqar Siddhu

If a pseudo code is memory wise efficient then

VuAnswers.com

Answer (Please select your correct option)

- Obviously it will be time wise efficient as well.
- Memory wise efficient codes cannot be time wise efficient
- Time wise efficient code can be memory wise efficient but vise versa is not true.
- It may be memory wise efficient but not necessary

correct

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

Answer (Please select your correct option)

VuAnswers.com

- The array elements form a heap
- Elements in each half of the array are sorted amongst themselves

correct

- Elements in the first half of the array are less than or equal to elements in the second half of the array
- None of the given options

Made by: Waqar Siddhu

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Search techniques of various algorithms look at____

Answer (Please select your correct option)

VuAnswers.com

- Many possible solutions

  correct

- Maximum 2 possible solutions
- Minimum 2 possible solutions
- Sorting solutions

Made by: Waqar Siddhu

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Using ASCII standard each character is represented by a fixed length codeword of _____

Answer (Please select your correct option)

VuAnswers.com

- 9 bits
- 16 bits
- 8 bits

correct

- 32 bits

Made by: Waqar Siddhu

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman encoding algorithm is a _____

Answer (Please select your correct option)

VuAnswers.com

- Dynamic and greedy algorithm
- Divide and conquer and greedy algorithm
- Geedy algorithm.

correct

- Dynamic programming algorithm

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Breadth first search is shortest path algorithm that works

VuAnswers.com

Answer (Please select your correct option)

- on un-weighted graphs **correct**
- on weighted graphs
- on both weighted and un-weighted graphs
- BFS cannot be used for shortest path problems

Made by: Waqar Siddhu

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Suppose that a graph $G = (V,E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?

VuAnswers.com

Answer (Please select your correct option)

- $O(|V|^2)$
- $O(|V| \, |E|)$
- $O(|V|^2|E|)$
- $O(|V| + |E|)$

Made by: Waqar Siddhu

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Non-optimal or greedy algorithm for money change takes_____

Answer (Please select your correct option)

VuAnswers.com

- O(kN)
- $O(2^k)$
- O(N)
- O(k) p 99

Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman algorithm finds a (n) _____ solution.

VuAnswers.com

Answer (Please select your correct option)

- Optimal
- Non-optimal
- Exponential
- Polynomial

Made by: Waqar Siddhu

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Using ASCII standard the string "abacdaacac" will be encoded with _____ bits.

VuAnswers.com

Answer (Please select your correct option)

- 80
- 160
- 320
- 100

Made by: Waqar Siddhu

Consider the string "abacdaacac" if the string is coded with ASCII codes using Huffman encoding scheme, the message length would be

VuAnswers.com

Answer (Please select your correct option)

- 8 bits
- 80 bits
- Less than 50 bits
- More than 50 bits

Made by: Waqar Siddhu

Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Random access machine or RAM is a/an

Answer (Please select your correct option)

VuAnswers.com

- Machine build by Al-Khwarizmi
- Mechanical machine
- Electronics machine
- Mathematical model

p 10

Made by: Waqar Siddhu

Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

In order to say anything meaningful about our algorithms, it will be important for us to settle on a _____.

Answer (Please select your correct option)

VuAnswers.com

- C++ program
- Java program
- Pseudo program
- Mathematical model of computation

p 10

Made by: Waqar Siddhu

Divide-and-conquer involves breaking the problem into a small number of

VuAnswers.com

Answer (Please select your correct option)

- Sub problems p 34
- Selection
- pivot
- Sieve

Made by: Waqar Siddhu

Quick sort procedure was invented by

VuAnswers.com

Answer (Please select your correct option)

- Hoare
- Sedgewick
- Mellroy
- Coreman

Made by: Waqar Siddhu

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

In Bucket sort, if there are duplicates then each bin can be replaced by a

Answer (Please select your correct option)

VuAnswers.com

- Linked list p 69
- Heap
- Hash table
- Stack

Made by: Waqar Siddhu

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

In in-place sorting algorithm is one that uses no _____ arrays for storage.

VuAnswers.com

Answer (Please select your correct option)

- two dimensional
- three dimensional
- n dimensional
- additional

Made by: Waqar Siddhu

A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. In particular, for $1 \leq i \leq p$ and $1 \leq j \leq r$,

VuAnswers.com

Answer (Please select your correct option)

- $C[i, j] = \sum_{k=1}^{q} A[i,k]B[k, j]$ p 84
- $C[i, j] = \sum_{k=1}^{q} A[k,i]B[k, j]$
- $C[i, j] = \sum_{k=1}^{q} A[k,i]B[j,k]$
- None of these

Made by: Waqar Siddhu

Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

For a given chain of matrices A1, A2, A3, A4, the product A1 A2 A3 A4 can be fully parenthesized in _____ distinct way(s).

VuAnswers.com

Answer (Please select your correct option)

- Five
- Four
- Three
- Two

Made by: Waqar Siddhu

Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Worst case running time of Quick Sort algorithm for an array with n elements is?

Answer (Please select your correct option)

VuAnswers.com

- n^2 p 49
- $n^{\frac{n}{2}}$
- n
- n^8

Made by: Waqar Siddhu

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

If we encode and compress text using ASCII standard each character is represented by

Answer (Please select your correct option)

VuAnswers.com

- Fixed length codeword of 4 bits
- Variable length codeword up to 4 bits
- Variable length codeword up to 8 bits
- Fixed length codeword of 8 bits. p 99

Made by: Waqar Siddhu

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman algorithm finds

Answer (Please select your correct option)

VuAnswers.com

- sometime optimal some time non optimal solution
- space wise optimal and time wise non optimal solution
- a non-optimal solution
- an optimal solution

Made by: Waqar Siddhu

Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The Huffman algorithm time complexity

Answer (Please select your correct option)

VuAnswers.com

- Can be improved up to O (n log n)
- Can be improved up to O($\sqrt{n}$ log n)
- Is always O (n^3)
- Is always O (n^2)

Made by: Waqar Siddhu

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Using ASCII standard the string "abacdaac"

Answer (Please select your correct option)

VuAnswers.com

- If we use variable code for ASCII it will be 32 bits
- If we use fixed code for ASCII it will be 256 bits
- If we use variable code for ASCII it will be will 64 bits
- If we use fixed code for ASCII it will be 64 bits

Made by: Waqar Siddhu

Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Using Huffman encoding technique the string "abc" will take ……..

VuAnswers.com

Answer (Please select your correct option)

- 5 bits
- 6 bits
- 24 bits
- 12 bits

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Using Huffman encoding technique the string "a@$a" will be encoded with _____bits

Answer (Please select your correct option)

VuAnswers.com

- 5
- 6
- 8
- Huffman encoding fail at this string

Made by: Waqar Siddhu

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

In fractional knapsack we sort the

Answer (Please select your correct option)

VuAnswers.com

- Value per unit weight in increasing order
- Weight per unit value in increasing order
- Value per unit weight in decreasing order
- Weight per unit value in decreasing order

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

In generic graph traversal algorithm we

Answer (Please select your correct option)

VuAnswers.com

- put vertices in the bag data structure
- put edges in the bag data structure
- put edges in stack data structure
- put vertices in the stack data structure

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

In time stamp traversal we can calculate

Answer (Please select your correct option)

VuAnswers.com

- whether the graph has Cycles
- total number of cycles on the bases of forward edges
- total number of cycles on the bases if back edges
- total no of paths of certain length

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

when the graph has relatively few edges

Answer (Please select your correct option)

VuAnswers.com

- Prim's algorithm is better than Kruskal's
- Kruskal's algorithm is better than Prim's
- No one has priority over each other
- the assumption few edges is not valid

Made by: Waqar Siddhu

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Kruskal's algorithm

Answer (Please select your correct option)

VuAnswers.com

- Choose the best tree edge
- Choose the vertex that gives the lightest weight
- Follow the dynamic programming rules for choosing edges
- Choose the best non-cycle edge

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

In Prim's algorithm we use

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Answer (Please select your correct option)

- Queue data structure
- Priority Queue data structure
- Stack data structure
- Both stack and Queue data structures

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Adding any edge to a free tree

VuAnswers.com

Answer (Please select your correct option)

- keeps it the free tree and increases the size of the tree
- creates a unique cycle
- it is not allowed to add the edge in free tree
- creates multiple cycles

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

An un-weighted graph can be considered as a graph in which every edge has

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Answer (Please select your correct option)

- no weight assigned to it
- by default weight of one unit p 153
- its own different keys of weights
- there are no such type of graphs in theory

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Kruskal's Algorithm is used for

Answer (Please select your correct option)

VuAnswers.com

- calculating shortest path problem
- calculating Minimum spanning tree
- shortest and Minimum Spanning tree both can be calculated by it
- single source shortest path problems

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Kruskal's Algorithm has time complexity

VuAnswers.com

Answer (Please select your correct option)

- overall Ø (E log E) and for sparse graph Ø (E log V)
- overall Ø (EV) and for sparse graph Ø (V^2)
- overall Ø (V log E)
- overall Ø (E log V) for sparse graph Ø (V log E)

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Bellman Ford algorithm applies relaxation to every

VuAnswers.com

Answer (Please select your correct option)

- edge of the graph and repeats exactly E-1 times
- edge but use the back edges for the completion
- edge of the graph and repeats exactly v-1 times
- vertex of the graph and repeats exactly E-1 times

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All algorithms having the time complexity $O(n^{10})$ and $O(n^{100})$ fall

VuAnswers.com

Answer (Please select your correct option)

- Non-Deterministic Polynomial class
- Deterministic Polynomial class
- $O(n^{10})$ in P class and $O(n^{100})$ in NP class
- $O(n^{10})$ in NP class and $O(n^{100})$ in P class

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

In NP-problems "NP" represents

Answer (Please select your correct option)

VuAnswers.com

- Non-deterministic Polynomials
- Null-polynomials
- Negative Polynomials
- Non-polynomials

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Space used by Floyd-Warshall algorithm is

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Answer (Please select your correct option)

- $\Theta(n^4)$
- $\Theta(n^3)$
- $\Theta(n^2)$ p 164
- $\Theta(2^n)$

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

If a problem is NP-complete

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Answer (Please select your correct option)

- there is no relation between NP and NP-complete
- it can be solved in P time
- it must be in P
- it must also be in NP

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Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

If a problem "S" is NP- complete it must be

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Answer (Please select your correct option)

- NP and NP-hard
- NP not necessarily NP-Hard
- NP-hard means it is NP complete as well
- in P and NP

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

In the 3-coloring problem, for two vertices to be in the same group, they must be not _____to each other.

Answer (Please select your correct option)

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- Apart from
- Far from
- Near to
- Adjacent to

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Algorithm's essential elements are

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Answer (Please select your correct option)

- Step wise solution
- Stepwise solution and finite time
- Step wise solution finite inputs
- Stepwise approach in which time and memory does not matter.

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Search techniques of various algorithms look at____

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Answer (Please select your correct option)

- Many possible solutions
- Maximum 2 possible solutions
- Minimum 2 possible solutions
- Sorting solutions

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Which traversal technique is look like propagating wave-front outward

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Answer (Please select your correct option)

- Generic Traversal
- Breadth First Traversal p 117
- Depth First Traversal
- Time Stamp Traversal

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CS502-Design and Analysis of Algorithm

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1. **Which method is preferable for dealing with chain matrix multiplication?**

- Divide and conquer strategy
- **Dynamic programming formulation**
- Graph theory
- Greedy Approach

2. **Huffman algorithm produces the…….prefix code tree.**

- Better
- **Optimal**
- Worst
- Best

3. **A….w is adjacent to vertex v if there is an edge from v to w.**

- Acyclic
- **Vertex**
- Loop
- Cycle

4. **Using ASCII standard the string "greedy" will be encoded with**

i. 44 bits
ii. 120 bits
iii. 40 bits
iv. 48 bits

5. **Find the maximum value of the items which can carry using knapsack weight capacity =50**

Item	weight	Value
	10	**70**
	20	**20**
	30	**80**
	70	**200**

i. 90
ii. 280
iii. 200
iv. 100

6. **In activity scheduling algorithm, each activity is represented by a**

i. Rectangle
ii. Square

iii. Circle
iv. Triangle

7. Those problems in which greedy finds good, but not always best is called a greedy…..

i. Heuristic
ii. Solution
iii. Result
iv. Algorithm

8. The knapsack problem belongs to the domain of…..Problems

i. Searching
ii. Sorting
iii. Linear solution
iv. Optimization

9. The general coin change problem can be solved using

i. Recursion
ii. Greedy algorithm
iii. Dynamic programming
iv. Divide and conquer

10. Huffman algorithm produces the ..... prefix code tree

i. Optimal
ii. Better
iii. Best
iv. Worst

11. Huffman algorithm generates an optimum.....code

i. Postfix
ii. Infix
iii. None of the given options
iv. Prefix

12 .....ways of representing graphics

i. 1
ii. 2
iii. 3
iv. 4

13. Knapsack word originates from ……language

i. German
ii. English
iii. French
iv. Norwegian

14. Graphs are important.....model for many application problems

i. Mathematical
ii. Unpredictable
iii. Haphazard
iv. Unsystematic

15. Which type of algorithm is harder to prove the correctness?

i. Dynamic

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ii. Greedy
iii. Divide and conquer
iv. Brute force

16 .....Items are not allowed in 0/1 knapsack problem

i. Fractional
ii. 0
iii. 1
iv. 0/1

17. Matrix multiplication is a(n)..... operation

i. Nether commutative nor associative
ii. Transitive
iii. Commutative
iv. Associative

18. For a Diagraph G= (V.E), Sum of in-degree (v) ----------.

- Not equal to sum of out-degree(v)
- = sum of out-degree(v) pg#115
- < sum of out-degree(v)
- > sum of out-degree(v)

19. DFS or BFS yields a ----------of the graph.

- Traversed tree
- Spanning tree pg#125
- Simple tree
- Free tree

20. Using ASCII code, each character is represented by a fixed-length code of---------- bits per character.

- 4
- 6
- 10
- 8 pg#100

21. In Knapsack Problem, the goal is to put items in the Knapsack such that the value of the items is---------

-------subject to weight limit of the Knapsack.

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- Minimized
- Decreased
- Maximized pg#109
- None of the above given

22. Consider the string"abcdaacac",if the string is coded with ASCII codes, the message length would be---

- 70 bits
- 60 bits
- 90 bits
- 72 bits

23. A graph is said to be acyclic if it contains --------.

- At least one cycle
- Exactly one cycle
- Always more than one cycle
- No cycles pg#116

24. The number of edges that come out of a vertex is called the---------- of that vertex in the digraph.

- Post-degree
- in-degree
- out-degree pg#114
- pre-degree

25. If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- 3x2
- 2x3
- 2x2
- 3x3pg#87

26. A/an--------- is one in which you want to find, not just a solution, but the best solution.

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- Optimization problem
- Divide and Conquer
- NP complete problem
- Best problem

27. Fractional Knapsack is founded on ----------method.

- Greedy page no109
- Recursive
- Divide and Conquer
- Dynamic programming

28. If the graph is represented using an adjacency matrix, then Breadth-first search takes----------time.

- O(E+1)
- O(V^2)
- O(V)
- O(E)

29. In inductive approach of Knapsack problem, we consider 2 cases, ----------or---------- --.

- Median, Mode
- Recursive, Iterative
- Leave object, Take object pg#93
- Sequentially, Parallel

30. A Greedy algorithm can NOT be used to solve all the---------- problems.

- Dynamic programming (Google)
- Memorization programming
- Edit-distance programming
- Storing value programming

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31. In Huffman encoding, the --------is the number of occurrences of a character divided by the total characters in the message.

- Counting
- Parsing
- Relative Probability pg#100
- Weight

32. The Binary Tree constructed by a Huffman Encoding is a:

- Full Binary Tree pg#102
- Partial Binary Tree
- Incomplete Binary Tree
- None of the given option

33. Following is not the application of Edit Distance Problem.

- Speech recognition pg#76
- Spelling correction
- Ascending order
- Computational Molecular Biology

34. Consider three Matrices X, Y, Z of dimensions 1x2, 2x3, 3x4 respectively. The number of multiplication of(XYZ) is:

- 18
- 32
- 24
- 30

35. In ---------- Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

a. 0

b. 1

c. 0/1 pg#91

d. Fractional

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36. An in-place sorting algorithm is one that ----------uses additional array for storage.

- Always
- Permanently
- Does not pg#54
- Sometime

37. If Matrix-A has dimensions "pxq" and Matrix-B has dimensions "qxr", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- P x q
- P x r page#84
- q x r
- q x p

38. Counting sort is suitable to sort the elements in range 1 to K.

- K is large
- K is small pg#71
- K may be large or small
- None

39. When matrix A of 5x 3 is multiply with matrix B of 3x 4 then the multiplication required is:

- 15
- 12
- 36
- 60

40 ---------- is a linear time sorting algorithm.

- Merge sort
- Quick sort

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- Bubble sort
- Radix sort

41. In Dynamic Programming approach, we do not store the solution to each sub problem in case if it reappears.

- True
- False page#75

42. Dynamic Programming approach is usually useful in solving optimization problem.

- True page#97
- False

43. Which of the following algorithm provides an optimal solution for the activity selection problem?

- Divide and Conquer
- Brute force
- Greedy pg#105
- Recursive

44. A graph is ---------- if every vertex can reach every other vertex.

- Connected pg#116
- Cycle
- Acyclic
- Loop

45. In a Huffman encoding when a new node is created by combining two nodes, the new node is placed in the _____.

- Priority queue pg#100
- Linked list
- Min heap tree
- Graph traversal

46. Huffman algorithm produces the_____prefix code

- Optimal pg#105

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- Best
- Worst
- Better

47. In_____algorithm, you hope that by choosing a local optimum at each step, you will end up a global

optimum.

- Simple
- Divide and conquer
- Greedy pg#97
- Brute Force

48. The string "Imncde" is coded with ASCII code, the message length would be_____bits.

- 24
- 36
- 48
- 60

49. For graph traversal, breadth-first search strategy _____

- Is always recursive
- Cannot br recursive
- Cannot be non-recursive
- Can be both recursive and non-recursive page 119

50. In activity scheduling algorithm , the width of a rectangle _____

- Is always ignored
- Directs towards recursion
- Should be maximized
- Indicates the duration of an activity pg#106

51. If the graph is represented using an adjacency list, then Breadth-first search takes ---------time

- O(V^2)

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- O(V)
- O(V+E) pg#138
- O(E+1)

52. Suppose you are given infinite coins of 1,2 ,3, and 4.Select the ways of the minimum number of coins that required to achieve a sum of 6:

- 1
- 2 Conceptual
- 3
- 4

53. sing ASCII standard the string “greedy” will be encoded with

- 48 bits Conceptual
- 120 bits
- 44 bits
- 40 bits

54. The Huffman codes provide a method of -------data efficiency.

- Reading/Writing
- Encoding/Decoding pg#99
- Divide/Conquer
- Inserting/Deleting

55. In the context of activity selection algorithm, time s dominated by sorting of the activities by------

- Start Times
- Finish Times pg#106
- Average Times

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- CPU Burst Times

56. Time complexity of the "0-1" knapsack algorithm depends on----

- Number of items
- Capacity of the knapsack
- Size of the Table
- Number of items and capacity of knapsack (confirm)

57. The greedy approach gives us an optimal solution when the coins are all powers of a -------- denomination

- Fixed pg#98
- Variable
- Constant
- Static

58. In Activity Selection, we say that two activities are non-interfering if their start-finish interval ------overlap

- Do
- Do not pg#105
- Sometimes
- Once

59. How many steps are involved to design the dynamic programming strategy?

- 2
- 3
- 1
- 4 pg#92

60. Bag is a.....

- type of algorithm pg#119
- data structure

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- program
- compiler

61. If a problem is in NP-complete, it must also be in NP.

► True page#170

► False

62. The Huffman algorithm finds a optimal solution.

► True pg#105

► False

63. The Huffman algorithm finds an exponential solution

► True

► False pg#105

64. The Huffman algorithm finds a polynomial solution

► True google

► **False**

65. The greedy part of the Huffman encoding algorithm is to first find two nodes with **smallest** frequency.

► True pg#100

► **False**

66. The code word assigned to characters by the Huffman algorithm have the property that no code word is the prefix of any other.

► True pg#101

► **False**

67. Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length B (T) of the encoded string.

► True

► **False pg#102**

68. Dijkestra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.

► True

► **False pg#154**

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69. The term "coloring" came form the original application which was in architectural design.

► True

► **False pg#176**

70. In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

► True pg#176

► **False**

71. Dijkstra's algorithm is operates by maintaining a subset of vertices

► True pg#155

► **False**

72. We do sorting to,

► keep elements in random positions

► keep the algorithm run in linear order

► keep the algorithm run in (log n) order

► **keep elements in increasing or decreasing order pg#40**

73. After partitioning array in Quick sort, pivot is placed in a position such that

► **Values smaller than pivot are on left and larger than pivot are on right pg#48**

► Values larger than pivot are on left and smaller than pivot are on right

➢ Pivot is the first element of array

► Pivot is the last element of array

74. Merge sort is stable sort, but not an in-place algorithm

► True pg#54

► **False**

75. A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p **.** r) total entries in C and each takes_____ to compute.

► O (q) pg#84

► O (1)

► O (n^2)

► O (n^3)

76. One of the clever aspects of heaps is that they can be stored in arrays without using any

_____.

- Pointers pg#40

- constants
- variables
- functions

77. Merge sort requires extra array storage,

- True pg#54
- False

78. The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.

- True
- False pg#99

79. Using ASCII standard the string abacdaacac will be encoded with_____bits.

- 80 pg#99
- 160
- 320
- 100

80. Using ASCII standard the string abacdaacac will be encoded with 160 bits.

- True
- False pg#99

81. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

- ❖ True
- ❖ False pg#99

82. The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency

- ❖ True
- ❖ False

83. Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

- ❖ True pg#100
- ❖ False

84. An optimization problem is one in which you want to find,
    - ► Not a solution
    - ► An algorithm
    - ► Good solution
    - ► The best solution pg#97

85. Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vert
    - ► **True pg#149**
    - ► False

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86. If a problem is in NP, it must also be in P.

► True

► False

► **unknown** pg#173

87. What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

► **Lists require less space than matrices but take longer to find the weight of an edge (v1,v2) pg#116**

► *Lists* require less space than *matrices* and they are faster to find the weight of an edge (v1,v2)

► *Lists* require more space than *matrices* and they take longer to find the weight of an edge (v1,v2)

► *Lists* require more space than *matrices* but are faster to find the weight of an edge (v1,

88. If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

► v edges.

► v – e + 5 edges

► v + e edges.

► **None of these**

89. Maximum number of vertices in a Directed Graph may be $|V^2|$

► True

► **False** pg#115

90. The Huffman algorithm finds a (n) _____solution.

► **Optimal pg#105**

► Non-optimal

► Exponential

► Polynomial

91. The Huffman algorithm finds an exponential solution

► True

► **False pg#115**

92. Edge (u, v) is a forward edge if

► u is a proper descendant of v in the tree

► v is a proper descendant of u in the tree pg#129

► None of these

93. After partitioning array in Quick sort, pivot is placed in a position such that

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► **Values smaller than pivot are on left and larger than pivot are on right (Page 48)**

► Values larger than pivot are on left and smaller than pivot are on right

► Pivot is the first element of array

► Pivot is the last element of array

94. Merge sort is stable sort, but not an in-place algorithm

► **True Page #54**

► False

95. In counting sort, once we know the ranks, we simply_____numbers to their final positions in an output array.

► Delete

► **copy Page# 57**

► Mark

► arrange

96. . Dynamic programming algorithms need to store the results of intermediate sub-problems.

► **True pg#75**

► False

97. . A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p **.** r) total entries in C and each takes_____ to compute.

► **O (q) pg#48**

► O (1)

► O (n^2)

► O (n^3)

98. ._____is a graphical representation of an algorithm

► Σ notation

► Θ notation

► **Flowchart**

► Asymptotic notation

99. **.**Which of the following is calculated with **big o notation**?

- ►Lower bounds
- **►Upper bounds pg#25**
- ►Both upper and lower bound
- ►Medium bounds

100. Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

- ►The array elements form a heap
- **►Elements in each half of the array are sorted amongst themselves**
- ►Elements in the first half of the array are less than or equal to elements in the second half of the array
- ►None of the above

101. What is the solution to the recurrence T(n) = T(n/2)+n, T(1) = 1

- ►O(logn)
- **►O(n) pg#37**
- ►O(nlogn)
- ►O(2n)

102. Consider the following Huffman Tree

The binary code for the string **TEA** is

- **►10 00 010**
- ►011 00 010
- ►10 00 110
- ►11 10 110

103. .A greedy algorithm does not work in phases.

► True

► **False pg#97**

104. Can an adjacency matrix for a directed graph ever not be square in shape?

► Yes

► **No pg#116**

105. 44.One of the clever aspects of heaps is that they can be stored in arrays without using any___.

► **Pointers pg#40**
► constants
► variables
► functions

106. .Merge sort requires extra array storage,

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►**True Page 54**
►False

107. .Non-optimal or greedy algorithm for money change takes_____

- ►**O(k) Pg #99**
- ►O(kN)
- ►O(2k)
- ►O(N)

108. The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.

- ►True
- ❖ ►**False Pg# 99**

109. Using ASCII standard the string abacdaacac will be encoded with_____bits.

- ►**80 pg# 99**
- ►160
- ►320
- ►100

110. Using ASCII standard the string abacdaacac will be encoded with 160 bits.

- ►True
- ❖ ►**False (Pg# 99)**

111. Using ASCII standard the string abacdaacac will be encoded with 320 bits.

❖ ►True
❖ ►**False (Pg# 99)**

112. **.**Using ASCII standard the string abacdaacac will be encoded with 100 bits.

❖ ►True
❖ ►**False (Pg# 99)**

113. Using ASCII standard the string abacdaacac will be encoded with 32 bytes

❖ ►True
❖ ►**False (Pg# 99) 8bytes**

114. The greedy part of the Huffman encoding algorithm is to first find two nodes with **smallest** frequency.

❖ ►**True (Pg# 100)**
❖ ►False

115. The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency

❖ ►True
❖ ►**False (Pg# 100)**

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116. .Huffman algorithm uses a greedy approach to generate an **antefix** code T that minimizes the expected length B (T) of the encoded string.

- ►True
- **►False (Pg# 102)**

117. Depth first search is shortest path algorithm that works on un-weighted graphs.

►True
►False (Pg# 159)

118. Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

►True (Pg# 162)
►Flase

119. .Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

- ►k
- **► d_{ij}^{k} (Pg# 164)**
- ►True
- ►Flase

120. .The term coloring came from the original application which was in map drawing.

- **►True (Pg# 176)**
- ►False

121. In the clique cover problem, for two vertices to be in the same group, they must be_____each other.

- ►Apart from
- ►Far from
- ►Near to
- **►Adjacent to (Pg# 176)**

122. . Fixed-length codes may not be efficient from the perspective of_____the total quantity of data.

Select correct option:

►Minimizing Pg# 99

►Averaging

►Maximizing

►Summing

123. In greedy algorithm, at each phase, you take the_____you can get right now, without regard for future consequences.

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► Worst

► Minimum

► Good

► Best Page #97

124. The difference between Prim s algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a same key.

► True

► False Page# 156

125. If a problem is in NP-complete, it must also be in NP.

► True Page# 178

► False

126. If there are n items, there are_____possible combinations of the items.

►2

►n

► 2^n Page# 92

► 3^n

127. .Using ASCII code, each character is represented by a fixed-length code word of _____bits per character.

►4

►6

►8 pg # 99

► 10

128. .In Knapsack Problem, the thief's goal is to put items in the bag such that the _____ of the items does not exceed the limit of the bag.

► Value Page #91

► Weight

► Length

► Balance

129. The knapsack problem does not belong to the domain of optimization problems.

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► True

► **False Page# 91**

130. .In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

► True

► **False Page 100**

131. Fixed-length codes are known for easy break up of a string into its individual characters.

► **True Page# 99**

► False

132. In_____Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items are not allowed.

►0

►1

► **0/1 Page# 91**

► Fractional

133. In Knapsack Problem, value and weight both are to be under consideration.

► **True page #91**

► False

134. Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is_____.

► log n

►n

► n2

► **n3 Page #90**

135. In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____approach.

► Subjective

► **Inductive Page #93**

► Brute force

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► Combination

136. A greedy algorithm sometimes works well for optimization problems.

► **True Page# 97**

► False

137. In Huffman encoding, frequency of each character can be determined by parsing the message and_____how many times each character (or symbol) appears.

► Printing

► Incrementing

► **Counting (Page 100)**

► Deleting

138. Greedy algorithm can do very poorly for some problems.

► **True Page# 97**

► False

139. The Huffman codes provide a method of_____data efficiently.

► Reading

► **Encoding Page# 99**

► Decoding

► Printing

140. In_____based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

► Brute force

► **Dynamic programming Page #93**

141. Those problems in which Greedy finds good, but not always best is called a greedy_.

► Algorithm

► Solution

► **Heuristic Page# 97**

► Result

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142. In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

► TRUE

► **FALSE Page# 97**

143. ._____problem, we want to find the best solution.

► Minimization

► Averaging

► **Optimization Page# 97**

► Maximization

144. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

► **True Page #101**
► False

145. In_____algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

► Simple

► Non Greedy

► **Greedy Page# 97**

► Brute force

146. Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

► **True Page# 102**
► False

147. Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

► **True Page# 97**

► False

148. Huffman algorithm generates an optimum prefix code.

► **True Page# 102**

► False

149. If the string "lmncde" is coded with ASCII code, the message length would be _____bits.

► 24

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► 36

► **48 6*8=48 page #99**

► 60

150. There are_____nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

►2

►**3 Page# 90**

►4

151. .A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

► **True Page# 105**

► False

152. The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an_____solution.

► Simple

► Sub optimal

► **Optimal Page# 105**

► Non optimal

153. The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

► **True (3*8=24) page#99**

► False

154. An application problem is one in which you want to find, not just a solution, but the _____solution.

► Simple

► **Good Page #113**

► Best

155. Suppose that a graph G = (V,E) is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?

► O(|V |^2)

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► O(|V | |E|)

► O(|V |^2|E|)

► O(|V | + |E|) pg #116

156. Which is true statement?

► Breadth first search is shortest path algorithm that works on un-weighted graphs Page #153

► Depth first search is shortest path algorithm that works on un-weighted graphs.

► Both of above are true.

► None of above are true.

157. .Forward edge is:

► (u, v) where u is a proper descendent of v in the tree.

► (u, v) where v is a proper descendent of u in the tree. Page #129

► (u, v) where v is a proper ancesstor of u in the tree.

► (u, v) where u is a proper ancesstor of v in the tree.

158. If you find yourself in maze the better traversal approach will be :

► BFS google

► BFS and DFS both are valid

► Level order

► DFS

159. .In digraph G=(V,E) ;G has cycle if and only if

► The DFS forest has forward edge.

► The DFS forest has back edge Page#131

► The DFS forest has both back and forward edge

► BFS forest has forward edge

160. .Back edge is:

► (u, v) where v is an ancestor of u in the tree. Page# 128

► (u,v) where u is an ancesstor of v in the tree.

► (u, v) where v is an predcessor of u in the tree.

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►None of above

161. Cross edge is :

► (u, v) where u and v are not ancestor of one another

► (u, v) where u is ancesstor of v and v is not descendent of u.

► (u, v) where u and v are not ancestor or descendent of one another Page# 129

► (u, v) where u and v are either ancestor or descendent of one another.

162. Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

► True google

► False

163. What algorithm technique is used in the implementation of Kruskal solution for the MST?

► Greedy Technique Page# 142

► Divide-and-Conquer Technique

► Dynamic Programming Technique

► The algorithm combines more than one of the above techniques

164. What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

► O (log E)

► (V)

► (V+E)

► O (log V) Page #152

165. The relationship between number of back edges and number of cycles in DFS is,

► Both are equal

► Back edges are half of cycles

► Back edges are one quarter of cycles

► There is no relationship between no. of edges and cycles Page# 131

166. You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T.?

► (V + E) Page# 138

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► (V E)

► (V)

► (V^2)

167. There is relationship between number of back edges and number of cycles in DFS

► Both are equal.

► Cycles are half of back edges.

► Cycles are one fourth of back edges.

► **There is no relationship between back edges and number of cycles. Page# 131**

168. A digraph is strongly connected under what condition?

► A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v .

► **A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa. Page# 135**

► A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.

► A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

169. In in-place sorting algorithm is one that uses arrays for storage :

► An additional array

► **No additional array Page#54**

► Both of above may be true according to algorithm

► More than 3 arrays of one dimension.

170. In stable sorting algorithm

► One array is used

► In which duplicating elements are not handled.

► More than one arrays are required.

► **Duplicating elements remain in same relative position after sorting. Page# 54**

171. Which sorting algorithm is faster :

► O(n^2)

► **O(nlogn) Page# 46**

► O(n+k)

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► O(n^3)

172. In Quick sort algorithm, constants hidden in T(n lg n) are

► Large

► Medium

► Not known

► **Small**

173. Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

► There is explicit combine process as well to conquer the solution.
► No work is needed to combine the sub-arrays, the array is already sorted
► Merging the sub arrays
► **None of above. Page# 51**

174. . Dijkstra's algorithm :

► Has greedy approach to find all shortest paths
► Has both greedy and Dynamic approach to find all shortest paths
► **Has greedy approach to compute single source shortest paths to all other vertices Page# 154**

► Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

175. Which may be stable sort:

► Bubble sort
► Insertion sort
► **Both of above page# 54**
► Selection sort

176. In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____series in the analysis,

► linear
► arithmetic
► **geometric page #37**
► exponent

177. How much time merge sort takes for an array of numbers?

► T(n^2)
► **T(n) Page# 40**
► T(log n)
► T(n log n)

178. Dijkstra's algorithm is operates by maintaining a subset of vertices

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❖ **True** **Page# 155**

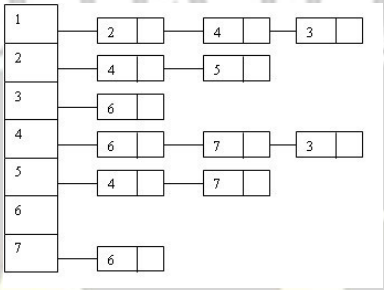
❖ False

179. The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a different key.

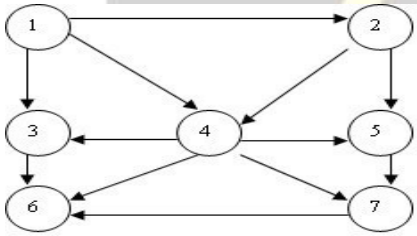
► **True** **Page# 156**

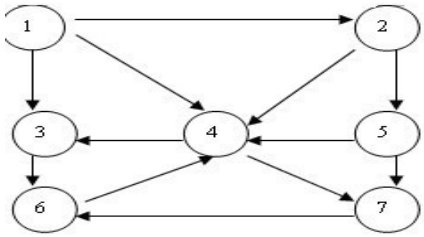
► False

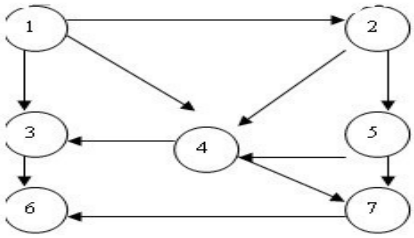
180. Consider the following adjacency list:



Which of the following graph(s) describe(s) the above adjacency list?

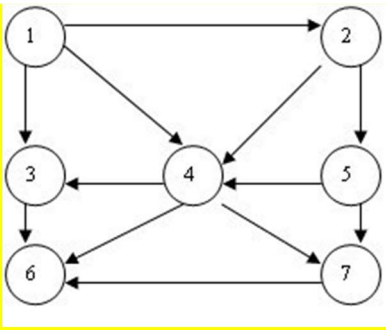






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► **Correct Option pg#116**

181. We do sorting to,

► keep elements in random positions

► keep the algorithm run in linear order

❖ ► keep the algorithm run in (log n) order

❖ ► **keep elements in increasing or decreasing order Page# 40**

182. . After partitioning array in Quick sort, pivot is placed in a position such that

❖ ► **Values smaller than pivot are on left and larger than pivot are on right Page# 48**

❖ ► Values larger than pivot are on left and smaller than pivot are on right

❖ ► Pivot is the first element of array

❖ ► Pivot is the last element of array

183. . Merge sort is stable sort, but not an in-place algorithm

❖ ► **True (Page# 54)**

❖ ► False

184. In counting sort, once we know the ranks, we simply_____numbers to their final positions in an output array.

❖ ► Delete

❖ ► **copy (Page# 57)**

❖ ► Mark

❖ ► arrange

185. Dynamic programming algorithms need to store the results of intermediate sub-problems.

❖ ► **True (Page# 75)**

❖ ► False

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186. A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p . r) total entries in C and each takes_____to compute.

- **► O (q) (Page# 84)**
- ► O (1)
- ► O (n^2)
- ► O (n^3)

187. _____ is a graphical representation of an algorithm

- ► Σ notation
- ► Θ notation
- **► Flowchart**
- ► Asymptotic notation

188. Which of the following is calculated with **big o notation**?

- ►Lower bounds
- **►Upper bounds (Page# 25)**
- ►Both upper and lower bound
- ►Medium bounds

189. Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

- ►The array elements form a heap
- **►Elements in each half of the array are sorted amongst themselves**
- ►Elements in the first half of the array are less than or equal to elements in the second half of the array
- ►None of the above

190. Non-optimal or greedy algorithm for money change takes_____

- **►O(k) (Page #99)**
- ►O(kN)
- ►O(2k)
- ►O(N)

191. The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.

- ►True
- **►False (Page# 99)**

192. Using ASCII standard the string abacdaacac will be encoded with _____ bits.

- **►80 (Page #99)**
- ►160

- ►320
- ►100

193. Using ASCII standard the string abacdaacac will be encoded with 160 bits.

- ►True
- **►False (Page #99)**

194. **u**sing ASCII standard the string abacdaacac will be encoded with 32 bytes

► True
► False (Page #99)

195. The greedy part of the Huffman encoding algorithm is to first find two nodes with **smallest** frequency.

► True (Page #100)
► False

196. The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency

► True
► False (Page# 100)

197. Huffman algorithm uses a greedy approach to generate an **antefix** code T that minimizes the expected length B (T) of the encoded string.

► True
► False (Page #102)

198. Depth first search is shortest path algorithm that works on un-weighted graphs.

► True
► False (Page# 153)

199. 12Dijkstras single source shortest path algorithm works if all edges weights are non negative and there are no negative cost cycles.

► True (Page#159)
► False

200. Dijkestra s single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.

► True
► False (Page #159)

201. Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

► True (Page #162)

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► False

202. Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

►k

► True

► False

203. the term coloring came from the original application which was in map drawing.

► True (Page #176)

► False

204. In the clique cover problem, for two vertices to be in the same group, they must be_____each other.

► Apart from

► Far from

► Near to

► Adjacent to (Page# 176)

205. Fixed-length codes may not be efficient from the perspective of _____the total quantity of data.

Select correct option:

► Minimizing (Page #99)

► Averaging

► Maximizing

► Summing

206. In greedy algorithm, at each phase, you take the_____you can get right now, without regard for future consequences.

► Worst

► Minimum

► Good

► Best (Page #97)

207. The difference between Prim s algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a same key.

► True

► False (Page# 156)

208. If a problem is in NP-complete, it must also be in NP.

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► **True (Page #178)**

► False

209. If there are n items, there are _____possible combinations of the items.

- ►2
- ►n

- **►2^n (Page# 92)**
- ►3^n

210. Using ASCII code, each character is represented by a fixed-length code word of____bits per character.

- ►4
- ►6
- **►8 (Page #99)**
- ►10

211. In Knapsack Problem, the thief's goal is to put items in the bag such that the _____of the items does not exceed the limit of the bag.

- **►Value (Page# 91)**
- ►Weight
- ►Length
- ►Balance

212. The knapsack problem does not belong to the domain of optimization problems.

- ►True
- **►False (Page# 91)**

213. In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

- ►True

- **►False (Page #100)**

214. Fixed-length codes are known for easy break up of a string into its individual characters.

► **True (Page# 99)**

► False

215. In_____Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items are not allowed.

- ►0
- ►1

- **►0/1 (Page# 91)**
- ►Fractional

216. The term "coloring" came from the original application which was in architectural design.

- ► True
- **► False (Page# 173)**

217. In Knapsack Problem, value and weight both are to be under consideration

- ►True (page# 91)

- ►False

218. Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is_____.

- ►log n
- ►n
- n2
- **►n3 (Page# 90)**

219. In DP based solution of knapsack problem, to compute entries of V we will imply a/an_____approach.

- ►Subjective
- **►Inductive (Page# 93)**
- ►Brute force
- ►Combination

220. A greedy algorithm sometimes works well for optimization problems.

- **► True (Page# 97)**
- ► False

221. In Huffman encoding, frequency of each character can be determined by parsing the message and_____how many times each character (or symbol) appears.

- ►Printing
- ►Incrementing
- **►Counting (Page# 100)**
- ►Deleting

222. .Greedy algorithm can do very poorly for some problems.

- **►True (Page# 97)**
- ►False

223. The Huffman codes provide a method of_____data efficiently.

- ►Reading
- **►Encoding (Page# 99)**
- ►Decoding
- ►Printing

224. In_____based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

- ►Brute force
- **►Dynamic programming (Page #93)**

225. Those problems in which Greedy finds good, but not always best is called a greedy_____.

- ►Algorithm
- ►Solution
- **►Heuristic (Page# 97)**
- ►Result

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226. In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

- ►TRUE
- **►FALSE (Page# 97)**

227. _____problem, we want to find the best solution.

- ►Minimization
- ►Averaging
- **►Optimization (Page#97)**
- ►Maximization

228. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

- **►True (Page# 101)**
- ►False

229. In_____algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

- ►Simple
- ►Non Greedy
- **►Greedy (Page# 97)**
- ►Brute force

230. Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

- **►True (Page #102)**
- ►False

231. How many elements do we eliminate in each time for the Analysis of Selection algorithm?

►n / 2 elements (Page #37)
►(n / 2) + n elements
►n / 4 elements
►2 n elements

232. Slow sorting algorithms run in,

►T(n^2) (Page #39)
►T(n)
►T(log n)
►T(n log n)

233. Counting sort is suitable to sort the elements in range 1 to k:

►K is large
►K is small (Page# 57)
►K may be large or small
►None

234. Heaps can be stored in arrays without using any pointers; this is due to the _____nature of the binary tree,

►left-complete (Page# 40)
►right-complete
►tree nodes
►tree leaves

235. Sieve Technique can be applied to selection problem?

►True (Page#35)
►False

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236. A heap is a left-complete binary tree that conforms to the _____

► increasing order only
► decreasing order only
► heap order (Page# 40)
► (log n) order

237. Divide-and-conquer as breaking the problem into a small number of
► pivot
► Sieve
► smaller sub problems (Page #34)
► Selection

238. In Sieve Technique we do not know which item is of interest
► True (Page# 34)
► False

239. The recurrence relation of Tower of Hanoi is given below T(n)={1 if n=1 and 2T(n-1) if n >1 In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

► 16
► 10
► 32
► 31

240. For the heap sort, access to nodes involves simple _____ operations.

► arithmetic (Page# 41)
► binary
► algebraic
► logarithmic

241. For the sieve technique we solve the problem,
► recursively (Page# 34)
► mathematically
► precisely
► accurately

❖ 242. The sieve technique works in_____as follows

► phases (Page #34)
► numbers
► integers
► routines

243. A (an)_____is a left-complete binary tree that conforms to the heap order
► heap (Page# 40)
► binary tree
► binary search tree
► array

244. The sieve technique is a special case, where the number of sub problems is just
►5
► many

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► **1 (Page #34)**
► few

245. Analysis of Selection algorithm ends up with,

► T(n)
► T(1 / 1 + n)
► T(n / 2)
► **T((n / 2) + n) (Page #37)**

246. For the heap sort we store the tree nodes in

► **level-order traversal (Page #40)**
► in-order traversal
► pre-order traversal
► post-order traversal

247. The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

► **divide-and-conquer (Page# 34)**
► decrease and conquer
► greedy nature
► 2-dimension Maxima

248. Theta asymptotic notation for T (n) :

► Set of functions described by: c1g(n)Set of functions described by c1g(n)>=f(n) for c1 s
► Theta for T(n)is actually upper and worst case comp
► **Set of functions described by:**
► c1g(n

249. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

► **n items (Page# 34)**
► phases
► pointers
► constant

250. Memorization is?

► To store previous results for future use
► **To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (Page# 47)**
► To make the process accurate
► None of the above

251. Quick sort is

► Stable & in place
► **Not stable but in place (Page#57)**
► Stable but not in place
► Some time stable & some times in place

252. One example of in place but not stable algorithm is

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► Merger Sort
► **Quick Sort (Page# 54)**
► Continuation Sort
► Bubble Sort

253. . Continuation sort is suitable to sort the elements in range 1 to k

► K is Large
► K is not known
► K may be small or large
► **K is small (Page# 57)**

254. Which may be a stable sort?

► Merger
► Insertion
► **Both above (Page #54)**
► None of the above

255. . An in place sorting algorithm is one that uses____arrays for storage

► Two dimensional arrays
► More than one array
► **No Additional Array (Page #54)**
► None of the above

256. Continuing sort has time complexity of ?

► **O(n**
► O(n+k)
► O(nlogn)
► O(k)

257. single item from a larger set of _____

► **n items (Page# 34)**
► phases

► pointers
► v constant

258. For the Sieve Technique we take time

► **T(nk) (Page# 34)**
► T(n / 3)
► n^2
► n/3

259. One Example of in place but not stable sort is

► **Quick (Page# 54)**
► Heap
► Merge
► Bubble

260. Consider the following Algorithm:

```
Factorial (n){

  if (n=1)

    return 1
```

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else

return (n * Factorial(n-1))

Recurrence for the following algorithm is:

- ► T(n) = T(n-1) +1
- ► T(n) = nT(n-1) +1
- ► T(n)= T(n-1) +n
- **► T(n)=T(n(n-1)) +1**

261. Due to left complete nature of binary tree, the heap can be stored in

- **► Arrays (Page #40)**
- ► Structures
- ► Link List
- ►Stack

262. What type of instructions Random Access Machine (RAM) can execute?

- ►Algebraic and logic
- ►Geometric and arithmetic
- **►Arithmetic and logic (Page# 10)**
- ►Parallel and recursive

263. What is the total time to heapify?

- **► O(log n) (Page 43)**
- ► O(n log n)
- ► $O(n^2 \log n)$
- ► $O(\log^2 n)$

264. word Algorithm comes from the name of the muslim author _____

- **►Abu Ja'far Mohammad ibn Musa al-Khowarizmi.**

265. Al-Khwarizmi's work was written in a book titled _____

- **►al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah**

266. Random access machine or RAM is a/an

- ► Machine build by Al-Khwarizmi
- ► Mechanical machine
- ► Electronics machine
- **► Mathematical model (Page# 10)**

267. A RAM is an idealized machine with_____ random-access memory.

- ► 256MB
- ► 512MB
- **► an infinitely large (Page #10)**
- ► 100GB

268. What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- ► n^2
- ► $n^{\frac{n}{2}}$
- **►** n **(Page# 14)**

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- ► n^8

269. Consider the following code:

```
For(j=1; j<n;j++)
        For(k=1; k<15;k++)
                For(l=5; l<n; l++)
                {
                        Do_something_constant();
                }
```

What is the order of execution for this code.

- **► O(*n*)**
- ► O(n^3)
- ► O(n^2 log n)
- ► O(n^2)

270. Is it possible to sort without making comparisons?

- **► Yes (Page# 57)**
- ► No

271. When we call heapify then at each level the comparison performed takes time

- **► It will take Θ (1) (Page# 43)**
- ► Time will vary according to the nature of input data
- ► It can not be predicted
- ► It will take Θ (log n)

272. In Quick sort, we don't have the control over the sizes of recursive calls

- **► True (Page# 40)**
- ► False
- ► Less information to decide
- ► Either true or false

273. For Chain Matrix Multiplication we cannot use divide and conquer approach because,

- **► We do not know the optimum k (Page# 86)**
- ► We use divide and conquer for sorting only
- ► We can easily perform it in linear time
- ► Size of data is not given

274. The Knapsack problem belongs to the domain of_____problems.

- **► Optimization (Page# 91)**
- ► NP Complete
- ► Linear Solution
- ► Sorting

275. Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. W = 50.

Item	Value	Weight
1	60	10
2	100	20
3	120	30

The optimal solution is to pick

- ► Items 1 and 2
- ► Items 1 and 3
- **► Items 2 and 3**
- ► None of these

276. who invented the quick sort

- **►C.A.R. Hoare**

277. main elements to a divide-and-conquer

- **►Divide, conquer, combine (Page# 27)**

278. Mergesort is a stable algorithm but not an in-place algorithm.

- **►True (Page# 54)**
- ►false

279. Counting sort the numbers to be sorted are in the range 1 to k where k is small.

- **►True (Page# 57)**
- ►False

280. In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

- **►Convergent geometric series (Page 37)**
- ►Divergent geometric series
- ►None of these

281. .In RAM model instructions are executed

- **►One after another (Page# 10)**
- ►Parallel

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- ►Concurrent
- ►Random

282. Due to left-complete nature of binary tree, heaps can be stored in

- ►Link list
- ►Structure
- **►Array (Page# 40)**
- ►None of above

283. The time assumed for each basic operation to execute on RAM model of computation is-----

- ►Infinite
- ►Continuous
- **►Constant (Page# 10)**
- ►Variable

284. If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

- ►True
- **►False (Page# 28)**

285. Brute-force algorithm uses no intelligence in pruning out decisions.

- **►True (Page #18)**
- ►False

286. In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

- **►True (Page #24)**
- ►False

287. For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large.

- ***►True (Page #14)***
- *►False*

288. In simple brute-force algorithm, we give no thought to efficiency.

- **►True (Page# 11)**
- ►False

289. The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

- ►True
- **►False (Page# 27)**

290. In 2d-space a point is said to be_____if it is not dominated by any other point in that space.

- ►Member
- ►Minimal
- **►Maximal (Page# 11)**
- ►Joint

291. An algorithm is a mathematical entity that is dependent on a specific programming language.

- ► True
- **►False (Page# 7)**

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292. The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.

- **►True (Page# 13)**
- ►False

293. F (n) and g (n) are asymptotically equivalent. This means that they have essentially the same_____ for large n.

- ►Results
- ►Variables
- ►Size
- **►Growth rates (Page#23)**

294. 8n2 + 2n - 3 will eventually exceed c2*(n) no matter how large we make c2.

- **►True (Page# 25)**
- ►False

295. If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a_____car.

- ►Fast
- ►Slow
- ►Expensive
- **►Cheap (Page# 11)**

296. The function f(n)= n(logn+1)/2 is asymptotically equivalent to n log n. Here Upper Bound means the function f(n) grows asymptotically_____faster than n log n.

- ► More
- ►Quiet
- **►Not (Page# 24)**
- ►At least

297. Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

- **►True (Page# 97)**
- ►False

298. Huffman algorithm generates an optimum prefix code.

- **►True (Page #102)**
- ►False

299. If the string "lmncde" is coded with ASCII code, the message length would be _____bits.

- ►24
- ►36
- **►48 (6*8=48)**
- ►60

300. There are _____nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

- ►2
- **►3 (Page# 90)**

- ►4
- ►5

301. Inductive approach to compute entries of V is implied in_____based solution of knapsack problem.

- ►Brute force
- **►Dynamic programming (Page #93)**

302. A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

- **►True (Page# 105)**
- ►False

303. The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an_____solution.

- ►Simple
- ►Sub optimal
- **►Optimal (Page# 105)**
- ►Non optimal

304. The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

- **►True (3*8=24)**
- ►False

305. An application problem is one in which you want to find, not just a solution, but the_____solution.

- ►Simple
- **►Good (Page# 113)**
- ►Best
- ►Worst

306. A free tree with n vertices has exactly _____edges.

a. n

b. n+1

c. n-1 page 142

1

307. Kruskal`s algorithm works by adding_____in increasing order of weight (lightest edge first)

a. Verticals

b. Edges page 147

- Trees
- Weights

308. Computing the strongly connected components of a digraphs is a/an _____ of the problem to determine whether a diagraph is strongly connected or not

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a. Size

b. Generalization page 135

- Optimization
- Connection

309. The process of updating estimates in Dijkstra`s algorithm is called _____

a. Updating

b. Amendment

c. Relaxation page 154

- Insertion

310. An un-weighted graph can be considered as a graph in which every edge has weight_____Unit.

a. 7
b. 5
c. 3

d. 1 page 153

311. The breadth-first-search algorithm is a shortest-path algorithm that works on_____graphs.

a. Weighted

b. Directed

c. Un-weighted page 153

d. Un-directed

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312. Which activity creates a unique cycle in a free tree:

a. adding any vertex
b. adding any sub tree
c. adding root

d. adding any edge page 142

313. The relationship between number of back edges and Number of cycles in DFS is,

a. Both are equal

b. Back edges are half of cycles

c. Back edges are one quarter of cycles

d. There is no relationship between no. of edges and cycles google

314. From given algorithms which one considered as best for finding the shortest-path:

- DFS
- Bellman-Ford algorithm
- **Dijkstra's algorithm page 154 / google**
- BFS

315. Overall time for Kruskal algorithm is

a. Θ(log E)

b. Θ(E log V) page 149 / google

c. Θ(E log E)

d. Θ(V log E)

316. Dijkstra's Algorithm is used to solve _____problems.

a. All-pair shortest path

b. Single-source shortest path page 154

- Multi-source shortest path
- Sorting & searching

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317. A digraph is strongly connected under what condition?

- A digraph is strongly connected if for every pair of vertices u, v e V, u cancreach v.

- **A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa.** **Page 135**

- A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.

- A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

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318. Floyd-Warshall Algorithm is based on _____

a. Dynamic Programming page 161

- Greedy Approach
- Divide and Conquer
- Complexity theory

319. . In strong components algorithm, vertices are sorted in_____order of finish times.

a. Any

b. Increasing

c. Decreasing page 141 / google

d. Strong

320. ._____is commonly the running time of Dijkstra's Algorithm using the binary heap method.

a. Θ (E log V) page 156

- Θ (V log)
- Θ (log E)
- Θ (Blog V)

321. Which technique is used in the implementation of Kruskal solution for the MST?

a. Greedy Technique page 142 / google

b. Divide-and-Conquer Technique

c. Dynamic Programming Technique

d. The algorithm combines more than one of the above techniques i.e. Divide-and-Conquer and Dynamic Programming

322. A fully connected undirected graph of 5 nodes will have_____edges.

- 4

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- 5

10 n(n-1)/2

d. 15

323. Prim's algorithm is based on_____strategy.

a. Greedy page 150 / google

b. Dynamic programming

c. Divide and Conquer

d. Exponential

324. We say that two vertices u and v are mutually_____if u can reach v and vice versa.

a. Crossed

b. Forward

c. Reachable page 135

d. Not Reachable

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325. In Dijkstra's algorithm, initially the estimated value from source vertex to any vertex v is:

a. Zero (0)

b. One (1)

c. Minus one (-1)

d. Infinity (∞) page 154

326. A strongly connected component only apply to:

a. Directed Graph page 135 / google

- Undirected Graph
- Minimum Spanning Tree
- Breadth First Search

327. A graph may contain_____

- Exactly one MST
- No MST
- One or zero MST
- **More than one MST google**

328. In _____algorithm(s), at any time, the subset of edges A forms a single tree.

a. Kruskals

b. Prim's page 149

c. kruskal's and Prim's

d. BFS

329. The_____given by DFS allow us to determine a number of things about a graph or

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Digraph

a. color stamps

b. time stamps page 130

c. line stamps

d. node stamps

330. . In Kruskal's algorithm, the vertices will be stored in_____.

a. Links

b. Sets page 147

- nodes
- Loops

331. Keeping in mind the shortest path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are_____. Edges are

_____which may be wired or wireless.

a. Routers, communication links page 153

- Internet, routers
- Communication links, routers
- Routers, internet

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332. In computing the strongly connected components of a digraph. vertices of the digraph

are_____into subsets

a. Joined

b. Partitioned page 135

c. Deleted

d. Created

333. In undirected graph, by convention all the edges are called_____edges.

a. Forward

b. Back page 130

c. Cross

d. Both forward and back

334. In Timestamped DFS, No back edges means_____

a. 1 cycle

b. no cycles page 131

c. DFS

d. BFS

335. . In Prim's algorithm, we start with the_____vertex r, it can be any vertex.

- Pivot
- Leaf
- negative

d. Root page 149

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336. . The ancestor and descendent relation can be nicely inferred by the_____lemma.

a. addition

b. division

c. parenthesis page 129

d. Node

337. Which of the following statement is true?

- Kruskal algorithm is multiple source technique for finding MST.
- Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is 0(EV)
- Both I and II
- **Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges.**

See option B of Sr 46

338. . The time complexity to compute Graph transposes G^T is (V+E), if you have ___ for G.

a. an adjacency list page 138

- Array list
- complete list
- stack

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339. . In Prim's algorithm, If there is no edge from u to a vertex in S. we set the key value to_____

- 0
- 1
- -1

d. ∞ page 151

340. . What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

- O(log E)
- O(V)
- O(V+E)

d. O(logV) page 152

341. . Edge weights can be interpreted as distance _____

- in breadth-First Search
- in Queue's

c. in the shortest-paths page 153

d. in depth-First Search

342. The_____given by DFS allow us to determine whether the graph contains any

cycles.

a. Order

b. Time stamps page 130

c. BFS traversing

d. Topological sort

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343. . Dijkstra's algorithm works on a weighted directed graph G(V, E) in which all weights

_____are non-negative.

a. Vertices

b. Edges page 154

- nodes
- links

344. . By breaking any edge on a cycle created in free tree, the free_____is restored.

a. Edge

b. Tree page 142

c. Cycle

d. Vertex

345. . Bellman-Ford algorithm is used to solve_____problems.

a. All pair shortest path

b. Single source shortest path google

c. Flow of networking

d. Double source shortest path

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346. . From the given options which one is correct regarding the time complexity of Dijkstra's algorithm

- O(N)
- O(N^3)
- O(N^2)

d. O(log N) google

347. . In which algorithm, information of shortest path is propagated sequentially along each

shortest path in the graph.

a. Bellman-Ford page 16

52) Brute-force technique

53) Dijkstra's

54) Prim's

348. In Timestamped DFS-cycles lemma, if edge (u, v) is a tree, forward or cross edge, then a. f[u] < f[v]

b. f[u] > f[v] page 130

52) f[u] ≤ f[v]

53) f[u] ≥ f[v]

349. Bellman-Ford Algorithm does not allow G(graph) to have _____

a. positive cost cycles

b. negative cost cycles page 159

c. negative weights edges

d. positive weights edges

350. Kruskal algorithm (choose best non cycle edge) is better than Prims (choose best tree edge) when the_____has relatively few_____

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a. tree, edges

b. graph edges see option D of Sr 23

- tree, branches
- graph, branches

351. Finding the faster result of the shortest path from u to v for every pair of vertices and we use _____

a. Single-pair shortest-paths problem

b. Two pair shortest the problem

c. All pair shortest paths problem page 153

d. both I and II

352. There are no_____edges in undirected graph.

a. Forward

b. Back

c. Cross page 130

d. Both forward and back

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353. Networks are_____in the sense that it is possible from any location in the network to

reach any other location in the digraph

a. Complete page 135

b. Incomplete

c. Not graphs

d. Transportation

354. . Dijkstra's algorithm:

a. Has greedy approach to find all shortest paths

b. Has both greedy and Dynamic approach to find all shortest paths

c. Has greedy approach to compute single source shortest paths to all other vertices page 154

e. Has both greedy and dynamic approach to compute single source shortest paths to all other vertices

355. Bellman-Ford algorithm is slower than

a. Brute-force technique

b. Dijkstra's page 159

58) Prim's

59) Graph Algorithm

356. Dijkstra's Algorithm cannot be applied on

a. directed and weighted graphs

b. graphs having negative weight function google

- unweighted graphs

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- undirected and unweighted graphs

357. . Which of the following is used in the data structure for implementing Dijkstra's Algorithm?

a. Max heap

b. Stack's

c. Circular queue

d. Priority queue Google

358. . In Generic approach determining of Greedy MST, we maintain a subset A of ____

a. Edges page 143

- Vertices
- Cycles
- Paths

359. In Dijkstra's algorithm the estimated value of source vertex d[s] is

a. Equal to 0 page 154

- Equal to 1
- Greater than 0
- Greater than 1

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360. Dijkstra's algorithm is a simple_____ algorithm for computing the single-source

shortest-paths to all other vertices.

a. Greedy page 154

c. Bellman-Ford

d. Divide and conquer

e. Brute-Force

361. In Timestamped DFS. If there is a back edge (u, v) then v is an ancestor of u and by following tree edge from v to u, we get_____.

a. Nothing

b. a cycle page 131

c. a line

d. a graph

362. There exists a unique path between any_____vertices of a free tree.

a. One

b. Two page 142

- Three
- Four

363. _____technique is look like propagating wave-front outward.

a. Generic traversal

b. Breadth first traversal page 117

c. Depth first traversal

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d. Time stamp traversal

364. You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T?

a. (V+E) page 138

- V.E
- V
- E

365. In the shortest-paths problem, we are given a weighted of_____G =(V, E).

a. Directed graph page 153

65) Line graph

66) Un-directed graph

67) Weighted graph

366. Equivalence relation partitions the vertices into_____classes of mutually reachable

vertices and these are the strong components

a. Variance

b. Equivalence page 136

c. Non equivalence

d. Non classes

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367. Overall Running time of Prim's algorithm is _____.

a. Θ(ElogE)

b. Θ(ElogV)

c. Θ((V+E)logV) page 152

d. Θ((V+E)logE)

368. For_____graphs, there is no distinction between forward and back edges.

a. large

b. directed

c. undirected page 130

d. Medium

369. For_____graphs, there is no distinction between forward and back edges.

a. Undirected page 130

- directed
- small
- large

370. As the Kruskal's algorithm runs, the edges in viable set A induce a_____on the

vertices.

a. Set

b. Graph

c. Tree

d. Forest page 147

371. In strong components algorithm, the form of graph is used in which all the _____of

original graph G have been reversed in direction.

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a. Vertices

b. Edges page 138

c. Both edges & vertices

d. Trees

372. In computing the_____components of a digraph, vertices of the digraph are partitioned into subsets.

a. weakly connected

b. strongly connected page 135

c. Best

d. Worst

373. Timestamp structure of_____is used in determining the strong components of a digraph.

a. DFS google

b. BFS

c. Both DFS & BFS

d. MST

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374. In Prim's algorithm, if the color of a vertex is_____then it is in S otherwise not.

a. White

b. Gray

c. Black page 151

d. Blue

375. Digraphs_____ in communication and transportation networks.

a. are not used

b. are used page 135

c. parts are used

d. final value is used

376. Once you enter a strong component, every vertex in the component is _____.

a. not reachable

b. reachable page 137

c. reachable some times

d. removed

377. In Kruskal's algorithm, the next edge is added to viable set A, if its adding does not induce a/an_____.

a. Vertex

b. Edge

c. Cycle page 147

d. Tree

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378. Problems such as the shortest route between cities can be solved efficiently by modelling the road map as a_____.

a. Tree

b. Graph page 153

c. Linked list

d. Stack

379. In Bellman-Ford Algorithm, relaxation applies to every edge of the graph and repeat this_____time.

a. E – 1

b. E + 1

c. V + 1

d. V – 1 page 159

380. A topological sort of a DAG is a_____ordering of the vertices of the DAG such

that for each edge (u, v), u appears before v in the ordering.

a. Linear page 134

b. Parallel

c. Sequence

d. Non-linear

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381. Adding any edge to a free tree creates a unique_____

a. Vertex

b. Edge

c. Cycle page 142

d. Strong component

382. Back edge is:

a. (u, v) where v is an ancestor of u in the tree. Page 128

b. (u, v) where u is an ancestor of v in the tree.

c. (u, v) where v is a predecessor of u in the tree.

d. (u, v) where u is a mid of v in the tree.

383. In Kruskal's algorithm, the next_____is not added to viable set A, if its adding induce a/an cycle.

a. Vertex

b. Edge page 147

c. Cycle

d. Tree

384. Which of the following statement is false about Dijkstra'sAlgorithm?

a. It can be applied on graphs having a negative weight function google

b. It is used to solve Single-source shortest path

c. It works on a weighted directed graph

d. Its implementation in data structure is possible through the priority queue

385. . In strong components algorithm, first of all DFS is run for getting _____ times of vertices.

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a. Start page 138

b. Finish

c. Both start & finish

d. Middle

386. . The tricky part of_____algorithm(s) is/are, how to detect whether the addition of an edge will create a cycle in viable set A.

a. Kruskal's page 149

b. Prim's

c. Both Kruskal's and Prim's

d. DFS

387. ._____components are not affected by reversal of all edges in terms of vertices reachability.

a. Strongly connected page 139

b. Weakly connected

c. First two

d. Last two

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388. In Prim's algorithm, we will make use of_____.

a. Stack

b. Priority Queue page 150

c. Array

d. List

389. The component digraph is necessarily_____.

a. straight

b. cyclic

c. Acyclic page 136

d. Strong

390. A free tree with n_____have exactly n - 1_____.

a. vertices, nodes

b. edges, vertices

c. nodes, vertices

d. vertices, edges page 142

391. If you find yourself in maze the better traversal approach will be:

a. BFS google (always return with short part)

c. Level order

d. DFS

392. Forward edge is :

a. (u, v) where u is proper descendent of v in the tree

b. (u, v) where v is proper descendent of u in the tree page 129

c. (u, v) where v is proper ancestor of u in the tree

d. (u, v) where u is proper ancestor of v in the tree

393. . cross edge is :

a. (u, v) where u and v are not ancestor of one another

b. (u, v) where u is ancestor of v and v is not descendent of u

c. (u, v) where u and v are not ancestor or descendent of one another page 129

d. (u, v) where u and v are either ancestor of descendent of one another

394. . The running time of Bellman-Ford algorithm is_____.

a. Θ(V + E)

b. Θ(E + E)

c. Θ(VE) page 159

d. Θ(V + V)

395. In Bellman-Ford Algorithm, relaxation applies to_____of the graph.

a. Every edge page 159

b. Every Vertices

c. Only First Vertices

d. Only First edge

396. If a subset of edges A is visible for building MST, it cannot contain a/an_____

a. Vertex

b. Edge

c. Cycle page 143

d. Graph

397. . In Timestamped DFS-cycles lemma, if edge (u, v) is a back edge, then _____.

a. f[u] <f[v]

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b. f[u] > f[v]

c. f[u] ≤ f[v] page 130

d. f[u] ≥ f[v]

398. In digraph G=(V,E) ;G has cycle if and only if

a. The DFS forest has forward edge.

b. The DFS forest has back edge page 131

c. The DFS forest has both back and forward edge

d. BFS forest has forward edge

399. . The key[u] is the weight of the_____going from u to any vertex in S.

a. lightest edge page 151

b. edge

c. lighter edge

d. heavier edge

400. In Bellman-Ford Algorithm, path consists of at most_____edges.

a. V + 1

b. V-1 page 160

c. E + 1

d. E-1

401. According to parenthesis lemma, vertex u is a descendent of v vertex it and only if;

a. [d[u], f[u]] ⊆ [d[v], f[v]] page 129

b. [d[u], f[u]] ⊇ [d[v], f[v]]

c. Unrelated

d. Disjoint

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402. Edge weights can be interpreted as distance_____.

a. in breadth-First Search

b. in Queue's

c. in the shortest-paths page 153

d. in depth-First Search

403. _____algorithm allows negative weights edges and no negative cost cycles.

a. Brute-force technique

b. Bellman-Ford page 159

c. Dijkstra's

d. Print

404. In Huffman encoding, the characters with smallest probabilities are placed at the ---------- depth of the tree.

a. Minimum

b. Average

- Maximum page 102
- Root

405. There are ---------- ways to representing graphs.

- 3
- 1
- 2 page 116

a. 4

406. Each time we traverse graph by Breadth-first search algorithm, we count the distance from ---------.

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a. Starting node

- Neighbours of the starting node page 117
- Right most node
- Left most node

407. In Activity Selection problem, intuitively---------- --.

- There are always short activities as input
- Short activities are not attractive
- Duration of the activities does not matter
- We do not like long activities page105

408. The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an ----------solution.

a. Simple

b. Sub-optimal

- Optimal page 105

c. Non optimal

409. The ---------- is a problem for which the greedy algorithm approach provides an optimal solution.

- Activity selection page 105
- Dynamic programming
- Knapsack problem
- NP complete problem

410. Which of the following ways can be used to represent a graph?

a. Adjacency list

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b. Incidence matrix

- Adjacency list , Adjacency Matrix page 116
- No way to represent

411.

Queue Data Structure work on ---------- principles.

4

- FIFO(first in first out) cs301
- LIFO(last in first out)
- JLO(just in last out)
- LOFI(last out first in)

412. Graphs can be represented by an---------- and ----------.

- queue , stack
- adjacency list , adjacency matrix page 116

a. adjacency right , adjacency left

b. Binary , linear

413. Identify a TRUE statement about Knapsack.

a. The Knapsack Problem does not belongs to the domain of optimization problems

- The Knapsack Problem belongs to the domain of optimization problems page 91

i. The Knapsack Problem cannot be solved by using Dynamic programming

- The Knapsack Problem is optimally solved by using Brute force algorithm

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414. Which of the following is true about graph?

a. A graph may contain no edges and many vertices

- A graph may contain many edges and no vertices

415. Which type of algorithm is harder to prove the correctness?

- Greedygoogle
- Brute force
- Dynamic programming
- Divide and Conquer

416. The running time of brute-force algorithm to solve Knapsack problem is------

- O(h1)
- O(n3)
- O (n!)
- O(2^n) pg#92

417. If a matrix has three rows and two columns, the dimension of matrix will be:

- 3x2 Conceptual
- 2x3
- 3x3
- 2x2

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418. The Probability in Huffman encoding is the number of occurrences of a character divided by the total-

----------in the message.

c. Numbers

d. Frequencies

e. Strings

f. Characters pg#100

419. In recursive formulation of Knapsack Problem: V [0, j] =---------- for j>=0

- -1
- 0 pg#93
- 1
- 2

420. The Knapsack Problem belongs to the domain of---------- problem.

g. Optimization pg#91

h. Sorting

i. Linear solution

j. Searching

421. An optimization problem is one in which you want to find the ---------- solution.

k. Simple

l. Good

m. Best pg#97

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- Worst

422. Huffman algorithm generates an optimum---------- code.

- Prefix page 102
- Postfix
- Infix
- None of the above

423. Which of the following algorithms solves the Fractional Knapsack Problem more effectively?

- Divide and Conquer
- Greedy algorithm page 109
- Dynamic programming
- Backtracking

424. One of the limitation in 0/1 Knapsack is that an item can either be --------- in the bag or not.

- Use
- Put page 91
- Move
- Store

425. In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object'i' then we gain ---------- --.

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a. W(Total weight of Knapsack)

b. V(Total value of all items)

- vi(value of object i) page 93
- None of the given option

426. An activity scheduling algorithm, the width of a rectangle ---------- --.

n. Is always ignored

o. Direct toward recursion

p. Should be maximized

- Indicates the duration of an activity page 106

427. The prefix code generated by Huffman algorithm ----------the expected length of the encoded string.

- Minimizes page 102
- Balances
- Maximizes
- Keeps constant

428. An optimization problem is one in which you want to find,
► Not a solution
► An algorithm
► Good solution
► The best solution (Page 97)

429. Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.
► True
► False

430. If a problem is in NP, it must also be in P.
► True
► False

► unknown (Page 173)

431. What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

► Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)

► Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2)

► Lists require more space than matrices and they take longer to find the weight of an edge (v1,v2)

► Lists require more space than matrices but are faster to find the weight of an edge (v1,v2)

432. If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

► v edges.

► v – e + 5 edges

► v + e edges.

► None of these

433. Maximum number of vertices in a Directed Graph may be |V2 |

► True

► False

434. The Huffman algorithm finds a (n) _____solution.

► Optimal

► Non-optimal

► Exponential

► Polynomial

435. The Huffman algorithm finds an exponential solution

► True

► False

436. The greedy part of the Huffman encoding algorithm is to first find two nodes with larger frequency.

► True

► False (Page 100)

437. one The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the postfix of any other.

► True (Page 101)

► False

438. Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length B (T) of the encoded string.

► True

► False (Page 102)

439. Shortest path problems can be solved efficiently by modeling the road map as a graph.
   - ► True (Page 153)
   - ► False

440. Dijkestra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.
   - ► True
   - ► False (Page 159)

441. Bellman-Ford allows negative weights edges and negative cost cycles.
   - ► True
   - ► False (Page 159)

442. The term "coloring" came form the original application which was in architectural design.
   - ► True
   - ► False (Page 176)

443. In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.
   - ► True (Page 176)
   - ► False

444. Dijkstra's algorithm is operates by maintaining a subset of vertices
   - ► True (Page 155)
   - ► False

445. The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a different key.
   - ► True (Page 156)
   - ► False

446. We do sorting to,
   - ► keep elements in random positions
   - ► keep the algorithm run in linear order
   - ► keep the algorithm run in (log n) order
   - ► keep elements in increasing or decreasing order (Page 40)

447. After partitioning array in Quick sort, pivot is placed in a position such that
   - ► Values smaller than pivot are on left and larger than pivot are on right (Page 48) ► Values larger than pivot are on left and smaller than pivot are on right
   - ► Pivot is the first element of array
   - ► Pivot is the last element of array

448. Merge sort is stable sort, but not an in-place algorithm
   - ► True (Page 54)
   - ► False

449. In counting sort, once we know the ranks, we simply_____numbers to their final positions in an output array.
- ► Delete
- ► copy (Page 57)
- ► Mark
- ► arrange

450. one Dynamic programming algorithms need to store the results of intermediate sub-problems.
- ► True (Page 75)
- ► False

451. A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p . r) total entries in C and each takes_____to compute.
- ► O (q) (Page 84)
- ► O (1)
- ► O (n 2)
- ► O (n3) –

452. _____is a graphical representation of an algorithm
- ► Σ notation
- ► Θ notation
- ► Flowchart
- ► Asymptotic notation

453. one Which of the following is calculated with big o notation?
- ►Lower bounds
- ►Upper bounds (Page 25)
- ►Both upper and lower bound
- ►Medium bounds

454. Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?
- ►The array elements form a heap
- ►Elements in each half of the array are sorted amongst themselves
- ►Elements in the first half of the array are less than or equal to elements in the second half of the array
- ►None of the above 7

455. Who invented Quick sort procedure?
- ►Hoare
- ►Sedgewick
- ►Mellroy
- ►Coreman

456. What is the solution to the recurrence T(n) = T(n/2)+n, T(1) = 1

- ►O(logn)
- ►O(n) (Page 37)
- ►O(nlogn)
- ►O(2n)

457. one Consider the following Huffman Tree The binary code for the string TEA is
- ►10 00 010
- ►011 00 010
- ►10 00 110
- ►11 10 110

458. A greedy algorithm does not work in phases.
►True
►False (Page 97)

459. Can an adjacency matrix for a directed graph ever not be square in shape?
- ►Yes
- ►No

460. One of the clever aspects of heaps is that they can be stored in arrays without using any_____.
- ►Pointers (Page 40)

►constants
- ►variables
- ►functions

461. Merge sort requires extra array storage, •
►True (Page 54) •
►False

462. Non-optimal or greedy algorithm for money change takes_____
- ►O(k) (Page 99)
- ►O(kN)
- ►O(2k)
- ►O(N)

463. The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.
- ►True
- ►False (Page 99)

464. Using ASCII standard the string abacdaacac will be encoded with _____ bits.
- ►80 (Page 99)
- ►160
- ►320

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- ►100

465. Using ASCII standard the string abacdaacac will be encoded with 160 bits.
   - ►True
   - ►False (Page 99)
466. one Using ASCII standard the string abacdaacac will be encoded with 320 bits.
   - ►True
   - ►False (Page 99)
467. Using ASCII standard the string abacdaacac will be encoded with 100 bits.
   - ►True
   - ►False (Page 99)
468. Using ASCII standard the string abacdaacac will be encoded with 32 bytes
   - ►True
   - ►False (Page 99)
469. The greedy part of the Huffman encoding algorithm is to first find two nodes with smallest frequency.
   - ►True (Page 100)
   - ►False
470. The greedy part of the Huffman encoding algorithm is to first find two nodes with character frequency
   - ►True
   - ►False (Page 100)
471. Huffman algorithm uses a greedy approach to generate an antefix code T that minimizes the expected length B (T) of the encoded string.
   - ►True
   - ►False (Page 102)
472. Depth first search is shortest path algorithm that works on un-weighted graphs.
   - ►True

   ►False (Page 153)
473. one Dijkestra s single source shortest path algorithm works if all edges weights are non negative and there are no negative cost cycles.
   - ►True (Page 159)
   - ►False
474. Dijkestra s single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.
   - ►True •

   ►False (Page 159
475. Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

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- ►True (Page 162) • ►Flase

476. Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for
- ►k
- ► k ij d (Page 164)
- ►True
- ►Flase

477. The term coloring came from the original application which was in map drawing.
- ►True (Page 176)
- ►False

478. In the clique cover problem, for two vertices to be in the same group, they must be_____each other.
- ►Apart from
- ►Far from
- ►Near to
- ►Adjacent to (Page 176)

479. Fixed-length codes may not be efficient from the perspective of _____the total quantity of data. Select correct option:
- ►Minimizing (Page 99)
- ►Averaging
- ►Maximizing
- ►Summing

480. In greedy algorithm, at each phase, you take the_____you can get right now, without regard for future consequences.
- ►Worst
- ►Minimum
- ►Good
- ►Best (Page 97)

481. one The difference between Prim s algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a same key.
- ►True
- ►False (Page 156)

482. If a problem is in NP-complete, it must also be in NP.
- ► True (Page 178)
- ► False

483. If there are n items, there are_____possible combinations of the items.
- ►2
- ►n
- ►2^n (Page 92)

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►3^n

484. Using ASCII code, each character is represented by a fixed-length code word of_____bits per character.
►4
►6
►8 (P)age 99)
►10

485. In Knapsack Problem, the thief's goal is to put items in the bag such that the _____of the items does not exceed the limit of the bag.
►Value (Page 91)
►Weight
►Length
►Balance

486. one The knapsack problem does not belong to the domain of optimization problems. ►True
►False (Page 91)

487. In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.
►True
►False (Page 100)

488. Fixed-length codes are known for easy break up of a string into its individual characters.
►True (Page 99)
►False

489. Question No: 8 (Marks: 1) - Please choose one In_____Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items are not allowed.
►0
►1
►0/1 (Page 91)
►Fractional

490. one The term "coloring" came from the original application which was in architectural design.
► True
► False (Page173)

491. In Knapsack Problem, value and weight both are to be under consideration.
►True (page 91)
►False

492. Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is_____.

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► log n
► n
► n2
► n3 (Page 90)

493. Qestion No: 12 (Marks: 1) - Please choose one In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____ approach.
► Subjective
► Inductive (Page 93)
► Brute force
► Combination

494. A greedy algorithm sometimes works well for optimization problems.
► True (Page 97)
► False

495. In Huffman encoding, frequency of each character can be determined by parsing the message and_____how many times each character (or symbol) appears. ►Printing
► Incrementing
► Counting (Page 100)
► Deleting

496. Greedy algorithm can do very poorly for some problems.
► True (Page 97)
► False 14

497. one The Huffman codes provide a method of_____data efficiently.
► Reading
► Encoding (Page 99)
► Decoding
► Printing

498. In_____based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.
► Brute force
► Dynamic programming (Page 93)

499. Those problems in which Greedy finds good, but not always best is called a greedy_____.
► Algorithm
► Solution
► Heuristic (Page 97)
► Result

500. one In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

► TRUE
► FALSE (Page 97)

501. _____problem, we want to find the best solution.
► Minimization
► Averaging
► Optimization (Page 97)
► Maximization

502. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.
• ►True (Page 101)
• ►False

503. In_____algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.
► Simple
► Non Greedy
► Greedy (Page 97)
► Brute force

504. one Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.
• ►True (Page 102)
• ►False

505. Counting Money problem is an example which cannot be optimally solved by greedy algorithm.
► True (Page 97)
► False

506. Huffman algorithm generates an optimum prefix code.
► True (Page 102)
► False

507. If the string "lmncde" is coded with ASCII code, the message length would be _____bits.
► 24
► 36
► 48 (6*8=48)
► 60

508. There are_____nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.
►2
► 3 (Page 90)
►4
►5

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509. Inductive approach to compute entries of V is implied in_____based solution of knapsack problem.
► Brute force
► Dynamic programming (Page 93)

510. A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.
► True (Page 105)
► False

511. The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an_____solution.
► Simple
► Sub optimal
► Optimal (Page 105)
► Non optimal

512. Question # 1 of 10 (Marks: 1) Please choose one The string |xyz|, if coded with ASCII code, the message length would be 24 bits.
► True
► False

513. An application problem is one in which you want to find, not just a solution, but the_____solution.
► Simple
► Good (Page 113)
► Best
► Worst

514. A dense undirected graph is:
► A graph in which E = O(V^2) click here 4 detail
► A graph in which E = O(V)
► A graph in which E = O(log V)
► All items above may be used to characterize a dense undirected graph

515. Suppose that a graph G = (V,E) is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?
► O(|V |^2)
► O(|V | |E|)
► O(|V |^2|E|)
► O(|V | + |E|) pg 116

516. Which is true statement?
► Breadth first search is shortest path algorithm that works on un-weighted graphs (Page 153)
► Depth first search is shortest path algorithm that works on un-weighted graphs. ►Both of above are true.

► None of above are true.

517. one Forward edge is:
► (u, v) where u is a proper descendent of v in the tree.
► (u, v) where v is a proper descendent of u in the tree. (Page 129)
► (u, v) where v is a proper ancesstor of u in the tree.
► (u, v) where u is a proper ancesstor of v in the tree.

518. What general property of the list indicates that the graph has an isolated vertex? ►There is Null pointer at the end of list.
► The Isolated vertex is not handled in list.
► Only one value is entered in the list.
► There is at least one null list.

519. one If you find yourself in maze the better traversal approach will be :
► BFS
► BFS and DFS both are valid
► Level order
► DFS

520. In digraph G=(V,E) ;G has cycle if and only if
► The DFS forest has forward edge.
► The DFS forest has back edge (Page 131)
► The DFS forest has both back and forward edge
► BFS forest has forward edge

521. Back edge is:
► (u, v) where v is an ancestor of u in the tree. (Page 128)
► (u,v) where u is an ancesstor of v in the tree.
► (u, v) where v is an predcessor of u in the tree.
► None of above

522. Which statement is true?
► If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.
► If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.
► Both of above
► None of above

523. Cross edge is :
► (u, v) where u and v are not ancestor of one another
► (u, v) where u is ancesstor of v and v is not descendent of u.
► (u, v) where u and v are not ancestor or descendent of one another (Page 129)
► (u, v) where u and v are either ancestor or descendent of one another.

524. Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.
► True click here 4 detail
► False

525. Which is true statement in the following?
► Kruskal algorithm is multiple source technique for finding MST. click here for detail ►Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is O(EV)
► Both of above
► Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges. click here 4 detail

526. Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.
► True click here 4 detail
► False

527. What algorithm technique is used in the implementation of Kruskal solution for the MST?
► Greedy Technique (Page 142)
► Divide-and-Conquer Technique
► Dynamic Programming Technique
► The algorithm combines more than one of the above techniques

528. What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?
► O (log E)
► (V)
► (V+E)
► O (log V) (Page 152)

529. The relationship between number of back edges and number of cycles in DFS is, ►Both are equal
► Back edges are half of cycles
► Back edges are one quarter of cycles
► There is no relationship between no. of edges and cycles (Page 131)

530. You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T.?
► (V + E) (Page 138)
► (V E)
► (V)

531. A digraph is strongly connected under what condition?

► A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v . ►A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa. (Page 135)

► A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.

► A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

532. In in-place sorting algorithm is one that uses arrays for storage :

► An additional array

► No additional array (Page 54)

► Both of above may be true according to algorithm

► More than 3 arrays of one dimension.

533. In stable sorting algorithm

► One array is uses

► In which duplicating elements are not handled.

► More then one arrays are required.

► Duplicating elements remain in same relative position after sorting. (Page 54)

534. Which sorting algorithm is faster :

► O(n^2)

► O(nlogn) (Page 46)

► O(n+k)

► O(n^3)

535. In Quick sort algorithm, constants hidden in T(n lg n) are

► Large

► Medium

► Not known

► Small

536. Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

► There is explicit combine process as well to conquer the solution.

► No work is needed to combine the sub-arrays, the array is already sorted

► Merging the sub arrays

► None of above. (Page 51

537. Dijkstra's algorithm :

► Has greedy approach to find all shortest paths

► Has both greedy and Dynamic approach to find all shortest paths

► Has greedy approach to compute single source shortest paths to all other vertices (Page 154)

► Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

538. Which may be stable sort:
► Bubble sort
► Insertion sort
► Both of above (page 54)
► Selection sort

539. Question # 1 of 10 (Marks: 1) Please choose one In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent_____series in the analysis,
► linear
► arithmetic
► geometric (page 37)
► exponent

540. How much time merge sort takes for an array of numbers?
► T(n^2)
► T(n) (Page 40)
► T(log n)
► T(n log n)

541. Counting sort has time complexity:
►O(n) ►O(n+k)
► O(k)
► O(nlogn)

542. The analysis of Selection algorithm shows the total running time is indeed _____in no
► arithmetic
► geometric
► linear (Page 37)
► orthogonal

543. Sorting is one of the few problems where provable_____bonds exits on how fast we can sort,
► upper
► lower (Page 39)
► average
► log n

544. In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as
► T(n)
► T(n / 2)
► log n (Page 37)

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► n / 2 + n / 4

545. number of nodes in a complete binary tree of height h is
► 2^(h+1) – 1 (Page 40)
► 2 * (h+1) – 1
► 2 * (h+1)
► ((h+1) ^ 2) – 1 25

546. How many elements do we eliminate in each time for the Analysis of Selection algorithm?
► n / 2 elements (Page 37)
► (n / 2) + n elements
► n / 4 elements
► 2 n elements

547. Slow sorting algorithms run in,
► T(n^2) (Page 39)
► T(n)
► T(log n)
► T(n log n)

548. Counting sort is suitable to sort the elements in range 1 to k:
► K is large
► K is small (Page 57)
► K may be large or small
► None

549. Heaps can be stored in arrays without using any pointers; this is due to the _____nature of the binary tree,
► left-complete (Page 40)
► right-complete
► tree nodes
► tree leaves

550. Sieve Technique can be applied to selection problem?
► True (Page 35)
► False

551. A heap is a left-complete binary tree that conforms to the _____
► increasing order only
► decreasing order only
► heap order (Page 40)
► (log n) order

552. Divide-and-conquer as breaking the problem into a small number of
► pivot
► Sieve
► smaller sub problems (Page 34)

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► Selection

553. In Sieve Technique we do not know which item is of interest
► True (Page 34)
► False

554. The recurrence relation of Tower of Hanoi is given below T(n)={1 if n=1 and 2T(n-1) if n >1 In order to move a tower of 5 rings from one peg to another, how many ring moves are required?
► 16
► 10
► 32
► 31 Click here 4 detail

555. one For the heap sort, access to nodes involves simple _____ operations.
► arithmetic (Page 41)
► binary
► algebraic
► logarithmic

556. For the sieve technique we solve the problem,
► recursively (Page 34)
► mathematically
► precisely
► accurately

557. The sieve technique works in_____as follows
► phases (Page 34)
► numbers
► integers
► routines

558. A (an)_____is a left-complete binary tree that conforms to the heap order
► heap (Page 40)
► binary tree
► binary search tree
► array

559. The sieve technique is a special case, where the number of sub problems is just
►5
► many
► 1 (Page 34)
► few

560. Analysis of Selection algorithm ends up with,
► T(n)
► T(1 / 1 + n)

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► T(n / 2)
► T((n / 2) + n) (Page 37)

561. For the heap sort we store the tree nodes in
► level-order traversal (Page 40)
► in-order traversal
► pre-order traversal
► post-order traversal

562. The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,
► divide-and-conquer (Page 34)
► decrease and conquer
► greedy nature
► 2-dimension Maxima

563. Theta asymptotic notation for T (n) :
► Set of functions described by: c1g(n)Set of functions described by c1g(n)>=f(n) for c1 s
► Theta for T(n)is actually upper and worst case comp
► Set of functions described by:
► c1g(n)

564. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____
► n items (Page 34)
► phases
► pointers
► constant

565. Memorization is?
► To store previous results for future use
► To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (Page 47)
► To make the process accurate
► None of the above

566. Quick sort is
► Stable & in place
► Not stable but in place (Page 57)
► Stable but not in place
► Some time stable & some times in place

567. One example of in place but not stable algorithm is
► Merger Sort
► Quick Sort (Page 54)
► Continuation Sort

► Bubble Sort

568. Continuation sort is suitable to sort the elements in range 1 to k
► K is Large
► K is not known
► K may be small or large
► K is small (Page 57)

569. Which may be a stable sort?
► Merger
► Insertion
► Both above (Page 54)
► None of the above

570. An in place sorting algorithm is one that uses____arrays for storage
► Two dimensional arrays
► More than one array
► No Additional Array (Page 54)
► None of the above

571. Continuing sort has time complexity of ?
► O(n)
► O(n+k)
► O(nlogn)
► O(k)

572. single item from a larger set of _____
► n items (Page 34)
► phases
► pointers
► vconstant

573. For the Sieve Technique we take time
► T(nk) (Page 34)
► T(n / 3)
► n^2
► n/3

574. One Example of in place but not stable sort is
► Quick (Page 54)
► Heap
► Merge
► Bubble

575. Consider the following Algorithm: Factorial (n){ if (n=1) return 1 else return (n * Factorial(n-1)) } Recurrence for the following algorithm is:
► T(n) = T(n-1) +1
► T(n) = nT(n-1) +1

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► T(n)= T(n-1) +n
► T(n)=T(n(n-1)) +1

576. Due to left complete nature of binary tree, the heap can be stored in
► Arrays (Page 40)
► Structures
► Link Lis
► Stack

577. What type of instructions Random Access Machine (RAM) can execute?
► Algebraic and logic
► Geometric and arithmetic
► Arithmetic and logic (Page 10)
► Parallel and recursive

578. What is the total time to heapify?
► O(log n) (Page 43)
► O(n log n)
► O(n2 log n)
► O(log2 n)

579. word Algorithm comes from the name of the muslim author _____
► Abu Ja'far Mohammad ibn Musa al-Khowarizmi.

580. al-Khwarizmi's work was written in a book titled _____
► al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah

581. Random access machine or RAM is a/an
► Machine build by Al-Khwarizmi
► Mechanical machine
► Electronics machine
► Mathematical model (Page 10)

582. A RAM is an idealized machine with_____random-access memory.
► 256MB
► 512MB
► an infinitely large (Page 10)
► 100GB

583. What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?
► 2 n
► 2 n n
► n (Page 14)
► 8 n

584. Due to left complete nature of binary tree, the heap can be stored in
► Arrays (Page 40)

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► Structures
► Link Lis
► Stack

585. one What type of instructions Random Access Machine (RAM) can execute?
► Algebraic and logic
► Geometric and arithmetic
► Arithmetic and logic (Page 10)
► Parallel and recursive

586. What is the total time to heapify?
► O(log n) (Page 43)
► O(n log n)
► O(n2 log n)
► O(log2 n)

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► 256MB
► 512MB
► an infinitely large (Page 10)
► 100GB

589. What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?
► 2 n
► 2 n n
► n (Page 14)
► 8 n

590. The sieve technique works where we have to find_____item(s) from a large input.
► Single (Page 34)
► Two
► Three
► Similar

591. In which order we can sort?
► increasing order only
► decreasing order only
► increasing order or decreasing order (Page 39)

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► both at the same time

592. For the heap sort we store the tree nodes in
► level-order traversal (Page 40)
► in-order traversal
► pre-order traversal
► post-order traversal

593. In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent_____series in the analysis, ►linear
► arithmetic
► geometric (Page 37)
► exponent

594. How much time merge sort takes for an array of numbers?
► T(n^2)
► T(n)
► T(log n)
► T(n log n) (Page 40)

595. Memoization is?
► To store previous results for future use
► To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (page 74)
► To make the process accurate
► None of the above

596. Cont sort is suitable to sort the elements in range 1 to k
► K is Large
► K is not known
► K may be small or large
► K is small (Page 57)

597. In place stable sorting algorithm.
► If duplicate elements remain in the same relative position after sorting (Page 54) ►One array is used
► More than one arrays are required
► Duplicating elements not handled

598. Sorting is one of the few problems where provable_____bonds exits on how fast we can sort, ►upper ►lower (Page 39) ►average ►log n

599. Question No: 65 (Marks: 1) - Please choose one Counting sort has time complexity: ►O(n) (Page 58)
► O(n+k)
► O(k)
► O(nlogn)

600. The running time of quick sort depends heavily on the selection of
- ► No of inputs
- ► Arrangement of elements in array
- ► Size o elements
- ► Pivot elements (Page 49)

601. Which may be stable sort:
- ► Bubble sort
- ► Insertion sort
- ► Both of above (Page 54)

602. In Quick Sort Constants hidden in T(n log n) are
- ► Large
- ► Medium
- ► Small
- ► Not Known

603. Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:
- ► There is explicit combine process as well to conquer the solution.
- ► No work is needed to combine the sub-arrays, the array is already sorted
- ► Merging the sub arrays
- ► None of above. (Page 51)

604. A point p in 2-dimensional space is usually given by its integer coordinate(s)_____
- ► p.x only
- ► p.y only
- ► p.x & p.z
- ► p.x & p.y (Page 10)

605. In_____we have to find rank of an element from given input.
- ► Merge sort algorithm
- ► Selection problem (Page 34)
- ► Brute force technique
- ► Plane Sweep algorithm

606. In Heap Sort algorithm, if heap property is violated _____
- ► We call Build heap procedure
- ► We call Heapify procedure
- ► We ignore
- ► Heap property can never be violated

607. Upper bound requires that there exist positive constants c2 and n0 such that f(n)_____c2n for all n <= n0(ye question ghalat lag raha hai mujhae
- ► Less than
- ► Equal to or Less than (Page 25)

► Equal or Greater than
► Greater than

608. A RAM is an idealized algorithm with takes an infinitely large random-access memory. ►True
► False (Page 10)

609. _is one of the few problems, where provable lower bounds exist onhow fast we can sort.
► Searching
► Sorting
► Both Searching & Sorting
► Graphing

610. Floor and ceiling are_____to calculate while analyzing algorithms.
► Very easy
► Usually considered difficult (Page 31)

611. In Heap Sort algorithm, the maximum levels an element can move upward is _____
► Theta (log n) (Page 43)
► Order (log n)
► Omega (log n)
► O (1) i.e. Constant time

612. A point p in 2-dimensional space is usually given by its integer coordinate(s)_____
► p.x only p.y
► only p.x & p.z
► p.x & p.y (Page 17)

613. In Heap Sort algorithm, the total running time for Heapify procedure is _____
► Theta (log n) (Page 43)
► Order (log n)
► Omega (log n)
► O (1) i.e. Constant time

614. Algorithm is a mathematical entity, which is independent of a specific machine and operating system.
► True
► False (Page 7)

615. While Sorting, the ordered domain means for any two input elements x and y _____satisfies only.
► x < y
► x > y
► x = y

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► All of the above (Page 39)

616. Quick sort is best from the perspective of Locality of reference.
► True (Page 9)
► False

617. In Heap Sort algorithm, we build_____for ascending sort.
► Max heap (Page 41)
► Min heap

618. In Sieve Technique, we know the item of interest.
► True
► False (Page 34)

619. While solving Selection problem, in Sieve technique we partition input data _____
► In increasing order
► In decreasing order
► According to Pivot (Page 35)
► Randomly

620. In pseudo code, the level of details depends on intended audience of the algorithm.
► True (Page 12)
► False

621. If the indices passed to merge sort algorithm are_____,then this means that there is only one element to sort. ►Small
► Large
► Equal (Page 28)
► Not Equal

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FINALTERM EXAMINATION
Spring 2010
CS502- Fundamentals of Algorithms (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
An optimization problem is one in which you want to find,
- ► Not a solution
- ► An algorithm
- ► Good solution
- ► **The best solution (Page 97)**

Question No: 2 (Marks: 1) - Please choose one
Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vertices.

- ► **True click here 4 detail**
- ► False

Question No: 3 (Marks: 1) - Please choose one
If a problem is in NP, it must also be in P.
- ► True
- ► False
- ► **unknown (Page 173)**

Question No: 4 (Marks: 1) - Please choose one
What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

- ► **Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)**
- ► *Lists* require less space than *matrices* and they are faster to find the weight of an edge (v1,v2)
- ► *Lists* require more space than *matrices* and they take longer to find the weight of an edge (v1,v2)
- ► *Lists* require more space than *matrices* but are faster to find the weight of an edge (v1,v2)

click here 4 detail

اللہ کا خوف سب سے بڑی دانائی ہے

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Question No: 5 (Marks: 1) - Please choose one
If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

- ► v edges.
- ► v – e + 5 edges
- ► v + e edges.
- ► **None of these**

Question No: 6 (Marks: 1) - Please choose one
Maximum number of vertices in a Directed Graph may be $|V^2|$

- ► True
- ► **False** **click here for details**

Question No: 7 (Marks: 1) - Please choose one
The Huffman algorithm finds a (n) _____ solution.

- ► **Optimal** **click here for detail**
- ► Non-optimal
- ► Exponential
- ► Polynomial

Question No: 8 (Marks: 1) - Please choose one
The Huffman algorithm finds an exponential solution

- ► True
- ► **False**

Question No: 9 (Marks: 1) - Please choose one
The Huffman algorithm finds a polynomial solution

- ► True
- ► **False**

Question No: 10 (Marks: 1) - Please choose one
The greedy part of the Huffman encoding algorithm is to first find two nodes with **larger** frequency.

- ► True
- ► **False** **(Page 100)**

Question No: 11 (Marks: 1) - Please choose one
The codeword assigned to characters by the Huffman algorithm have the property that no codeword is the postfix of any other.

- ► **True** **(Page 101)**
- ► False

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 12 (Marks: 1) - Please choose one
Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length B (T) of the encoded string.

► True
► **False (Page 102)**

Question No: 13 (Marks: 1) - Please choose one
Shortest path problems can be solved efficiently by modeling the road map as a graph.

► **True (Page 153)**
► False

Question No: 14 (Marks: 1) - Please choose one
Dijkestra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.

► True
► **False (Page 159)**

Question No: 15 (Marks: 1) - Please choose one
Bellman-Ford allows negative weights edges and negative cost cycles.

► True
► **False (Page 159)**

Question No: 16 (Marks: 1) - Please choose one
The term "coloring" came form the original application which was in architectural design.

► True
► **False (Page 176)**

Question No: 17 (Marks: 1) - Please choose one
In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

► **True (Page 176)**
► False

Question No: 18 (Marks: 1) - Please choose one
Dijkstra's algorithm is operates by maintaining a subset of vertices

► **True (Page 155)**
► False

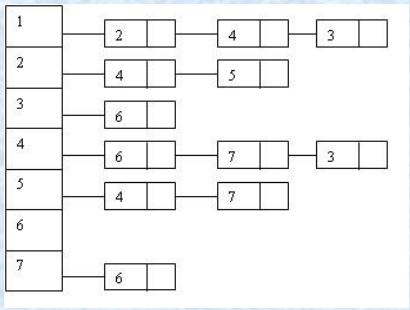
Question No: 19 (Marks: 1) - Please choose one
The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a different key.

►**True (Page 156)**
► False

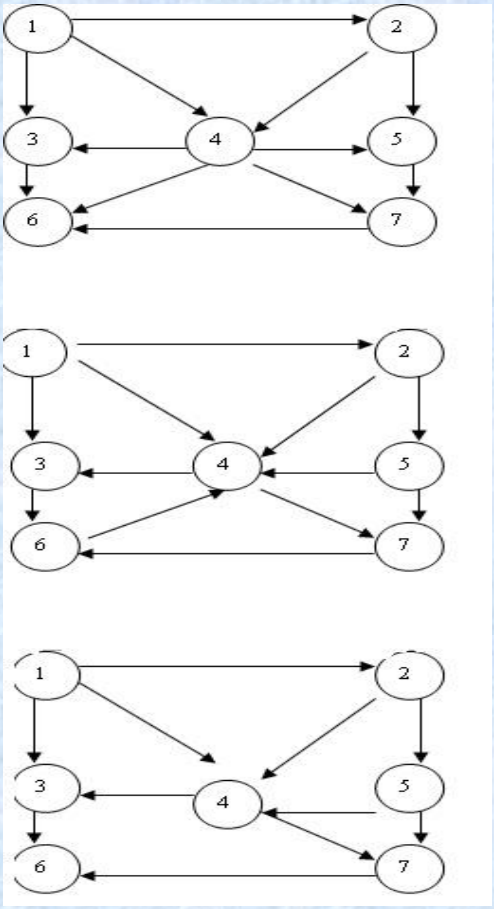
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Question No: 20 (Marks: 1) - Please choose one

Consider the following adjacency list:

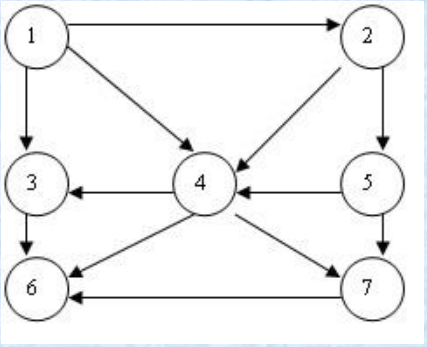


Which of the following graph(s) describe(s) the above adjacency list?



بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

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► **Correct Option**

Question No: 21 (Marks: 1) - Please choose one

We do sorting to,

► keep elements in random positions
► keep the algorithm run in linear order
► keep the algorithm run in (log n) order
► **keep elements in increasing or decreasing order (Page 40)**

Question No: 22 (Marks: 1) - Please choose one

After partitioning array in Quick sort, pivot is placed in a position such that

► **Values smaller than pivot are on left and larger than pivot are on right (Page 48)**
► Values larger than pivot are on left and smaller than pivot are on right
► Pivot is the first element of array
► Pivot is the last element of array

Question No: 23 (Marks: 1) - Please choose one

Merge sort is stable sort, but not an in-place algorithm

► **True (Page 54)**
► False

Question No: 24 (Marks: 1) - Please choose one

In counting sort, once we know the ranks, we simply _____ numbers to their final positions in an output array.

► Delete
► **copy (Page 57)**
► Mark
► arrange

Question No: 25 (Marks: 1) - Please choose one

Dynamic programming algorithms need to store the results of intermediate sub-problems.

► **True (Page 75)**
► False

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Question No: 26 (Marks: 1) - Please choose one

A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p . r) total entries in C and each takes _____ to compute.

► **O (q) (Page 84)**
► O (1)
► O (n^2)
► O (n^3)

FINALTERM EXAMINATION
Fall 2008
CS502- Fundamentals of Algorithms (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

_____ is a graphical representation of an algorithm

► Σ notation
► Θ notation
► **Flowchart** **Click here for detail**
► Asymptotic notation

Question No: 2 (Marks: 1) - Please choose one

Which of the following is calculated with **big o notation**?

►Lower bounds
►**Upper bounds** **(Page 25)**
►Both upper and lower bound
►Medium bounds

Question No: 3 (Marks: 1) - Please choose one

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

►The array elements form a heap
►**Elements in each half of the array are sorted amongst themselves** **click here 4 detail**
►Elements in the first half of the array are less than or equal to elements in the second half of the array
►None of the above

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 4 (Marks: 1) - Please choose one
Who invented Quick sort procedure?

►**Hoare click here 4 detail**
►Sedgewick
►Mellroy
►Coreman

Question No: 5 (Marks: 1) - Please choose one
What is the solution to the recurrence T(n) = T(n/2)+n, T(1) = 1

►O(logn)
►**O(n) (Page 37)**
►O(nlogn)
►O(2n)

Question No: 6 (Marks: 1) - Please choose one
Consider the following Huffman Tree
The binary code for the string **TEA** is

►**10 00 010 click here 4 detail**
►011 00 010
►10 00 110
►11 10 110

Question No: 7 (Marks: 1) - Please choose one
A greedy algorithm does not work in phases.

►True
►**False (Page 97)**

Question No: 8 (Marks: 1) - Please choose one
Can an adjacency matrix for a directed graph ever not be square in shape?
►Yes
►**No** click here 4 detail

Question No: 9 (Marks: 1) - Please choose one
One of the clever aspects of heaps is that they can be stored in arrays without using any_____.
►**Pointers (Page 40)**
►constants
►variables
►functions

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Question No: 10 (Marks: 1) - Please choose one
Merge sort requires extra array storage,

►**True (Page 54)**
►False

Question No: 11 (Marks: 1) - Please choose one
Non-optimal or greedy algorithm for money change takes_____

►**O(k) (Page 99)**
►O(kN)
►O(2k)
►O(N)

Question No: 12 (Marks: 1) - Please choose one
The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.
►True
►**False (Page 99)**

Question No: 13 (Marks: 1) - Please choose one
Using ASCII standard the string abacdaacac will be encoded with _____ bits.

►**80 (Page 99)**
►160
►320
►100

Question No: 14 (Marks: 1) - Please choose one
Using ASCII standard the string abacdaacac will be encoded with 160 bits.

►True
►**False (Page 99)**

Question No: 15 (Marks: 1) - Please choose one
Using ASCII standard the string abacdaacac will be encoded with 320 bits.
►True
►**False (Page 99)**

Question No: 16 (Marks: 1) - Please choose one
Using ASCII standard the string abacdaacac will be encoded with 100 bits.
►True
►**False (Page 99)**

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Question No: 17 (Marks: 1) - Please choose one
Using ASCII standard the string abacdaacac will be encoded with 32 bytes

►True
►**False (Page 99)**

Question No: 18 (Marks: 1) - Please choose one
The greedy part of the Huffman encoding algorithm is to first find two nodes with **smallest** frequency.
►**True (Page 100)**
►False

Question No: 19 (Marks: 1) - Please choose one
The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency

►True
►**False (Page 100)**

Question No: 20 (Marks: 1) - Please choose one
Huffman algorithm uses a greedy approach to generate an **antefix** code T that minimizes the expected length B (T) of the encoded string.

►True
►**False (Page 102)**

Question No: 21 (Marks: 1) - Please choose one
Depth first search is shortest path algorithm that works on un-weighted graphs.
►True
►**False (Page 153)**

Question No: 22 (Marks: 1) - Please choose one
Dijkestra s single source shortest path algorithm works if all edges weights are non negative and there are no negative cost cycles.

►**True (Page 159)**
►False

Question No: 23 (Marks: 1) - Please choose one
Dijkestra s single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.

►True
►**False (Page 159)**

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Question No: 24 (Marks: 1) - Please choose one
Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

►**True (Page 162)**
►Flase

Question No: 25 (Marks: 1) - Please choose one
Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

►k
► d_{ij}^{k} **(Page 164)**
►True
►Flase

Question No: 26 (Marks: 1) - Please choose one
The term coloring came from the original application which was in map drawing.

►**True (Page 176)**
►False

Question No: 27 (Marks: 1) - Please choose one
In the clique cover problem, for two vertices to be in the same group, they must be_____each other.

►Apart from
►Far from
►Near to
►**Adjacent to (Page 176)**

Question No: 28 (Marks: 1) - Please choose one
Fixed-length codes may not be efficient from the perspective of _____the total quantity of data.
Select correct option:

►**Minimizing (Page 99)**
►Averaging
►Maximizing
►Summing

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Question No: 29 (Marks: 1) - Please choose one
In greedy algorithm, at each phase, you take the_____ you can get right now, without regard for future consequences.

►Worst
►Minimum
►Good
►**Best (Page 97)**

Question No: 30 (Marks: 1) - Please choose one
The difference between Prim s algorithm and Dijkstra s algorithm is that Dijkstra s algorithm uses a same key.

►True
►**False (Page 156)**

FINALTERM EXAMINATION
Spring 2007
CS502- Fundamentals of Algorithms (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
If a problem is in NP-complete, it must also be in NP.

► **True (Page 178)**
► False

Question No: 2 (Marks: 1) - Please choose one
If there are n items, there are _____ possible combinations of the items.

►2
►n
►**2^n (Page 92)**
►3^n

Question No: 3 (Marks: 1) - Please choose one
Using ASCII code, each character is represented by a fixed-length code word of _____ bits per character.

►4
►6
►**8 (P)age 99)**
►10

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Question No: 4 (Marks: 1) - Please choose one
In Knapsack Problem, the thief's goal is to put items in the bag such that the _____ of the items does not exceed the limit of the bag.

►**Value (Page 91)**
►Weight
►Length
►Balance

Question No: 5 (Marks: 1) - Please choose one
The knapsack problem does not belong to the domain of optimization problems.

►True
►**False (Page 91)**

Question No: 6 (Marks: 1) - Please choose one
In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

►True
►**False (Page 100)**

Question No: 7 (Marks: 1) - Please choose one
Fixed-length codes are known for easy break up of a string into its individual characters.

►**True (Page 99)**
►False

Question No: 8 (Marks: 1) - Please choose one
In _____ Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items are not allowed.

►0
►1
►**0/1 (Page 91)**
►Fractional

Question No: 9 (Marks: 1) - Please choose one
The term "coloring" came from the original application which was in architectural design.

► True
► **False (Page 173)**

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Question No: 10 (Marks: 1) - Please choose one
In Knapsack Problem, value and weight both are to be under consideration.

►**True (page 91)**
►False

Question No: 11 (Marks: 1) - Please choose one
Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is _____ .

►log n
►n
►n2
►**n3 (Page 90)**

Question No: 12 (Marks: 1) - Please choose one
In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____ approach.

►Subjective
►**Inductive (Page 93)**
►Brute force
►Combination

Question No: 13 (Marks: 1) - Please choose one
A greedy algorithm sometimes works well for optimization problems.

►**True (Page 97)**
►False

Question No: 14 (Marks: 1) - Please choose one
In Huffman encoding, frequency of each character can be determined by parsing the message and _____ how many times each character (or symbol) appears.

►Printing
►Incrementing
►**Counting (Page 100)**
►Deleting

Question No: 15 (Marks: 1) - Please choose one
Greedy algorithm can do very poorly for some problems.

►**True (Page 97)**
►False

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Question No: 16 (Marks: 1) - Please choose one
The Huffman codes provide a method of _____ data efficiently.

►Reading
►**Encoding (Page 99)**
►Decoding
►Printing

Question No: 17 (Marks: 1) - Please choose one
In _____ based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

►Brute force
►**Dynamic programming (Page 93)**

Question No: 18 (Marks: 1) - Please choose one
Those problems in which Greedy finds good, but not always best is called a greedy_____.

►Algorithm
►Solution
►**Heuristic (Page 97)**
►Result

Question No: 19 (Marks: 1) - Please choose one
In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

►TRUE
►**FALSE (Page 97)**

Question No: 20 (Marks: 1) - Please choose one
_____ problem, we want to find the best solution.

►Minimization
►Averaging
►**Optimization (Page 97)**
►Maximization

Question No: 21 (Marks: 1) - Please choose one
Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

►**True (Page 101)**
►False

دنیا کی سب سے بڑی فتح نفس پر قابو رکھنا ہے

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Question No: 22 (Marks: 1) - Please choose one
In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

►Simple
►Non Greedy
►**Greedy (Page 97)**
►Brute force

Question No: 23 (Marks: 1) - Please choose one
Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

►**True (Page 102)**
►False

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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CS502 – Quiz No.2 (26 – June - 2013)

Question # 1 of 10 (Marks: 1) Please choose one
Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

►**True (Page 97)**
►False

Question # 1 of 10 (Marks: 1) Please choose one
Huffman algorithm generates an optimum prefix code.

►**True (Page 102)**
►False

Question # 1 of 10 (Marks: 1) Please choose one
If the string "lmncde" is coded with ASCII code, the message length would be _____ bits.

►24
►36
►**48 (6*8=48)**
►60

Question # 1 of 10 (Marks: 1) Please choose one
There are _____ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

►2
►**3 (Page 90)**
►4
►5

Question # 1 of 10 (Marks: 1) Please choose one
Inductive approach to compute entries of V is implied in _____ based solution of knapsack problem.

►Brute force
►**Dynamic programming (Page 93)**

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question # 1 of 10 (Marks: 1) Please choose one
A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

►**True (Page 105)**
►False

Question # 1 of 10 (Marks: 1) Please choose one
The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an _____solution.

►Simple
►Sub optimal
►**Optimal (Page 105)**
►Non optimal

Question # 1 of 10 (Marks: 1) Please choose one
The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

►**True (3*8=24)**
►False

Question # 1 of 10 (Marks: 1) Please choose one
An application problem is one in which you want to find, not just a solution, but the ____ solution.

►Simple
►**Good (Page 113) not sure**
►Best
►Worst

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Quiz No.3(January 28, 2013)

Question # 1 of 10 (Marks: 1) Please choose one
A dense undirected graph is:

► **A graph in which E = O(V^2) click here 4 detail**
►A graph in which E = O(V)
►A graph in which E = O(log V)
►All items above may be used to characterize a dense undirected graph

Question # 1 of 10 (Marks: 1) Please choose one
Suppose that a graph G = (V,E) is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G?

►O(|V |^2)
►O(|V | |E|)
►O(|V |^2|E|)
►**O(|V | + |E|) pg 116**

Question # 1 of 10 (Marks: 1) Please choose one
Which is true statement?

►**Breadth first search is shortest path algorithm that works on un-weighted graphs (Page 153)**
►Depth first search is shortest path algorithm that works on un-weighted graphs.
►Both of above are true.
►None of above are true.

Question # 1 of 10 (Marks: 1) Please choose one
Forward edge is:

► (u, v) where u is a proper descendent of v in the tree.
► **(u, v) where v is a proper descendent of u in the tree. (Page 129)**
► (u, v) where v is a proper ancesstor of u in the tree.
► (u, v) where u is a proper ancesstor of v in the tree.

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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Question # 1 of 10 (Marks: 1) Please choose one
What general property of the list indicates that the graph has an isolated vertex?

►There is Null pointer at the end of list.
►The Isolated vertex is not handled in list.
► Only one value is entered in the list.
►There is at least one null list.

Question # 1 of 10 (Marks: 1) Please choose one
If you find yourself in maze the better traversal approach will be :

►BFS Click here for detail
►BFS and DFS both are valid
►Level order
►DFS

Question # 1 of 10 (Marks: 1) Please choose one
In digraph G=(V,E) ;G has cycle if and only if

►The DFS forest has forward edge.
►The DFS forest has back edge (Page 131)
►The DFS forest has both back and forward edge
►BFS forest has forward edge

Question # 1 of 10 (Marks: 1) Please choose one
Back edge is:

► (u, v) where v is an ancestor of u in the tree. (Page 128)
► (u,v) where u is an ancesstor of v in the tree.
► (u, v) where v is an predcessor of u in the tree.
►None of above

Question # 1 of 10 (Marks: 1) Please choose one
Which statement is true?

►If a dynamic-programming problem satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.
►If a greedy choice property satisfies the optimal-substructure property, then a locally optimal solution is globally optimal.
► Both of above
►None of above

بد صورت چہرہ بد صورت دماغ سے بہتر ہے

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Question # 1 of 10 (Marks: 1) Please choose one
Cross edge is :

► (u, v) where u and v are not ancestor of one another
► (u, v) where u is ancesstor of v and v is not descendent of u.
► **(u, v) where u and v are not ancestor or descendent of one another (Page 129)**
► (u, v) where u and v are either ancestor or descendent of one another.

Quiz No.4(February 5, 2013)

Question # 1 of 10 (Marks: 1) Please choose one
Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

►**True click here 4 detail**
►False

Question # 1 of 10 (Marks: 1) Please choose one
Which is true statement in the following?

►**Kruskal algorithm is multiple source technique for finding MST. click here for detail**
►Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is O(EV)
►Both of above
►**Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges. click here 4 detail**

Question # 1 of 10 (Marks: 1) Please choose one
What algorithm technique is used in the implementation of Kruskal solution for the MST?

►**Greedy Technique (Page 142)**
►Divide-and-Conquer Technique
►Dynamic Programming Technique
►The algorithm combines more than one of the above techniques

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question # 1 of 10 (Marks: 1) Please choose one
What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

►O (log E)
► (V)
► (V+E)
►O (log V) (Page 152)

Question # 1 of 10 (Marks: 1) Please choose one
The relationship between number of back edges and number of cycles in DFS is,

►Both are equal
►Back edges are half of cycles
►Back edges are one quarter of cycles
►There is no relationship between no. of edges and cycles (Page 131)

Question # 1 of 10 (Marks: 1) Please choose one
You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T.?

► (V + E) (Page 138)
► (V E)
► (V)
► (V^2)

Question # 1 of 10 (Marks: 1) Please choose one
There is relationship between number of back edges and number of cycles in DFS

►Both are equal.
►Cycles are half of back edges.
►Cycles are one fourth of back edges.
►There is no relationship between back edges and number of cycles. (Page 131)

Question # 1 of 10 (Marks: 1) Please choose one
A digraph is strongly connected under what condition?

►A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v .
►A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and vice versa. (Page 135)
►A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and vice versa.
►A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v and vice versa.

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CS502 - Fundamentals of Algorithms
Quiz No.5 Dated FEB 15TH 2013

Question # 1 of 10 (Marks: 1) Please choose one
In in-place sorting algorithm is one that uses arrays for storage :

►An additional array
►**No additional array (Page 54)**
►Both of above may be true according to algorithm
►More than 3 arrays of one dimension.

Question # 1 of 10 (Marks: 1) Please choose one
In stable sorting algorithm

►One array is used
►In which duplicating elements are not handled.
►More then one arrays are required.
►**Duplicating elements remain in same relative position after sorting. (Page 54)**

Question # 1 of 10 (Marks: 1) Please choose one
Which sorting algorithm is faster :

►O(n^2)
►**O(nlogn) (Page 46)**
►O(n+k)
►O(n^3)

Question # 1 of 10 (Marks: 1) Please choose one
In Quick sort algorithm, constants hidden in T(n lg n) are

►Large
►Medium
►Not known
►**Small** **Click here for detail**

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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Question # 1 of 10 (Marks: 1) Please choose one
Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

►There is explicit combine process as well to conquer the solution.
►No work is needed to combine the sub-arrays, the array is already sorted
►Merging the sub arrays
►**None of above. (Page 51)**
Ref: - random choices for the pivot element and each choice have an equal probability of 1/n of occurring. So we can modify the above recurrence to compute an average rather than a max

Question # 1 of 10 (Marks: 1) Please choose one
Dijkstra's algorithm :

►Has greedy approach to find all shortest paths
►Has both greedy and Dynamic approach to find all shortest paths
►**Has greedy approach to compute single source shortest paths to all other vertices (Page 154)**
►Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

Question # 1 of 10 (Marks: 1) Please choose one
Which may be stable sort:

►Bubble sort
►Insertion sort
►**Both of above (page 54)**
►Selection sort

Question # 1 of 10 (Marks: 1) Please choose one
In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis,

►linear
►arithmetic
►**geometric (page 37)**
►exponent

Question # 1 of 10 (Marks: 1) Please choose one
How much time merge sort takes for an array of numbers?

►T(n^2)
►**T(n) (Page 40)**
►T(log n)
►T(n log n)

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Question # 1 of 10 (Marks: 1) Please choose one
Counting sort has time complexity:

►**O(n)** **Click here for detail**
►O(n+k)
►O(k)
►O(nlogn)

Question # 1 of 10 (Marks: 1) Please choose one
The analysis of Selection algorithm shows the total running time is indeed _____in n,

►arithmetic
►geometric
►**linear (Page 37)**
►orthogonal

Question # 1 of 10 (Marks: 1) Please choose one
Sorting is one of the few problems where provable _____ bonds exits on how fast we can sort,

►upper
►**lower (Page 39)**
►average
►log n

Question # 1 of 10 (Marks: 1) Please choose one
In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,

►T(n)
►T(n / 2)
►**log n (Page 37)**
►n / 2 + n / 4

Question # 1 of 10 (Marks: 1) Please choose one
The number of nodes in a complete binary tree of height h is

►**2^(h+1) – 1 (Page 40)**
►2 * (h+1) – 1
►2 * (h+1)
► ((h+1) ^ 2) – 1

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Question # 1 of 10 (Marks: 1) Please choose one
How many elements do we eliminate in each time for the Analysis of Selection algorithm?
►**n / 2 elements (Page 37)**
► (n / 2) + n elements
►n / 4 elements
►2 n elements

Question # 1 of 10 (Marks: 1) Please choose one
Slow sorting algorithms run in,
►**T(n^2) (Page 39)**
►T(n)
►T(log n)
►T(n log n)

Question # 1 of 10 (Marks: 1) Please choose one
Counting sort is suitable to sort the elements in range 1 to k:

►K is large
►**K is small (Page 57)**
►K may be large or small
►None

Question # 1 of 10 (Marks: 1) Please choose one
Heaps can be stored in arrays without using any pointers; this is due to the _____ nature of the binary tree,

►**left-complete (Page 40)**
►right-complete
►tree nodes
►tree leaves

Question # 1 of 10 (Marks: 1) Please choose one
Sieve Technique can be applied to selection problem?
►**True (Page 35)**
►False

Question # 1 of 10 (Marks: 1) Please choose one
A heap is a left-complete binary tree that conforms to the _____

►increasing order only
►decreasing order only
►**heap order (Page 40)**
► (log n) order

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Question # 1 of 10 (Marks: 1) Please choose one
Divide-and-conquer as breaking the problem into a small number of
►pivot
►Sieve
►smaller sub problems (Page 34)
►Selection

Question # 1 of 10 (Marks: 1) Please choose one
In Sieve Technique we do not know which item is of interest
►True (Page 34)
►False

Question # 1 of 10 (Marks: 1) Please choose one
The recurrence relation of Tower of Hanoi is given below T(n)={1 if n=1 and 2T(n-1) if n >1 In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

►16
►10
►32
►31 Click here 4 detail

Question # 1 of 10 (Marks: 1) Please choose one
For the heap sort, access to nodes involves simple _____ operations.

►arithmetic (Page 41)
►binary
►algebraic
►logarithmic

Question # 1 of 10 (Marks: 1) Please choose one
For the sieve technique we solve the problem,
►recursively (Page 34)
►mathematically
►precisely
►accurately

Question # 1 of 10 (Marks: 1) Please choose one
The sieve technique works in _____ as follows

►phases (Page 34)
►numbers
►integers
►routines

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Question # 1 of 10 (Marks: 1) Please choose one
A (an) _____ is a left-complete binary tree that conforms to the heap order
►**heap (Page 40)**
►binary tree
►binary search tree
►array

Question # 1 of 10 (Marks: 1) Please choose one
The sieve technique is a special case, where the number of sub problems is just
►5
►many
►**1 (Page 34)**
►few

Question # 1 of 10 (Marks: 1) Please choose one
Analysis of Selection algorithm ends up with,

►T(n)
►T(1 / 1 + n)
►T(n / 2)
►**T((n / 2) + n) (Page 37)**

Question # 1 of 10 (Marks: 1) Please choose one
For the heap sort we store the tree nodes in
►**level-order traversal (Page 40)**
►in-order traversal
►pre-order traversal
►post-order traversal

Question # 1 of 10 (Marks: 1) Please choose one
The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,
►**divide-and-conquer (Page 34)**
►decrease and conquer
►greedy nature
►2-dimension Maxima

Question # 1 of 10 (Marks: 1) Please choose one
Theta asymptotic notation for T (n) :

►Set of functions described by: c1g(n)Set of functions described by c1g(n)>=f(n) for c1 s
►Theta for T(n)is actually upper and worst case comp
►**Set of functions described by:**
►c1g(n)

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Question # 1 of 10 (Marks: 1) Please choose one
Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

►**n items (Page 34)**
►phases
►pointers
►constant

Question # 1 of 10 (Marks: 1) Please choose one
Memorization is?

►To store previous results for future use
►**To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (Page 47)**
►To make the process accurate
►None of the above

Question # 1 of 10 (Marks: 1) Please choose one
Quick sort is

►Stable & in place
►**Not stable but in place (Page 57)**
►Stable but not in place
►Some time stable & some times in place

Question # 1 of 10 (Marks: 1) Please choose one
One example of in place but not stable algorithm is

►Merger Sort
►**Quick Sort (Page 54)**
►Continuation Sort
►Bubble Sort

Question # 1 of 10 (Marks: 1) Please choose one
Continuation sort is suitable to sort the elements in range 1 to k

►K is Large
►K is not known
►K may be small or large
►**K is small (Page 57)**

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

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Question # 1 of 10 (Marks: 1) Please choose one
Which may be a stable sort?

►Merger
►Insertion
► Both above (Page 54)
►None of the above

Question # 1 of 10 (Marks: 1) Please choose one
An in place sorting algorithm is one that uses ___ arrays for storage

►Two dimensional arrays
►More than one array
►No Additional Array (Page 54)
►None of the above

Question # 1 of 10 (Marks: 1) Please choose one
Continuing sort has time complexity of ?

►O(n) **Click here fir detail**
►O(n+k)
►O(nlogn)
►O(k)

Question # 1 of 10 (Marks: 1) Please choose one
single item from a larger set of _____

►n items (Page 34)
►phases
►pointers
►vconstant

Question # 1 of 10 (Marks: 1) Please choose one
For the Sieve Technique we take time

►T(nk) (Page 34)
►T(n / 3)
►n^2
►n/3

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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Question # 1 of 10 (Marks: 1) Please choose one
One Example of in place but not stable sort is

►**Quick (Page 54)**
►Heap
►Merge
►Bubble

Question # 1 of 10 (Marks: 1) Please choose one
Consider the following Algorithm:

```
Factorial (n){
  if (n=1)
    return 1
  else
     return (n * Factorial(n-1))
```

Recurrence for the following algorithm is:

► T(n) = T(n-1) +1
► T(n) = nT(n-1) +1
► T(n)= T(n-1) +n
► **T(n)=T(n(n-1)) +1**

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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Some More MCQs

Question No: 1 (Marks: 1) - Please choose one
Due to left complete nature of binary tree, the heap can be stored in

► **Arrays (Page 40)**
► Structures
► Link Lis
►Stack

Question No: 2 (Marks: 1) - Please choose one
What type of instructions Random Access Machine (RAM) can execute?

►Algebraic and logic
►Geometric and arithmetic
►**Arithmetic and logic (Page 10)**
►Parallel and recursive

Question No: 3 (Marks: 1) - Please choose one
What is the total time to heapify?

► **O(log n) (Page 43)**
► O(n log n)
► $O(n^2 \log n)$
► $O(\log^2 n)$

Question No: 4 (Marks: 1) - Please choose one
word Algorithm comes from the name of the muslim author _____

►**Abu Ja'far Mohammad ibn Musa al-Khowarizmi**.

Question No: 5 (Marks: 1) - Please choose one
al-Khwarizmi's work was written in a book titled _____

►**al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah**

خاموشی غصے کا بہترین علاج ہے

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Question No: 6 (Marks: 1) - Please choose one
Random access machine or RAM is a/an

- ► Machine build by Al-Khwarizmi
- ► Mechanical machine
- ► Electronics machine
- ► **Mathematical model (Page 10)**

Question No: 7 (Marks: 1) - Please choose one
A RAM is an idealized machine with _____ random-access memory.

- ► 256MB
- ► 512MB
- ► **an infinitely large (Page 10)**
- ► 100GB

Question No: 8 (Marks: 1) - Please choose one
What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- ► n^2
- ► $n^{\frac{n}{2}}$
- ► n **(Page 14)**
- ► n^8

Question No: 9 (Marks: 1) - Please choose one
Consider the following code:

```
For(j=1; j<n;j++)
        For(k=1; k<15;k++)
                For(l=5; l<n; l++)
                {
                        Do_something_constant();
                }
```

What is the order of execution for this code.

- ► **O(*n*)**
- ► O(n^3)
- ► O(n^2 log n)
- ► O(n^2)

Question No: 10 (Marks: 1) - Please choose one
Is it possible to sort without making comparisons?

- ► **Yes (Page 57)**
- ► No

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Question No: 11 (Marks: 1) - Please choose one
When we call heapify then at each level the comparison performed takes time

- **It will take Θ (1) (Page 43)**
- Time will vary according to the nature of input data
- It can not be predicted
- It will take Θ (log n)

Question No: 12 (Marks: 1) - Please choose one
In Quick sort, we don't have the control over the sizes of recursive calls

- **True (Page 40)**
- False
- Less information to decide
- Either true or false

Question No: 13 (Marks: 1) - Please choose one
If there are Θ (n^2) entries in edit distance matrix then the total running time is

- Θ (1)
- **Θ (n^2)** **Click here for detail**
- Θ (n)
- Θ (n log n)

Question No: 14 (Marks: 1) - Please choose one
For Chain Matrix Multiplication we can not use divide and conquer approach because,

- **We do not know the optimum k (Page 86)**
- We use divide and conquer for sorting only
- We can easily perform it in linear time
- Size of data is not given

Question No: 15 (Marks: 1) - Please choose one
The Knapsack problem belongs to the domain of _____ problems.

- **Optimization (Page 91)**
- NP Complete
- Linear Solution
- Sorting



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Question No: 16 (Marks: 1) - Please choose one
Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. W = 50.

Item	Value	Weight
1	60	10
2	100	20
3	120	30

The optimal solution is to pick

- ► Items 1 and 2
- ► Items 1 and 3
- ► **Items 2 and 3 (correct)**
- ► None of these

Question No: 17 (Marks: 1) - Please choose one
who invented the quick sort

►**C.A.R. Hoare** Click here for detail

Question No: 18 (Marks: 1) - Please choose one
main elements to a divide-and-conquer

►**Divide, conquer, combine (Page 27)**

Question No: 19 (Marks: 1) - Please choose one
Mergesort is a stable algorithm but not an in-place algorithm.

►**True (Page 54)**
►false

Question No: 20 (Marks: 1) - Please choose one
Counting sort the numbers to be sorted are in the range 1 to k where k is small.

►**True (Page 57)**
►False

Question No: 21 (Marks: 1) - Please choose one
In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

►**Convergent geometric series (Page 37)**
►Divergent geometric series
►None of these

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Question No: 22 (Marks: 1) - Please choose one
If an algorithm has a complexity of $\log_2 n + n\log_2 n + n$. we could say that it has complexity

►O(n)
►O(n $\log_2$ n)
►O(3)
►O($\log_2$ ($\log_2$ n))
►O ($\log_2$ n)

Question No: 23 (Marks: 1) - Please choose one
In RAM model instructions are executed

►**One after another (Page 10)**
►Parallel
►Concurrent
►Random

Question No: 24 (Marks: 1) - Please choose one
Due to left-complete nature of binary tree, heaps can be stored in

►Link list
►Structure
►**Array (Page 40)**
►None of above

Question No: 25 (Marks: 1) - Please choose one
The time assumed for each basic operation to execute on RAM model of computation is-----
►Infinite
►Continuous
►**Constant (Page 10)**
►Variable

Question No: 26 (Marks: 1) - Please choose one
If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.
►True
►**False (Page 28)**

Question No: 27 (Marks: 1) - Please choose one
Brute-force algorithm uses no intelligence in pruning out decisions.

►**True (Page 18)**
►False

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Question No: 28 (Marks: 1) - Please choose one
In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

►**True (Page 24)**
►False

Question No: 29 (Marks: 1) - Please choose one
For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large.

►**True (Page 14)**
►Fast

Question No: 30 (Marks: 1) - Please choose one
The array to be sorted is not passed as argument to the merge sort algorithm.

►True
►**False**

Question No: 31 (Marks: 1) - Please choose one
In simple brute-force algorithm, we give no thought to efficiency.

►**True (Page 11)**
►False

Question No: 32 (Marks: 1) - Please choose one
The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

►True
►**False (Page 27) [Divide and Conquer]**

Question No: 33 (Marks: 1) - Please choose one
In 2d-space a point is said to be _____if it is not dominated by any other point in that space.

►Member
►Minimal
►**Maximal (Page 11)**
►Joint

Question No: 34 (Marks: 1) - Please choose one
An algorithm is a mathematical entity that is dependent on a specific programming language.
► True
►**False (Page 7)**

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Question No: 35 (Marks: 1) - Please choose one
The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.

►**True (Page 13)**
►False

Question No: 36 (Marks: 1) - Please choose one
F (n) and g (n) are asymptotically equivalent. This means that they have essentially the same _____ for large n.

►Results
►Variables
►Size
►**Growth rates (Page 23)**

Question No: 37 (Marks: 1) - Please choose one
8n2 + 2n - 3 will eventually exceed c2*(n) no matter how large we make c2.

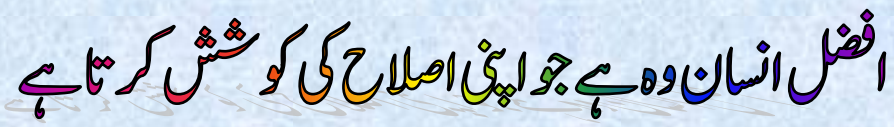
►**True (Page 25)**
►False

Question No: 38 (Marks: 1) - Please choose one
If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

►Fast
►Slow
►Expensive
►**Cheap (Page 11)**

Question No: 39 (Marks: 1) - Please choose one
The function f(n)= n(logn+1)/2 is asymptotically equivalent to n log n. Here Upper Bound means the function f(n) grows asymptotically _____ faster than n log n.

►More
►Quiet
►**Not (Page 24)**
►At least



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Question No: 40 (Marks: 1) - Please choose one
After sorting in merge sort algorithm, merging process is invoked.

► **True (Page 28)**
►False

Question No: 41 (Marks: 1) - Please choose one
Asymptotic growth rate of the function is taken over_____ case running time.

►Best
►Average
►**Worst (Page 14)**
►Normal

Question No: 42 (Marks: 1) - Please choose one
In analysis of f (n) =n (n/5) +n-10 log n, f (n) is asymptotically equivalent to _____.
►n
►2n
►n+1
►**n2 (Page 23)**

Question No: 43 (Marks: 1) - Please choose one
Algorithm is concerned with.....issues.
►Macro
►Micro
►**Both Macro & Micro (Page 8)**
►Normal

Question No: 44 (Marks: 1) - Please choose one
We cannot make any significant improvement in the running time which is better than that of brute-force algorithm.

►True
►**False (Page 18)**

Question No: 45 (Marks: 1) - Please choose one
In addition to passing in the array itself to Merge Sort algorithm, we will pass in _____other arguments which are indices.

►**Two (Page 28)**
►Three
►Four
►Five

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Question No: 46 (Marks: 1) - Please choose one
In analysis, the Lower Bound means the function grows asymptotically at least as fast as its largest term.

►**True (Page 24)**
►False

Question No: 47 (Marks: 1) - Please choose one
Efficient algorithm requires less computational…….

►Memory
►Running Time
►**Memory and Running Time (Page 9)**
►Energy

Question No: 48 (Marks: 1) - Please choose one
The O-notation is used to state only the asymptotic _____bounds.

►Two
►Lower
►**Upper (Page 25)**
►Both lower & upper

Question No: 49 (Marks: 1) - Please choose one
For the worst-case running time analysis, the nested loop structure containing one "for" and one "while" loop, might be expressed as a pair of _____nested summations.

►**1**
►**2 (Page 16)**
►3
►4

Question No: 50 (Marks: 1) - Please choose one
Before sweeping a vertical line in plane sweep approach, in start sorting of the points is done in increasing order of their _____coordinates.

►**X (Page 18)**
►Y
►Z
►X & Y

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question No: 51 (Marks: 1) - Please choose one
Brute-force algorithm for 2D-Maxima is operated by comparing _____ pairs of points.

►Two
►Some
►Most
►**All (Page 18)**

Question No: 52 (Marks: 1) - Please choose one
The function f(n)=n(logn+1)/2 is asymptotically equivalent to nlog n. Here Lower Bound means function f(n) grows asymptotically at _____ as fast as nlog n.

►Normal
►**Least (Page 23)**
►Most
►All

Question No: 53 (Marks: 1) - Please choose one
In plane sweep approach, a vertical line is swept across the 2d-plane and _____structure is used for holding the maximal points lying to the left of the sweep line.

►Array
►Queue
►**Stack (Page 18)**
►Tree

Question No: 54 (Marks: 1) - Please choose one
Algorithm analysts know for sure about efficient solutions for NP-complete problems.

►True
►**False (Page 9)**

Question No: 55 (Marks: 1) - Please choose one
The analysis of Selection algorithm shows the total running time is indeed _____in n,

►arithmetic
►geometric
►**linear (Page 37)**
►orthogonal

اس سے پہلے کہ تمہیں شہوت فتنے میں ڈالے نکاح کر لو

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Question No: 56 (Marks: 1) - Please choose one
The sieve technique works where we have to find _____ item(s) from a large input.

►**Single (Page 34)**
►Two
►Three
►Similar

Question No: 57 (Marks: 1) - Please choose one
In which order we can sort?

►increasing order only
►decreasing order only
►**increasing order or decreasing order (Page 39)**
►both at the same time

Question No: 58 (Marks: 1) - Please choose one
For the heap sort we store the tree nodes in

►**level-order traversal (Page 40)**
►in-order traversal
►pre-order traversal
►post-order traversal

Question No: 59 (Marks: 1) - Please choose one
In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis,

►linear
►arithmetic
►**geometric (Page 37)**
►exponent

Question No: 60 (Marks: 1) - Please choose one
How much time merge sort takes for an array of numbers?

►T(n^2)
►T(n)
►T(log n)
►**T(n log n) (Page 40)**

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

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Question No: 61 (Marks: 1) - Please choose one
Memoization is?

►To store previous results for future use
►**To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (page 74)**
►To make the process accurate
►None of the above

Question No: 62 (Marks: 1) - Please choose one
Cont sort is suitable to sort the elements in range 1 to k

►K is Large
►K is not known
►K may be small or large
►**K is small (Page 57)**

Question No: 63 (Marks: 1) - Please choose one
In place stable sorting algorithm.

►**If duplicate elements remain in the same relative position after sorting (Page 54)**
►One array is used
►More than one arrays are required
►Duplicating elements not handled

Question No: 64 (Marks: 1) - Please choose one
Sorting is one of the few problems where provable _____ bonds exits on how fast we can sort,

►upper
►**lower (Page 39)**
►average
►log n

Question No: 65 (Marks: 1) - Please choose one
Counting sort has time complexity:

►**O(n) (Page 58)**
►O(n+k)
►O(k)
►O(nlogn)

اپنی مرضی اور اللہ کی مرضی میں فرق کا نام غم ہے

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Question No: 66 (Marks: 1) - Please choose one
The running time of quick sort depends heavily on the selection of

►No of inputs
►Arrangement of elements in array
►Size o elements
►**Pivot elements (Page 49)**

Question No: 67 (Marks: 1) - Please choose one
Which may be stable sort:

►Bubble sort
►Insertion sort
►**Both of above (Page 54)**

Question No: 68 (Marks: 1) - Please choose one
In Quick Sort Constants hidden in T(n log n) are

►Large
►Medium
►**Small** Click here for detail
►Not Known

Question No: 69 (Marks: 1) - Please choose one
Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

►There is explicit combine process as well to conquer the solution.
►No work is needed to combine the sub-arrays, the array is already sorted
►Merging the sub arrays
►**None of above. (Page 51)**
Ref: - random choices for the pivot element and each choice have an equal probability of 1/n of occurring. So we can modify the above recurrence to compute an average rather than a max

Question No: 70 (Marks: 1) - Please choose one
A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

► p.x only
► p.y only
► p.x & p.z
► **p.x & p.y (Page 10)**

وہ لوگ مبارک ہیں جو الفاظ سے نصیحت نہیں کرتے بلکہ عمل سے کرتے ہیں

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Question No: 71 (Marks: 1) - Please choose one
In _____ we have to find rank of an element from given input.

►Merge sort algorithm
►**Selection problem (Page 34)**
►Brute force technique
►Plane Sweep algorithm

Question No: 72 (Marks: 1) - Please choose one
In Heap Sort algorithm, if heap property is violated _____

►We call Build heap procedure
►**We call Heapify procedure**
►We ignore
►Heap property can never be violated

Question No: 73 (Marks: 1) - Please choose one
Upper bound requires that there exist positive constants c2 and n0 such that f(n) ____ c2n for all n <= n0(ye question ghalat lag raha hai mujhae

►Less than
►**Equal to or Less than (Page 25)**
►Equal or Greater than
►Greater than

Question No: 74 (Marks: 1) - Please choose one
A RAM is an idealized algorithm with takes an infinitely large random-access memory.

►True
►**False (Page 10)**

Question No: 75 (Marks: 1) - Please choose one
_____ is one of the few problems, where provable lower bounds exist on how fast we can sort.
►Searching
►**Sorting (Page)**
►Both Searching & Sorting
►Graphing

Question No: 76 (Marks: 1) - Please choose one
Floor and ceiling are _____ to calculate while analyzing algorithms.

►Very easy
►**Usually considered difficult (Page 31)**

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Question No: 77 (Marks: 1) - Please choose one
In Heap Sort algorithm, the maximum levels an element can move upward is _____

►**Theta (log n) (Page 43)**
►Order (log n)
►Omega (log n)
►O (1) i.e. Constant time

Question No: 78 (Marks: 1) - Please choose one
A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

►p.x only p.y
►only p.x & p.z
► **p.x & p.y (Page 17)**

Question No: 79 (Marks: 1) - Please choose one
In Heap Sort algorithm, the total running time for Heapify procedure is _____

► **Theta (log n) (Page 43)**
►Order (log n)
►Omega (log n)
►O (1) i.e. Constant time

Question No: 80 (Marks: 1) - Please choose one
Algorithm is a mathematical entity, which is independent of a specific machine and operating system.

►True
►**False (Page 7)**

Question No: 81 (Marks: 1) - Please choose one
While Sorting, the ordered domain means for any two input elements x and y _____ satisfies only.

►x < y
►x > y
►x = y
►**All of the above (Page 39)**

Question No: 82 (Marks: 1) - Please choose one
Quick sort is best from the perspective of Locality of reference.

►**True (Page 9)**
►False

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Question No: 83 (Marks: 1) - Please choose one
In Heap Sort algorithm, we build _____ for ascending sort.

►**Max heap (Page 41)**
►Min heap

Question No: 84 (Marks: 1) - Please choose one
In Sieve Technique, we know the item of interest.

►True
►**False (Page 34)**

Question No: 85 (Marks: 1) - Please choose one
While solving Selection problem, in Sieve technique we partition input data _____

►In increasing order
►In decreasing order
►**According to Pivot (Page 35)**
►Randomly

Question No: 86 (Marks: 1) - Please choose one
In pseudo code, the level of details depends on intended audience of the algorithm.

►**True (Page 12)**
►False

Question No: 87 (Marks: 1) - Please choose one
If the indices passed to merge sort algorithm are _____,then this means that there is only one element to sort.

►Small
►Large
►**Equal (Page 28)**
►Not Equal

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CS502 Fundamentals Of Algorithms
Update MCQS For FinalTerm
Solve By Vu Topper RM

80 To 100% Marks

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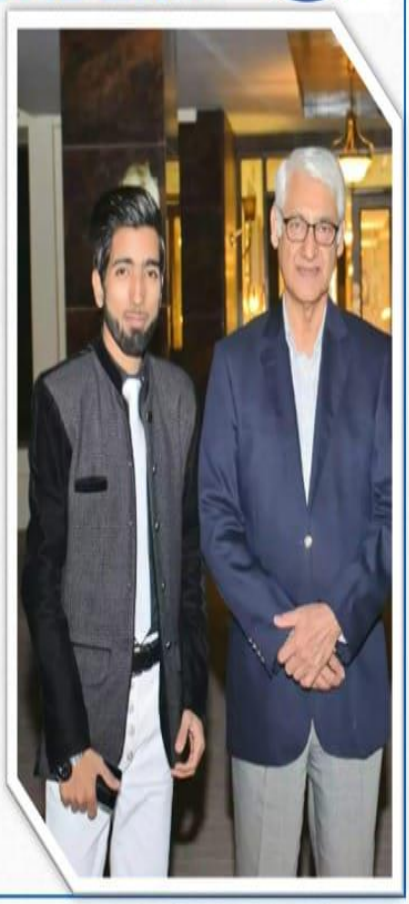
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Question No:01 **(Marks:01)** **Vu-Topper RM**

If we encode and compress text using ASCII standard each character is represented b

A. Fixed length code word of 4 bits
B. Variable length code word up to 4 bits
C. Fixed length code word of 8 bits
D. Variable length code word up to 8 bits

Question No:02 **(Marks:01)** **Vu-Topper RM**

G power T for graph can be computed in

A. Theta(VE)
B. Theta(V log E)
C. Theta(E log V)
D .Theta (V+E)

Question No:03 **(Marks:01)** **Vu-Topper RM**

Adding any edge to free tree

A .Keep it free tree and increase the size of the tree
B. Create a unique cycle
C. Not allow to add edge to free tree
D. Create multiple cycles

Question No:04 **(Marks:01)** **Vu-Topper RM**

Which sorting algorithn is faster :

A. O(n^2)
B. O(nlogn)
C. **O(n+k)**
D. O(n^3)

Question No:05 **(Marks:01)** **Vu-Topper RM**

There are no _____ edges in undirected graph.

A . Forward
B. Back
C. Cross
D. Both forward and back

Question No:06 **(Marks:01)** **Vu-Topper RM**

Adding any edge to a free tree creates a unique _____ .

A. Vertex
B. cycle
C. Edge

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Vu Topper RM

D.Strong component

Question No:07 (Marks:01) Vu-Topper RM

Networks are complete in the sense that it is possible from any location in the network to reach any other location in the digraph.

A. True

B. False

Question No:08 (Marks:01) Vu-Topper RM

Runtime complexity of Prim's algorithm is _____.

A. V log V

B. E log V

C. log V

D. None of the above

Question No:09 (Marks:01) Vu-Topper RM

In Prim's algorithm, we start with the _____ vertex r; it can be any vertex.

A. First

B. Leaf

C. root

D. Mid

Question No:10 (Marks:01) Vu-Topper RM

Adding any edge to a free tree creates a unique cycle.

A. true

B. false

Question No:11 (Marks:01) Vu-Topper RM

For undirected graph, there is no distinction between forward and back edges.

A. true

B. false

Question No:12 (Marks:01) Vu-Topper RM

in strong components algorithm, first of all DFS is run for computing finish times of vertices.

A. true

B. false

Question No:13 (Marks:01) Vu-Topper RM

Which method is preferable for dealing with chain matrix multiplication?

A. Divide and conquer strategy

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B. Dynamic programming formulation

C. Graph theory

D. Greedy Approach

Question No:14 (Marks:01) Vu-Topper RM

Huffman algorithm produces the…….prefix code tree.

A. Better

B. Optimal

C. Worst

D. Best

Question No:15 (Marks:01) Vu-Topper RM

A….w is adjacent to vertex v if there is an edge from v to w.

A. Acyclic

B. Vertex

C. Loop

D. Cycle

Question No:16 (Marks:01) Vu-Topper RM

Using ASCII standard the string "greedy" will be encoded with

A. 44 bits

B. 120 bits

C. 40 bits

D. 48 bits

Question No:17 (Marks:01) Vu-Topper RM

In activity scheduling algorithm, each activity is represented by a

A. Rectangle

B. Square

C. Circle

D. Triangle

Question No:18 (Marks:01) Vu-Topper RM

Those problems in which greedy finds good, but not always best is called a greedy…..

A. **Heuristic**

B. Solution

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C. Result
D. Algorithm

Question No:19 (Marks:01) Vu-Topper RM

The knapsack problem belongs to the domain of…..Problems

A. Searching
B. Sorting
C. Linear solution
D. Optimization

Question No:20 (Marks:01) Vu-Topper RM

The general coin change problem can be solved using

A. Recursion
B. Greedy algorithm
C. Dynamic programming
D. Divide and conquer

Question No:21 (Marks:01) Vu-Topper RM

Huffman algorithm generates an optimum.....code

A. Postfix
B. Infix
C. None of the given options
D. Prefix

Question No:22 (Marks:01) Vu-Topper RM

.....ways of representing graphics

A. 1
B. 2
C. 3
D. 4

Question No:23 (Marks:01) Vu-Topper RM

Knapsack word originates from……language

A. German
B. English
C. French
D. Norwegian

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Question No:24 (Marks:01) Vu-Topper RM

Graphs are important .....model for many application problems

A. Mathematical

B. Unpredictable

C. Haphazard

D. Unsystematic

Question No:25 (Marks:01) Vu-Topper RM

Which type of algorithm is harder to prove the correctness?

A. Dynamic

B. Greedy

C. Divide and conquer

D. Brute force

Question No:26 (Marks:01) Vu-Topper RM

Items are not allowed in 0/1 knapsack problem

A. Fractional

B. 0

C. 1

D. 0/1

Question No:27 (Marks:01) Vu-Topper RM

Matrix multiplication is a(n).....operation

A. Nether commutative nor associative

B. Transitive

C. Commutative

D. Associative

Question No:28 (Marks:01) Vu-Topper RM

For a Diagraph G= (V.E), Sum of in-degree (v) --.

A. Not equal to sum of out-degree(v)

B. = sum of out-degree(v)

C. < sum of out-degree(v)

D. > sum of out-degree(v)

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Question No:29 **(Marks:01)** **Vu-Topper RM**

DFS or BFS yields a of the graph.

A. Traversed tree
B. Spanning tree **Page 125**
C. Simple tree
D. Free tree

Question No:30 **(Marks:01)** **Vu-Topper RM**

Using ASCII code, each character is represented by a fixed-length code of bits per character.

A. 4
B. 6
C. 10
D. 8

Question No:31 **(Marks:01)** **Vu-Topper RM**

In Knapsack Problem, the goal is to put items in the Knapsacksuch that the value of the items is----------subject to weight limit of the Knapsack.

A. Minimized
B. Decreased
C. Maximized
D. None of the above given

Question No:32 **(Marks:01)** **Vu-Topper RM**

A graph is said to be acyclic if it contains --.

A. At least one cycle
B. Exactly one cycle
C. Always more than one cycle
D. No cycles

Question No:33 **(Marks:01)** **Vu-Topper RM**

The number of edges that come out of a vertex is called the of that vertex in the digraph.

A. Post-degree
B. in-degree
C. out-degree
D. pre-degree

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Question No:34 **(Marks:01)** **Vu-Topper RM**

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

A. 3x2
B. 2x3
C. 2x2
D. 3x3

Question No:35 **(Marks:01)** **Vu-Topper RM**

A/an is one in which you want to find, not just a solution, but the best solution.

A. Optimization problem
B. Divide and Conquer
C. NP complete problem
D. Best problem

Question No:36 **(Marks:01)** **Vu-Topper RM**

Fractional Knapsack is founded on ----------method.

A. Greedy page
B. Recursive
C. Divide and Conquer
D. Dynamic programming

Question No:37 **(Marks:01)** **Vu-Topper RM**

Find the maximum value of the items which can carry using knapsack weight capacity **=50**

ITEM WEIGHT	10	20	30	70
VALUE	70	20	80	200

A. 90
B. 280
C. 200
D. 100

Question No:38 **(Marks:01)** **Vu-Topper RM**

If the graph is represented using an adjacency matrix, then Breadth-first search takes----------time.

A. O(E+1)
B. O(V^2)
C. O(V)

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D. O(E)

Question No:39 (Marks:01) Vu-Topper RM

In inductive approach of Knapsack problem, we consider 2 cases, ---------- or --.Median, Mode

A. Recursive, Iterative

B. Leave object, Take object

C. Sequentially, Parallel

Question No:40 (Marks:01) Vu-Topper RM

A Greedy algorithm can NOT be used to solve all the problems.

A. Dynamic programming

B. Memorization programming

C. Edit-distance programming

D. Storing value programming

Question No:41 (Marks:01) Vu-Topper RM

In Huffman encoding, the is the number of occurrences of a character divided by the total characters in the message.

A. Counting

B. Parsing

C. Relative Probability

D. Weight

Question No:42 (Marks:01) Vu-Topper RM

The Binary Tree constructed by a Huffman Encoding is a:

A. Full Binary Tree

B. Partial Binary Tree

C. Incomplete Binary Tree

D. None of the given option

Question No:43 (Marks:01) Vu-Topper RM

Following is not the application of Edit Distance Problem.

A. Speech recognition

B. Spelling correction

C. Ascending order

D. Computational Molecular Biology

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Question No:44 (Marks:01) Vu-Topper RM

Consider three Matrices X, Y, Z of dimensions 1x2, 2x3, 3x4 respectively. The numberof multiplication of(XYZ) is:

A. 18
B. 32
C. 24
D. 30

Question No:45 (Marks:01) Vu-Topper RM

In-----Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

A. 0
B. 1
C. 0/1
D. Fractional

Question No:46 (Marks:01) Vu-Topper RM

An in-place sorting algorithm is one that uses additional array for storage.

A. Always
B. Permanently
C. Does not
D. Sometime

Question No:47 (Marks:01) Vu-Topper RM

If Matrix-A has dimensions “pxq” and Matrix-B has dimensions “qxr”, then multiplication ofMatrix-A and Matrix-B will result a new Matrix-C having dimensions.

A. P x q
B. P x r
C. q x r
D. q x p

Question No:48 (Marks:01) Vu-Topper RM

Counting sort is suitable to sort the elements in range 1 to K.

A. K is large
B. K is small
C. K may be large or small
D. None

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Question No:49 (Marks:01) Vu-Topper RM

When matrix A of 5x 3 is multiply with matrix B of 3x 4 then the multiplication required is:

A. 15
B. 12
C. 36
D. 60

Question No:50 (Marks:01) Vu-Topper RM

---------- is a linear time sorting algorithm.

Merge sort
Quick sort
Bubble sort
Radix sort

Question No:51 (Marks:01) Vu-Topper RM

In Dynamic Programming approach, we do not store the solution to each sub problem in caseif it reappears.

A. True
B. False

Question No:52 (Marks:01) Vu-Topper RM

Dynamic Programming approach is usually useful in solving optimization problem.

A. True
B. False

Question No:53 (Marks:01) Vu-Topper RM

Which of the following algorithm provides an optimal solution for the activity selectionproblem?

A. Divide and Conquer
B. Brute force
C. Greedy
D. Recursive

Question No:54 (Marks:01) Vu-Topper RM

A graph is if every vertex can reach every other vertex.

A. Connected
B. Cycle
C. Acyclic
D. Loop

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Question No:55 (Marks:01) Vu-Topper RM

In a Huffman encoding when a new node is created by combining two nodes, the new nodeis placed in the __.

A. Priority queue
B. Linked list
C. Min heap tree
D. Graph traversal

Question No:56 (Marks:01) Vu-Topper RM

In___algorithm, you hope that by choosing a local optimum at each step, you will endup a global optimum.

A. Simple
B. Divide and conquer
C. Greedy
D. Brute Force

Question No:57 (Marks:01) Vu-Topper RM

The string "Imncde" is coded with ASCII code, the message length would be___bits.

A. 24
B. 36
C. 48
D. 60

Question No:58 (Marks:01) Vu-Topper RM

For graph traversal, breadth-first search strategy _____

A. Is always recursive
B. Cannot br recursive
C. Cannot be non-recursive
D. Can be both recursive and non-recursive

Question No:59 (Marks:01) Vu-Topper RM

In activity scheduling algorithm , the width of a rectangle ____

A. Is always ignored
B. Directs towards recursion
C. Should be maximized
D. Indicates the duration of an activity

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Question No:60 (Marks:01) Vu-Topper RM

If the graph is represented using an adjacency list, then Breadth-first search take time

A. O(V^2)
B. O(V)
C. O(V+E)
D.O(E+1)

Question No:61 (Marks:01) Vu-Topper RM

Suppose you are given infinite coins of 1,2 ,3, and 4.Select the ways of the minimum numberof coins that required to achieve a sum of 6:

A. 1
B. 2
C. 3
D. 4

Question No:62 (Marks:01) Vu-Topper RM

sing ASCII standard the string "greedy" will be encoded with

A. 48 bitS
B. 20 bits
C. 44 bits
D.40 bits

Question No:63 (Marks:01) Vu-Topper RM

The Huffman codes provide a method of data efficiency.

A. Reading/Writing
B.Encoding/Decoding
B. Divide/Conquer
C. Inserting/Deleting

Question No:64 (Marks:01) Vu-Topper RM

In the context of activity selection algorithm, time s dominated by sorting of the activities by------.

A. Start Times
B. Finish Times
C. Average Times
D.CPU Burst Times

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Question No:65 **(Marks:01)** **Vu-Topper RM**

Time complexity of the "0-1" knapsack algorithm depends on----

A. Number of items
B. Capacity of the knapsack
C. Size of the Table
D. Number of items and capacity of knapsack

Question No:66 **(Marks:01)** **Vu-Topper RM**

The greedy approach gives us an optimal solution when the coins are all powers of a --------denomination

A. Fixed
B. Variable
C. Constant
D.Static

Question No:67 **(Marks:01)** **Vu-Topper RM**

In Activity Selection, we say that two activities are non-interfering if their start-finishinterval overlap

A. Do
B. Do not
C. Sometimes
D.Once

Question No:68 **(Marks:01)** **Vu-Topper RM**

How many steps are involved to design the dynamic programming strategy?

A. 2
B. 3
C. 1
D. 4

Question No:69 **(Marks:01)** **Vu-Topper RM**

Bag is a.....

A. type of algorithm
B. data structure
C. program
D.compiler

Question No:70 **(Marks:01)** **Vu-Topper RM**

If a problem is in NP-complete, it must also be in NP.

A. True
B.False

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Question No:71 (Marks:01) Vu-Topper RM

The Huffman algorithm finds a optimal solution.

A. True

B. false

Question No:72 (Marks:01) Vu-Topper RM

The Huffman algorithm finds an exponential solution

A. True

B. False

Question No:73 (Marks:01) Vu-Topper RM

The Huffman algorithm finds a polynomial solution

A. True

B. False

Question No:74 (Marks:01) Vu-Topper RM

The greedy part of the Huffman encoding algorithm is to first find two nodes withs sallest frequency.

A. True

B. False

Question No:75 (Marks:01) Vu-Topper RM

The code word assigned to characters by the Huffman algorithm have the property that nocode word is the prefix of any other.

A. True

B.False

Question No:76 (Marks:01) Vu-Topper RM

Dijkestra's single source shortest path algorithm works if all edges weights are non-negativeand there are negative cost cycles.

A. True

B. False

Question No:77 (Marks:01) Vu-Topper RM

The term "coloring" came form the original application which was in architectural design.

A. True

B. False

Question No:78 (Marks:01) Vu-Topper RM

In the clique cover problem, for two vertices to be in the same group, they must be adjacent toeach other.

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A. True
B. False

Question No:79 (Marks:01) Vu-Topper RM
Dijkstra's algorithm is operates by maintaining a subset of vertices
A. True
B.False

Question No:80 (Marks:01) Vu-Topper RM
We do sorting to,
A. keep elements in random positions
B. keep the algorithm run in linear order
C. keep the algorithm run in (log n) order
D. keep elements in increasing or decreasing order

Question No:81 (Marks:01) Vu-Topper RM
After partitioning array in Quick sort, pivot is placed in a position such that
A. Values smaller than pivot are on left and larger than pivot are on right
B. Values larger than pivot are on left and smaller than pivot are on right
C. Pivot is the first element of array
D. Pivot is the last element of array

Question No:82 (Marks:01) Vu-Topper RM
Merge sort is stable sort, but not an in-place algorithm
A. True
B.False

Question No:83 (Marks:01) Vu-Topper RM
A p × q matrix A can be multiplied with a q × r matrix B. The result will be a p × r matrix C. There are (p . r) total entries in C and each takes____ to compute.
A. O (q)
B. O (1)
C. O (n^2)
D. O (n^3)

Question No:84 (Marks:01) Vu-Topper RM
One of the clever aspects of heaps is that they can be stored in arrays without using any ---------.
A. Pointers
B. constants

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C. variables
D.functions

Question No:85 (Marks:01) Vu-Topper RM
Merge sort requires extra array storage,
A. True
B.False

Question No:86 (Marks:01) Vu-Topper RM
The Huffman codes provide a method of encoding data **inefficiently** when coded using ASCII standard.
A. True
B. False

Question No:87 (Marks:01) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with___bits.
A. 80
B. 160
C. 320
D. 100

Question No:88 (Marks:01) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with 160 bits.
A. True
B. False

Question No:89 (Marks:01) Vu-Topper RM
Using ASCII standard the string abacdaacac will be encoded with 10 bytes.
A. True
B. False

Question No:90 (Marks:01) Vu-Topper RM
The greedy part of the Huffman encoding algorithm is to first find two nodes with **character** frequency
A. True
B. False

Question No:91 (Marks:01) Vu-Topper RM
Huffman algorithm uses a greedy approach to generate an prefix code T that minimizesth expected length B (T) of the encoded string.
A. True
B. False

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Question No:92 (Marks:01) Vu-Topper RM

An optimization problem is one in which you want to find,

A. Not a solution
B. An algorithm
C. Good solution
D. The best solution

Question No:93 (Marks:01) Vu-Topper RM

Although it requires more complicated data structures, Prim's algorithm for a minimumspanning tree is better than Kruskal's when the graph has a large number of vert

A. True
B.False

Question No:94 (Marks:01) Vu-Topper RM

If a problem is in NP, it must also be in P.

A. True
B. False
C.unknown

Question No:95 (Marks:01) Vu-Topper RM

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

A. Lists require less space than matrices but take longer to find the weight of an edge(v1,v2)
B.Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2)
C. Lists require more space than matrices and they take longer to find the weight of an edge(v1,v2)
D. Lists require more space than matrices but are faster to find the weight of an edge (v1,V2)

Question No:96 (Marks:01) Vu-Topper RM

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

A. v edges.
B. v – e + 5 edges
C. v + e edges.
D. None of these

Vu Topper RM

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Question No:97 (Marks:01) Vu-Topper RM

Maximum number of vertices in a Directed Graph may be $|V^2|$

A. True

B. False

Question No:98 (Marks:01) Vu-Topper RM

The Huffman algorithm finds a (n) _solution.

A. Optimal

B. Non-optimal

C. Exponential

D.Polynomial

Question No:99 (Marks:01) Vu-Topper RM

Edge (u, v) is a forward edge if

A. u is a proper descendant of v in the tree

B. v is a proper descendant of u in the tree pg#129

C.None of these

Question No:100 (Marks:01) Vu-Topper RM

In counting sort, once we know the ranks, we simply__numbers to their finalpositions in an output array.

A. Delete

B. copy

C. Mark

D.arrange

Question No:101 (Marks:01) Vu-Topper RM

Dynamic programming algorithms need to store the results of intermediate sub-problems.

A. True

B.False

Question No:102 (Marks:01) Vu-Topper RM

_____is a graphical representation of an algorithm

A. notation

B. notation

C. Flowchart

D.Asymptotic notation

Vu Topper RM

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Question No:103 (Marks:01) Vu-Topper RM

Which of the following is calculated with **big o notation**?

A. Lower bounds

B. Upper bounds

C. Both upper and lower bound

D.Medium bounds

Question No:104 (Marks:01) Vu-Topper RM

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

A. The array elements form a heap

B. Elements in each half of the array are sorted amongst themselves

C. Elements in the first half of the array are less than or equal to elements in the second half of the array

D. None of the above

Question No:105 (Marks:01) Vu-Topper RM

What is the solution to the recurrence T(n) = T(n/2)+n, T(1) = 1

A. O(logn)

B.O(n)

C. O(nlogn)

D. O(2n)

Question No:106 (Marks:01) Vu-Topper RM

Consider the following Huffman Tree The binary code for the string **TEA** is

A. 00 010

B. 011 00 010

C. 10 00 110

D. 11 10 110

Question No:107 (Marks:01) Vu-Topper RM

A greedy algorithm does not work in phases.

A. True

B. False

Question No:108 (Marks:01) Vu-Topper RM

Can an adjacency matrix for a directed graph ever not be square in shape?

A. Yes

B. No

Vu Topper RM

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Question No:109 (Marks:01) Vu-Topper RM

One of the clever aspects of heaps is that they can be stored in arrays without usingany_____.

A. Pointers
B. constants
C. variables
D.functions

Question No:110 (Marks:01) Vu-Topper RM

Non-optimal or greedy algorithm for money change takes____

A. O(k)
B. O(kN)
C. O(2k)
D. O(N)

Question No:111 (Marks:01) Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 320 bits.

A. True
B. False

Question No:112 (Marks:01) Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 100 bits.

A. True
B. False

Question No:113 (Marks:01) Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 32 bytes

A. True
B. False

Question No:114 (Marks:01) Vu-Topper RM

Huffman algorithm uses a greedy approach to generate an **antefix** code T that minimizes the expected length B (T) of the encoded string.

A. True
B. False

Question No:115 (Marks:01) Vu-Topper RM

Depth first search is shortest path algorithm that works on un-weighted graphs.

A. True
B.False

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Question No:116 (Marks:01) Vu-Topper RM

Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of thealgorithm is in the clever recursive formulation of the shortest path problem.

A. True

B. Flase

Question No:117 (Marks:01) Vu-Topper RM

Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursiveevaluation by generating a table for

A. k

B. d^k

C. True

D. Flase

Question No:118 (Marks:01) Vu-Topper RM

The term coloring came from the original application which was in map drawing.

A. True

B. False

Question No:119 (Marks:01) Vu-Topper RM

In the clique cover problem, for two vertices to be in the same group, they mustbe_____each other.

A. Apart from

B. Far from

C. Near to

D. Adjacent to

Question No:120 (Marks:01) Vu-Topper RM

Fixed-length codes may not be efficient from the perspective of___the total quantity of data.

A. Minimizing

B. Averaging

C. Maximizing

D. Summing

Question No:121 (Marks:01) Vu-Topper RM

In greedy algorithm, at each phase, you take the__you can get right now, without regard for future consequences.

A. Worst

B. Minimum

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C. Good
D.Best

Question No:122 (Marks:01) Vu-Topper RM
The difference between Prim s algorithm and Dijkstra s algorithm is that Dijkstra salgorithm uses a same key.
A. True
B.False

Question No:123 (Marks:01) Vu-Topper RM
If there are n items, there are__possible combinations of the items.
A. 2
B. N
C. 2^n
D. 3^n

Question No:124 (Marks:01) Vu-Topper RM
In Knapsack Problem, the thief's goal is to put items in the bag such that the _____of the items does not exceed the limit of the bag.
A. Value
B.Weight
C. Length
D.Balance

Question No:125 (Marks:01) Vu-Topper RM
The knapsack problem does not belong to the domain of optimization problems.
A. True
B.False

Question No:126 (Marks:01) Vu-Topper RM
In Huffman encoding, for a given message string, the frequency of occurrence(relative probability) of each character in the message is determined last.
A. True
B.False

Question No:127 (Marks:01) Vu-Topper RM
Fixed-length codes are known for easy break up of a string into its individual characters.
A. True
B. False

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Question No:128 (Marks:01) Vu-Topper RM

In Knapsack Problem, value and weight both are to be under consideration.

A. True

B. False

Question No:129 (Marks:01) Vu-Topper RM

Multiplication is__.

A. log n

B. n

C. n2

D. n3

Question No:130 (Marks:01) Vu-Topper RM

In DP based solution of knapsack problem, to compute entries of V we will imply a/an_approach.

A. Subjective

B. Inductive

C.Brute force

D.Combination

Question No:131 (Marks:01) Vu-Topper RM

A greedy algorithm sometimes works well for optimization problems.

A. True

B. False

Question No:132 (Marks:01) Vu-Topper RM

In Huffman encoding, frequency of each character can be determined by parsing themessage and_how many times each character (or symbol) appears.

A. Printing

B. Incrementing

C. Counting

D. Deleting

Question No:133 (Marks:01) Vu-Topper RM

Greedy algorithm can do very poorly for some problems.

A. True

B. False

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Question No:134 (Marks:01) Vu-Topper RM

The Huffman codes provide a method of__data efficiently.

A. Reading
B. Encoding
C. Decoding
D. Printing

Question No:135 (Marks:01) Vu-Topper RM

In___based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

A. Brute force
B. Dynamic programming

Question No:136 (Marks:01) Vu-Topper RM

In brute force based solution of knapsack problem, we consider 2 cases, Leave objectOr Take object.

A. TRUE
B. False

Question No:137 (Marks:01) Vu-Topper RM

_____problem, we want to find the best solution.

A. Minimization
B. Averaging
C. Optimization
D. Maximization

Question No:138 (Marks:01) Vu-Topper RM

Counting Money problem is an example which cannot be optimally solved by greedyalgorithm.

A. True
B. False

Question No:139 (Marks:01) Vu-Topper RM

Huffman algorithm generates an optimum prefix code.

A. True
B. False

Vu Topper RM

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Question No:140 (Marks:01) Vu-Topper RM

If the string "lmncde" is coded with ASCII code, the message length would be __bits.

A. 24
B. B36
C. 48 6*8=48
D. 60

Question No:141 (Marks:01) Vu-Topper RM

There are__nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

A. 2
B. 3
C. 4

Question No:142 (Marks:01) Vu-Topper RM

A number of lectures are to be given in a single lecture hall. Optimum scheduling forthis is an example of Activity selection.

A. True
B. False

Question No:143 (Marks:01) Vu-Topper RM

The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an___solution.

A. Simple
B. Sub optimal
C. Optimal
D. Non optimal

Question No:144 (Marks:01) Vu-Topper RM

The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

A. True (3*8=24)
B. False

Question No:145 (Marks:01) Vu-Topper RM

An application problem is one in which you want to find, not just a solution, but the ___solution.

A. Simple
B. Good
C. Best

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Question No:146 (Marks:01) Vu-Topper RM

Suppose that a graph G = (V,E) is implemented using adjacency lists. What is thecomplexity of a breadth-first traversal of G?

A. O(|V |^2)
B. O(|V | |E|)
C. O(|V |^2|E|)
D. O(|v|+|E|)

Question No:147 (Marks:01) Vu-Topper RM

Which is true statement?

A. Breadth first search is shortest path algorithm that works on un-weighted graphS
B. Depth first search is shortest path algorithm that works on un-weighted graphs.
C. Both of above are true.
D. None of above are true.

Question No:148 (Marks:01) Vu-Topper RM

Using ASCII standard the string “abacdaacacwe” will be encoded with _____ bits

A. 64
B. 128
C. 96
D. 120

Question No:149 (Marks:01) Vu-Topper RM

Forward edge is:

A. (u, v) where u is a proper descendent of v in the tree.
B. (u, v) where v is a proper descendent of u in the tree.
C. (u, v) where v is a proper ancesstor of u in the tree.
D. (u, v) where u is a proper ancesstor of v in the tree.

Question No:150 (Marks:01) Vu-Topper RM

In digraph G=(V,E) ;G has cycle if and only if

A. The DFS forest has forward edge.
B. The DFS forest has back edge
C. The DFS forest has both back and forward edge
D. BFS forest has forward edge

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Question No:151 (Marks:01) Vu-Topper RM

Back edge is:

A. (u, v) where v is an ancestor of u in the tree.

B. (u,v) where u is an ancesstor of v in the tree.

C. (u, v) where v is an predcessor of u in the tree.

D. None of above

Question No:152 (Marks:01) Vu-Topper RM

Cross edge is :

A. (u, v) where u and v are not ancestor of one another

B. (u, v) where u is ancesstor of v and v is not descendent of u.

C.(u, v) where u and v are not ancestor or descendent of one another

C. (u, v) where u and v are either ancestor or descendent of one another.

Question No:153 (Marks:01) Vu-Topper RM

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose besttree edge) when the graph has relatively few edges.

A. True

B. False

Question No:154 (Marks:01) Vu-Topper RM

What algorithm technique is used in the implementation of Kruskal solution for theMST?

A. Greedy Technique

B. Divide-and-Conquer Technique

C. Dynamic Programming Technique

D. The algorithm combines more than one of the above techniques

Question No:155 (Marks:01) Vu-Topper RM

The _____ given by DFS allow us to determine whether the graph contains any cycles.

A. Order

B. **Time stamps**

C. BFS traversing

D. Topological sort

Question No:156 (Marks:01) Vu-Topper RM

In Prim's algorithm, we will make use of_____.

A. Priority Queue

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Question No:157 (Marks:01) Vu-Topper RM

_____ technique is look like propagating wave-front outward.

Breath first traversal

Question No:158 (Marks:01) Vu-Topper RM

What is the time complexity to extract a vertex from the priority queue in Prim'salgorithm?

A. O (log E)
B. (V)
C. (V+E)
D. O(log V)

Question No:159 (Marks:01) Vu-Topper RM

The relationship between number of back edges and number of cycles in DFS is,

A. Both are equal
B. Back edges are half of cycles
C. Back edges are one quarter of cycles
D. There is no relationship between no. of edges and cycles

Question No:160 (Marks:01) Vu-Topper RM

You have an adjacency list for G, what is the time complexity to compute Graphtranspose G^T.?

A. (V + E)
B. (V E)
C. (V)
D. (V^2)

Question No:161 (Marks:01) Vu-Topper RM

There is relationship between number of back edges and number of cycles in DFS

A. Both are equal.
B.Cycles are half of back edges.
C. Cycles are one fourth of back edges.
D. There is no relationship between back edges and number of cycles.

Question No:162 (Marks:01) Vu-Topper RM

A digraph is strongly connected under what condition?

A. A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v .

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B.A digraph is strongly connected if for every pair of vertices u, v e V, u can reach v and viceversa.

B. A digraph is strongly connected if for at least one pair of vertex u, v e V, u can reach v and viceversa.

C. A digraph is strongly connected if at least one third pair of vertices u, v e V, u can reach v andvice versa.

Question No:163 (Marks:01) Vu-Topper RM

In in-place sorting algorithm is one that uses arrays for storage :

A. An additional array

B. No additional array

C.Both of above may be true according to algorithm

D.More than 3 arrays of one dimension.

Question No:164 (Marks:01) Vu-Topper RM

In stable sorting algorithm

A. One array is used

B. In which duplicating elements are not handled.

C. More than one arrays are required.

D. Duplicating elements remain in same relative position after sorting.

Question No:165 (Marks:01) Vu-Topper RM

Which sorting algorithm is faster :

A. O(n^2)

B. O(nlogn)

C. O(n+k)

D. O(n^3)

Question No:166 (Marks:01) Vu-Topper RM

In Quick sort algorithm, constants hidden in T(n lg n) are

A. Large

B. Medium

C. Not known

D. Small

Question No:167 (Marks:01) Vu-Topper RM

Quick sort is based on divide and conquer paradigm; we divide the problem on base ofpivot element and:

A. There is explicit combine process as well to conquer the solution.

B. No work is needed to combine the sub-arrays, the array is already sorted

C. Merging the sub arrays

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D.None of the above

Question No:168 (Marks:01) Vu-Topper RM

Dijkstra's algorithm :

A. Has greedy approach to find all shortest paths
B. Has both greedy and Dynamic approach to find all shortest paths
C. Has greedy approach to compute single source shortest paths to all other vertices
D. Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

Question No:169 (Marks:01) Vu-Topper RM

Which may be stable sort:

A. Bubble sort
B. Insertion sort
C. Selection sort
D.Both of above

Question No:170 (Marks:01) Vu-Topper RM

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent __ series in the analysis,

A. linear
B. arithmetic
C. exponent
D. Geometric

Question No:171 (Marks:01) Vu-Topper RM

Which may be stable sort:

A. Bubble sort
B. Insertion sort
C. Selection sort
D. Both of above

Question No:172 (Marks:01) Vu-Topper RM

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent __ series in the analysis,

A. linear
B. arithmetic
C. exponent
D. Geometric

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Question No:173 (Marks:01) Vu-Topper RM

How much time merge sort takes for an array of numbers?

A. T(n^2)

B. T(n)

C. T(log n)

D. T(n log n)

Question No:174 (Marks:01) Vu-Topper RM

We do sorting to,

A. keep elements in random positions

B. keep the algorithm run in linear order

C. keep the algorithm run in (log n) order

D.keep elements in increasing or decreasing order

Question No:175 (Marks:01) Vu-Topper RM

Dynamic programming algorithms need to store the results of intermediate sub-problems.

A. True

B. False

Question No:176 (Marks:01) Vu-Topper RM

Dijkstras single source shortest path algorithm works if all edges weightsare non negative and there are no negative cost cycles.

A. False

B. True

Question No:177 (Marks:01) Vu-Topper RM

Dijkestra s single source shortest path algorithm works if all edges weights arenegative and there are no negative cost cycles.

A. True

B. False

Question No:178 (Marks:01) Vu-Topper RM

In the clique cover problem, for two vertices to be in the same group, they must be____each other.

A. Apart from

B. Far from

C. Near to

D.Adjacent to

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Question No:179 (Marks:01) Vu-Topper RM

Fixed-length codes may not be efficient from the perspective of ___the total quantity of data.

A. Minimizing

B.Averaging

C. Maximizing

D. Summing

Question No:180 (Marks:01) Vu-Topper RM

In___based solution of knapsack problem, we consider 2 cases, Leaveobject Or Take object.

A. Brute force

B. Dynamic programming

Question No:181 (Marks:01) Vu-Topper RM

In brute force based solution of knapsack problem, we consider 2 cases, Leaveobject Or Take object.

A. TRUE

B. False

Question No:182 (Marks:01) Vu-Topper RM

_____problem, we want to find the best solution.

A. Minimization

B. Averaging

C. Optimization

D. Maximization

Question No:183 (Marks:01) Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

A. True

B. False

Question No:184 (Marks:01) Vu-Topper RM

How many elements do we eliminate in each time for the Analysis of Selectionalgorithm?

A. (n / 2) + n elements

B. n / 4 elements

C. 2 n elements

D.n / 2 elements

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Question No:185 (Marks:01) Vu-Topper RM

Slow sorting algorithms run in,

A. T(n^2)
B. T(n)
C. T(log n)
D. T(n log n)

Question No:186 (Marks:01) Vu-Topper RM

Counting sort is suitable to sort the elements in range 1 to k:

A. K is large
B. K is small
C. K may be large or small
D. None

Question No:187 (Marks:01) Vu-Topper RM

Heaps can be stored in arrays without using any pointers; this is due to the _____nature of the binary tree,

A. left-complete
B. right-complete
C. tree nodes
D. tree leaves

Question No:188 (Marks:01) Vu-Topper RM

A heap is a left-complete binary tree that conforms to the ____

A. increasing order only
B. **heap order**
C. decreasing order only
D. (log n) order

Question No:189 (Marks:01) Vu-Topper RM

Divide-and-conquer as breaking the problem into a small number of

A. pivot
B. Sieve
C. smaller sub problems
D. Selection

Question No:190 (Marks:01) Vu-Topper RM

In Sieve Technique we do not know which item is of interest

A. True
B. False

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Question No:191 (Marks:01) Vu-Topper RM

The recurrence relation of Tower of Hanoi is given below T(n)={1 if n=1 and 2T(n-1) if n >1 In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

A. 16
B. 10
C. 32
D. 31

Question No:192 (Marks:01) Vu-Topper RM

For the heap sort, access to nodes involves simple _____ operations.

A. arithmetic
B. binary
C. algebraic
D. logarithmic

Question No:193 (Marks:01) Vu-Topper RM

For the sieve technique we solve the problem,

A. recursively
B. mathematically
C. precisely
D. accurately

Question No:194 (Marks:01) Vu-Topper RM

The sieve technique works in_as follows

A. phases
B. numbers
C. integers
D. routines

Question No:195 (Marks:01) Vu-Topper RM

A (an)____is a left-complete binary tree that conforms to the heaporder

A. heap
B. binary tree
C. binary search tree
D. array

Vu Topper RM

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Question No:196 (Marks:01) Vu-Topper RM

The sieve technique is a special case, where the number of sub problems is just

A. 5
B. Many
C. 1
D. Few

Question No:197 (Marks:01) Vu-Topper RM

Analysis of Selection algorithm ends up with,

A. T(n)
B. T(1 / 1 + n)
C. T(n / 2)
D. T((n / 2) + n)

Question No:198 (Marks:01) Vu-Topper RM

For the heap sort we store the tree nodes in

A. level-order traversal
B. in-order traversal
C. pre-order traversal
D. post-order traversal

Question No:199 (Marks:01) Vu-Topper RM

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

A. divide-and-conquer
B. decrease and conquer
C. greedy nature
D. 2-dimension Maxima

Question No:200 (Marks:01) Vu-Topper RM

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

A. n items
B. phases
C. pointers
D. constant

Question No:201 (Marks:01) Vu-Topper RM

Memorization is?

A. To store previous results for future use

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B. **To avoid this unnecessary repetitions by writing down the results of recursive calls andlooking them up again if we need them later**
C. To make the process accurate
D. None of the above

Question No:202 (Marks:01) Vu-Topper RM

Quick sort is

A. Not stable but in place
B. Stable & in place
C. Stable but not in place
D. Some time stable & some times in place

Question No:203 (Marks:01) Vu-Topper RM

One example of in place but not stable algorithm is

A. Merger Sort
B.Quick Sort
B. Continuation Sort
C. Bubble Sort

Question No:204 (Marks:01) Vu-Topper RM

Continuation sort is suitable to sort the elements in range 1 to k

A. K is Large
B. K is not known
C. K may be small or large
D.K is small

Question No:205 (Marks:01) Vu-Topper RM

Which may be a stable sort?

A. Merger
B. Insertion
C. None of the above
D. Both above

Question No:206 (Marks:01) Vu-Topper RM

An in place sorting algorithm is one that uses____arrays for storage

A. Two dimensional arrays
B. More than one array
C. **No Additional Array**
D. None of the above

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Question No:207 (Marks:01) Vu-Topper RM

Continuing sort has time complexity of ?

A. O(n)
B. O(n+k)
C. O(nlogn)
D. O(k)

Question No:208 (Marks:01) Vu-Topper RM

single item from a larger set of _____

A. phases
B. **n items**
C. pointers
D. constant

Question No:209 (Marks:01) Vu-Topper RM

For the Sieve Technique we take time

A. T(nk)
B. T(n / 3)
C. n^2
D. n/3

Question No:210 (Marks:01) Vu-Topper RM

Due to left complete nature of binary tree, the heap can be stored in

A. Arrays
B. Structures
C. Link List
D. Stack

Question No:211 (Marks:01) Vu-Topper RM

What type of instructions Random Access Machine (RAM) can execute?

A. Algebraic and logic
B. Geometric and arithmetic
C. **Arithmetic and logic**
D. Parallel and recursive

Question No:212 (Marks:01) Vu-Topper RM

What is the total time to heapify?

A. O(log n)
B. O(n log n)
C. $O(n^2 \log n)$
D. $O(\log^2 n)$

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Question No:213 (Marks:01) Vu-Topper RM

word Algorithm comes from the name of the muslim author __

A. Abu Ja'far Mohammad ibn Musa al-Khowarizmi.

Question No:214 (Marks:01) Vu-Topper RM

Al-Khwarizmi's work was written in a book titled ____

A. al Kitab al-mukhatasar fi hisab al-jabr wa'l-muqabalah

Question No:215 (Marks:01) Vu-Topper RM

Random access machine or RAM is a/an

A. Machine build by Al-Khwarizmi
B. Mechanical machine
C. Electronics machine
D. Mathematical model

Question No:216 (Marks:01) Vu-Topper RM

A RAM is an idealized machine with_ random-accessmemory.

A. 256MB
B. 512MB
C. 100GB
D. an infinitely large

Question No:217 (Marks:01) Vu-Topper RM

What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

A. 2 n
B. 2 n n
C. n
D. 8 n

Question No:218 (Marks:01) Vu-Topper RM

Consider the following code:For(j=1; j<n;j++)
For(k=1; k<15;k++)
For(l=5; l<n; l++)
{
Do_something_constant();
}

What is the order of execution for this code.

A. O(*n*)
B. O(n^3)

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C. O(n^2 log n)
D. O(n^2)

Question No:219 (Marks:01) Vu-Topper RM
Is it possible to sort without making comparisons?
A. Yes
B. No

Question No:220 (Marks:01) Vu-Topper RM
When we call heapify then at each level the comparison performed takes time
A. It will take Θ (1)
B. Time will vary according to the nature of input data
C. It can not be predicted
D. It will take Θ (log n)

Question No:221 (Marks:01) Vu-Topper RM
In Quick sort, we don't have the control over the sizes of recursive calls
A. True
B. False
C. Less information to decide
D. Either true or false

Question No:222 (Marks:01) Vu-Topper RM
For Chain Matrix Multiplication we cannot use divide and conquer approach because,
A. We do not know the optimum k
B. We use divide and conquer for sorting only
C. We can easily perform it in linear time
D. Size of data is not given

Question No:223 (Marks:01) Vu-Topper RM
Suppose we have three items as shown in the following table, and suppose thecapacity of the knapsack is 50 i.e. W = 50.

Item	Value	Weigh t
1	60	10
2	100	20
3	120	30

The optimal solution is to pick
A. Items 1 and 2

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B. Items 1 and 3
C. Items 2 and 3
D. None of these

Question No:224 (Marks:01) Vu-Topper RM
who invented the quick sort
A. C.A.R. Hoare

Question No:225 (Marks:01) Vu-Topper RM
main elements to a divide-and-conquer
A. Divide, conquer, combine

Question No:226 (Marks:01) Vu-Topper RM
Mergesort is a stable algorithm but not an in-place algorithm.
A. True
B. False

Question No:227 (Marks:01) Vu-Topper RM
Counting sort the numbers to be sorted are in the range 1 to k where k is small.
A. True
B. False

Question No:228 (Marks:01) Vu-Topper RM
In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the
A. Convergent geometric series
B. Divergent geometric series
C. None of these

Question No:229 (Marks:01) Vu-Topper RM
In RAM model instructions are executed
A. One after another
B. Parallel

Question No:230 (Marks:01) Vu-Topper RM
Due to left-complete nature of binary tree, heaps can be stored in
A. Link list
B. Structure
C. Array
D. None of above

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Question No:231 (Marks:01) Vu-Topper RM

The time assumed for each basic operation to execute on RAM model of computation is-----

A. Constant
B. Infinite
C. Continuous
D. Variable

Question No:232 (Marks:01) Vu-Topper RM

If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

A. True
B.False

Question No:233 (Marks:01) Vu-Topper RM

Brute-force algorithm uses no intelligence in pruning out decisions.

A. True
B. False

Question No:234 (Marks:01) Vu-Topper RM

In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

A. True
B. False

Question No:235 (Marks:01) Vu-Topper RM

For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large

A. True .
B.False

Question No:236 (Marks:01) Vu-Topper RM

In simple brute-force algorithm, we give no thought to efficiency.

A. True
B. False

Question No:237 (Marks:01) Vu-Topper RM

The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

A. True
B. False

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Question No:238 (Marks:01) Vu-Topper RM

If the indices passed to merge sort algorithm are___,then this means thatthere is only one element to sort.

A. Small
B. Large
C. Equal
D. Not Equal

Question No:239 (Marks:01) Vu-Topper RM

In pseudo code, the level of details depends on intended audience of the algorithm.

A. True
B. False

Question No:240 (Marks:01) Vu-Topper RM

In 2d-space a point is said to be_____if it is not dominated by any other point in that space.

A. Member
B. Minimal
C. Maximal
D. Joint

Question No:241 (Marks:01) Vu-Topper RM

An algorithm is a mathematical entity that is dependent on a specific programming language.

A. True
B. False

Question No:242 (Marks:01) Vu-Topper RM

The running time of an algorithm would not depend upon the optimization bythe compiler but that of an implementation of the algorithm would depend on it.

A. TRUE
B. False

Question No:243 (Marks:01) Vu-Topper RM

F (n) and g (n) are asymptotically equivalent. This means that they have essentially the same_____ for large n.

A. Results
B. Variables
C. **Growth rates**

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D. Size

Question No:244 (Marks:01) Vu-Topper RM

8n2 + 2n - 3 will eventually exceed c2*(n) no matter how large we make c2.

A. TRUE

B. False

Question No:245 (Marks:01) Vu-Topper RM

If we associate (x, y) integers pair to cars where x is the speed of the car and yis the negation of the price. High y value for a car means a__car.

A. Fast

B. Slow

C. Cheap

D. Expensive

Question No:246 (Marks:01) Vu-Topper RM

While solving Selection problem, in Sieve technique we partition input data

A. In increasing order

B. In decreasing order

C. According to Pivot

D. Randomly

Question No:247 (Marks:01) Vu-Topper RM

In Sieve Technique, we know the item of interest.

A. True

B. False

Question No:248 (Marks:01) Vu-Topper RM

In Heap Sort algorithm, we build____for ascending sort.

A. Max heap

B. Min heap

Question No:249 (Marks:01) Vu-Topper RM

Quick sort is best from the perspective of Locality of reference.

A. True

B. False

Question No:250 (Marks:01) Vu-Topper RM

While Sorting, the ordered domain means for any two input elements x and y _____satisfies only.

A. x < y

B. x > y

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C. x = y
D. All of the above

Question No:251 (Marks:01) Vu-Topper RM
Algorithm is a mathematical entity, which is independent of a specificmachine and operating system.
A. True
B. False

Question No:252 (Marks:01) Vu-Topper RM
In Heap Sort algorithm, the total running time for Heapify procedure is
A. Theta (log n)
B. Order (log n)
C. Omega (log n)
D. O (1)

Question No:253 (Marks:01) Vu-Topper RM
A point p in 2-dimensional space is usually given by its integercoordinate(s)_
A. p.x only p.y
B. only p.x & p.z
C. p.x & p.y

Question No:254 (Marks:01) Vu-Topper RM
In Heap Sort algorithm, the maximum levels an element can move upward is
A. Theta (log n)
B. Order (log n)
C. Omega (log n)
D. O

Question No:255 (Marks:01) Vu-Topper RM
Floor and ceiling are____to calculate while analyzing algorithms.
A. Very easy
B. Usually considered difficult

Question No:256 (Marks:01) Vu-Topper RM
_____is one of the few problems, where provable lower bounds exist onhow fast we can sort.
A. Searching
B. Sorting
C. Both Searching & Sorting
D. Graphing

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Question No:257 (Marks:01) Vu-Topper RM

A RAM is an idealized algorithm with takes an infinitely large random-access memory.

A. True
B. False

Question No:258 (Marks:01) Vu-Topper RM

Upper bound requires that there exist positive constants c2 and n0 such thatf(n) _____c2n for all n <= n0(ye question ghalat lag raha hai mujhae

A. Less than
B. Equal to or Less than
C. Equal or Greater than
D. Greater than

Question No:259 (Marks:01) Vu-Topper RM

In Heap Sort algorithm, if heap property is violated _____

A. We call Build heap procedure
B. We call Heapify procedure
C. We ignore
D. Heap property can never be violated

Question No:260 (Marks:01) Vu-Topper RM

In___we have to find rank of an element from given input.

A. Merge sort algorithm
B. Selection problem
C. Brute force technique
D. Plane Sweep algorithm

Question No:261 (Marks:01) Vu-Topper RM

A point p in 2-dimensional space is usually given by its integercoordinate(s)_

A. p.x only
B. p.y only
C. p.x & p.z
D. p.x & p.y

Question No:262 (Marks:01) Vu-Topper RM

In Quick Sort Constants hidden in T(n log n) are

A. Large
B. Medium
C. Small
D. Not Known

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Question No:263 (Marks:01) Vu-Topper RM

The running time of quick sort depends heavily on the selection of

A. No of inputs
B. Arrangement of elements in array
C. Size o elements
D. Pivot elements

Question No:264 (Marks:01) Vu-Topper RM

Choose one Counting sort has timecomplexity:

A. O(n) (Page 58)
B. O(n+k)
C. O(k)
D. O(nlogn)

Question No:265 (Marks:01) Vu-Topper RM

In place stable sorting algorithm.

A. If duplicate elements remain in the same relative position after sorting
B. One array is used
C. More than one arrays are required
D. Duplicating elements not handled

Question No:266 (Marks:01) Vu-Topper RM

Cont sort is suitable to sort the elements in range 1 to k

A. K is Large
B. K is not known
C. K may be small or large
D. K is small

Question No:267 (Marks:01) Vu-Topper RM

The sieve technique works where we have to find_item(s) from alarge input.

A. Single
B. Two
C. Three
D. Similar

Question No:268 (Marks:01) Vu-Topper RM

A RAM is an idealized machine with_random-accessmemory.

A. 256MB
B. 512MB
C. an infinitely large

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D. 100GB

Question No:269 (Marks:01) Vu-Topper RM

A RAM is an idealized machine with_random-accessmemory.

A. 256MB
B. 512MB
C. an infinitely large
D. 100GB

Question No:270 (Marks:01) Vu-Topper RM

For the sieve technique we solve the problem,

A. recursively
B. mathematically
C. precisely
D. accurately

Question No:271 (Marks:01) Vu-Topper RM

number of nodes in a complete binary tree of height h is

A. 2^(h+1) – 1
B. 2 * (h+1) – 1
C. 2 * (h+1)
D. ((h+1) ^ 2) – 1 25

Question No:272 (Marks:01) Vu-Topper RM

Heaps can be stored in arrays without using any pointers; this is due to the _____nature of the binary tree,

A. left-complete
B. right-complete
C. tree nodes
D. tree leaves

Question No:273 (Marks:01) Vu-Topper RM

Sorting is one of the few problems where provable_bonds exits onhow fast we can sort,

A. upper
B. lower
C. average
D. log n

Vu Topper RM

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Question No:274 (Marks:01) Vu-Topper RM

The analysis of Selection algorithm shows the total running time is indeed _____in no

A. arithmetic
B. geometric
C. linear
D. orthogonal

Question No:275 (Marks:01) Vu-Topper RM

What algorithm technique is used in the implementation of Kruskal solution for the MST?

A. Greedy Technique
B. Divide-and-Conquer Technique
C. Dynamic Programming Technique
D. The algorithm combines more than one of the above techniques

Question No:276 (Marks:01) Vu-Topper RM

one If you find yourself in maze the better traversal approach will be :

A. BFS
B. BFS and DFS both are valid
C. Level order
D. DFS

Question No:277 (Marks:01) Vu-Topper RM

In digraph G=(V,E) ;G has cycle if and only if

A. The DFS forest has forward edge.
B. The DFS forest has back edge (Page 131)
C. The DFS forest has both back and forward edge
D. BFS forest has forward edge

Question No:278 (Marks:01) Vu-Topper RM

Suppose that a graph G = (V,E) is implemented using adjacency lists. What isthe complexity of a breadth-first traversal of G?

A. O(|V |^2)
B. O(|V | |E|)
C. O(|V |^2|E|)
D. O(|V | + |E|)

Question No:279 (Marks:01) Vu-Topper RM

There are___nested loops in DP based algorithm for computing theminimum cost of chain matrix multiplication.

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A. 2
B. 3
C. 4
D. 5

Question No:280 (Marks:01) Vu-Topper RM

Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

A. True
B. False

Question No:281 (Marks:01) Vu-Topper RM

Who invented Quick sort procedure?

A. Hoare
B. Sedgewick
C. Mellroy
D. Coreman

Question No:282 (Marks:01) Vu-Topper RM

An activity scheduling algorithm, the width of a rectangle--.

A. Is always ignored
B. Direct toward recursion
C. Should be maximized
D. Indicates the duration of an activity

Question No:283 (Marks:01) Vu-Topper RM

In recursive formulationof Knapsack Problem: V[0, j] = for j>=0

A. -1
B. 0
C. 1
D. 2

Question No:284 (Marks:01) Vu-Topper RM

The Probability in Huffman encoding is the number of occurrences of a character divided by the total----------in the message.

A. Numbers
B. Frequencies
C. Strings
D. Characters

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Question No:285 (Marks:01) Vu-Topper RM

Graphs can be represented by an---------- and --.

A. queue , stack

B.adjacency list , adjacency matrix

C.adjacency right , adjacency left

D.Binary , linear

Question No:286 (Marks:01) Vu-Topper RM

Edge weights can be interpreted as distance_.

A. in breadth-First Search

B. in Queue's

C. **in the shortest-paths**

D. in depth-First Search

Question No:287 (Marks:01) Vu-Topper RM

_____algorithm allows negative weights edges and no negative costcycles.

A. Brute-force technique

B. **Bellman-Ford**

C. Dijkstra's

D. Print

Question No:288 (Marks:01) Vu-Topper RM

According to parenthesis lemma vertex u is a descendent of v vertex if and only if:

A. f [d[u], f[u]] ⊆ [d[v], f[v]] (Page 129)

Question No:289 (Marks:01) Vu-Topper RM

There exist a unique path between any _____ vertices of a free tree.

A. Two

Question No:290 (Marks:01) Vu-Topper RM

Keeping in mind the shortest-path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are_____and Edges are _____which may be wired or wireless.

A. Routers, communication links

Question No:291 (Marks:01) Vu-Topper RM

In undirected graph, by convention all the edges are called _____ edges.

A. back

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Question No:292 (Marks:01) Vu-Topper RM

Bellman-Ford algorithm is slower than_____.

A. Dijkstra's

Question No:293 (Marks:01) Vu-Topper RM

From the given option's which one is correct regarding the time complexity of Dijkstra's algorithm:

A. O(N^2)

Question No:294 (Marks:01) Vu-Topper RM

In the shortest-paths problem, we are given a weighted of_____G = (V, E).

A. Un-directed graph

Question No:295 (Marks:01) Vu-Topper RM

You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T ?

A. (V+E)

Question No:296 (Marks:01) Vu-Topper RM

If you find yourself in maze the better traversel approach will be :

A. BFS

B. BFS and DFS both are valid

C. Level order

D. DFS

Question No:297 (Marks:01) Vu-Topper RM

In strong components algorithm, first of all DFS is run for computing finish times of vertices.

A. [d[u], f[u]]⊆[d[v], f[v]]

B. [d[u], f[u]]⊇[d[v], f[v]]

C. **unrelated**

D. Disjoint

Question No:298 (Marks:01) Vu-Topper RM

The relationship between number of back edges and number of cycles in DFS is,

A. Both are equal

B. Back edges are half of cycles

C. Back edges are one quarter of cycles

D. **There is no relationship between no. of edges and cycles**

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Question No:299 (Marks:01) Vu-Topper RM

What general property of the list indicates that the graph has an isolated vertex?

A. There is Null pointer at the end of list.
B. **The Isolated vertex is not handled in list.**
C. Only one value is entered in the list.
D. There is at least one null list.

Question No:300 (Marks:01) Vu-Topper RM

As the Kruskal's runs, the edges in viable set A induce a _____ on the vertices.

A. Set
B. Graph
C. Tree
D. **Fore**

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بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

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